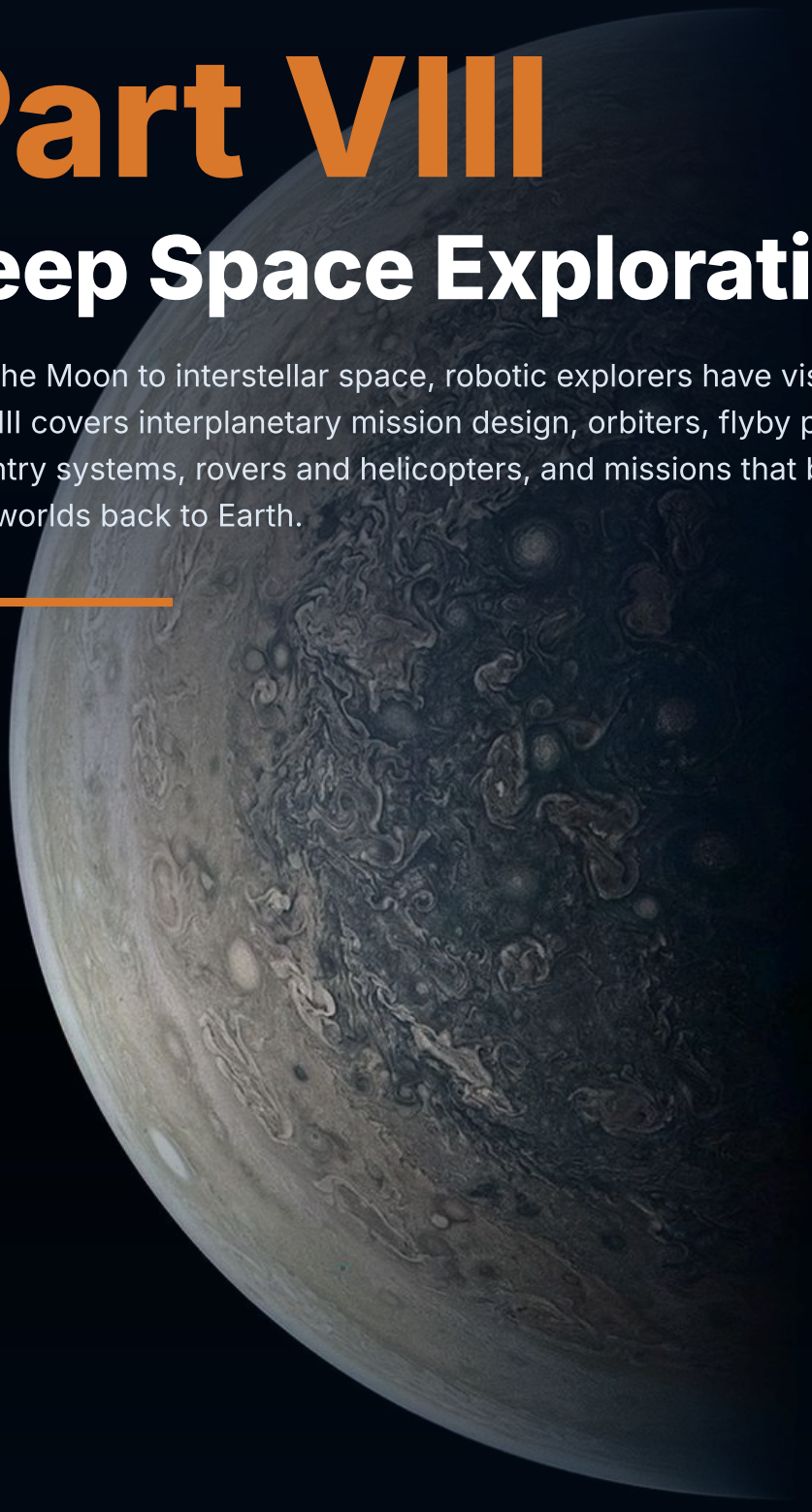
Space Exploration & Aerospace Engineering

Part VIII

Deep Space Exploration

From the Moon to interstellar space, robotic explorers have visited every planet. Part VIII covers interplanetary mission design, orbiters, flyby probes, landers and entry systems, rovers and helicopters, and missions that bring samples of other worlds back to Earth.



Chapters 24–29 · pages 430–525 of the complete manual
First Edition · Current to October 2026 · Written in plain language with deep technical detail

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First Edition · Information current to October 2026
A4 format · Fourteen Parts · Fifty-three chapters · Ten appendices
Version 1.0 · October 2026

About this edition

Space Exploration & Aerospace Engineering — An Engineering Reference Manual. First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

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Contents of Part VIII

PART VIII · DEEP SPACE EXPLORATION

24 Deep-Space Mission Design 431

24.1 The Challenge of Deep Space 431
24.2 Interplanetary Trajectories 432
24.3 Gravity Assists 435
24.4 Low-Thrust Trajectories 436
24.5 Arrival: Orbit Insertion, Aerobraking and Aerocapture 437
24.6 Cislunar Space and Libration Points 438
24.7 Navigating in Deep Space 438
24.8 Communications, Power and Thermal Design 440
24.9 Planetary Protection 442
24.10 From Idea to Arrival: The Mission Life Cycle 443
24.11 The Deep-Space Fleet in 2026 443

25 Planetary Orbiters 446

25.1 Why Orbit? 446
25.2 Designing Orbits Around Other Worlds 447
25.3 Getting Into Orbit 450
25.4 What Orbiters Carry 453
25.5 Orbiters World by World 455
25.6 Ending an Orbiter Mission 460

26 Space Probes and Flyby Missions 462

26.1 Kinds of Probes 463
26.2 Sixty Years of Flybys 463
26.3 The Grand Tour: Pioneer and Voyager 464
26.4 Engineering a Flyby 468
26.5 Power and Communications Far from the Sun 471
26.6 Atmospheric Entry Probes 473
26.7 Impactors and Planetary Defence 475
26.8 Comets, Asteroids and Interstellar Visitors 478
26.9 Into Interstellar Space 478
26.10 Probes in 2026 480

27 Landers and Entry, Descent & Landing 482

27.1 Why Land? 483
27.2 The Landing Problem 483
27.3 Landing on the Moon 484
27.4 Entry, Descent and Landing at Mars 489
27.5 Venus, Titan and Small Bodies 495
27.6 Surviving on the Surface 497

28 Planetary Rovers and Aerial Explorers 499

28.1 Sixty Years of Rovers 500
28.2 Engineering a Rover 503
28.3 What Rovers Have Found 507
28.4 Aerial Explorers 508
28.5 Rovers at the Lunar Poles 511
28.6 The Next Generation 512

29 Sample-Return Missions 514

29.1 Sample-Return Missions So Far 515

29.2 Engineering a Sample Return 520
29.3 Mars Sample Return and the Next Decade 524

This volume is one Part of *Space Exploration & Aerospace Engineering — An Engineering Reference Manual*. Page numbers, chapter numbers and figure numbers are the same as in the complete manual, so cross-references to other chapters (for example “Chapter 24”) and to the appendices refer to the complete book.



PART VIII

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From the Moon to interstellar space, robotic explorers have visited every planet. Part VIII covers interplanetary mission design, orbiters, flyby probes, landers and entry systems, rovers and helicopters, and missions that bring samples of other worlds back to Earth.

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29 Sample-Return Missions

Image: NASA/JPL (NASA/JPL-Caltech/SwRI/MSSS)

24 CHAPTER Deep-Space Mission Design

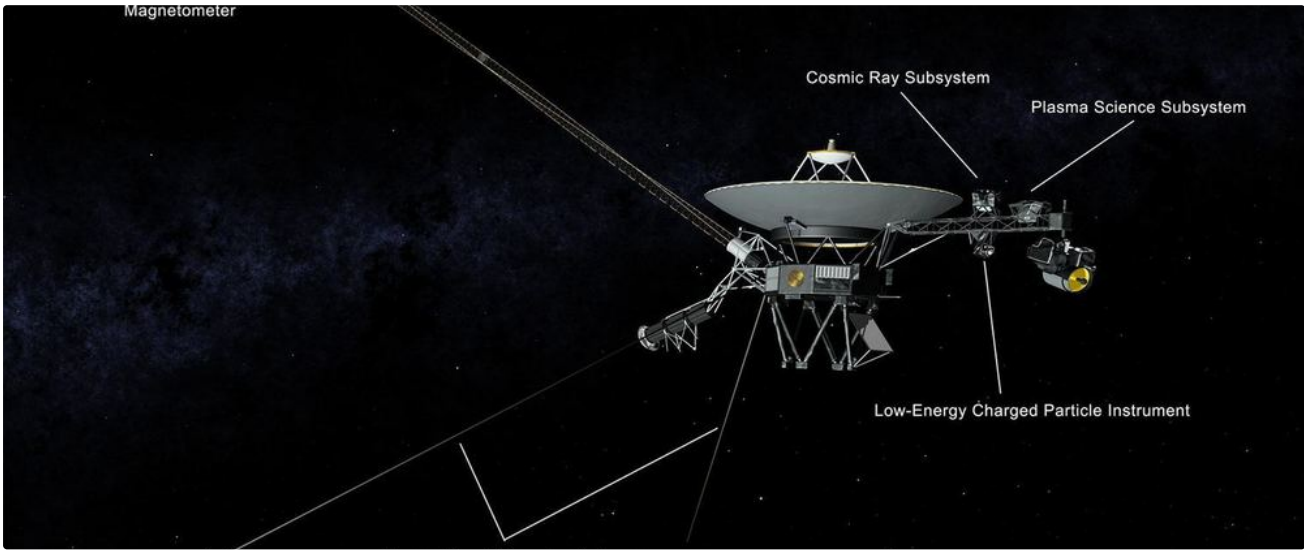


Image: NASA/JPL (NASA/JPL-Caltech)

Sending a spacecraft to another world is an exercise in patience and precision. Launch windows open only every few years; journeys last months to decades; signals take minutes to hours to arrive; sunlight fades; radiation, dust and extreme temperatures test every component. This chapter explains how deep-space missions are designed: interplanetary trajectories and launch windows — with launch opportunities to Mars computed for 2026–2041 — gravity assists, low-thrust electric propulsion, orbit insertion and aerobraking, cislunar and libration-point orbits, deep-space navigation and communications, power and thermal design across the Solar System, planetary protection, and the life cycle and cost of a flagship mission. It ends with the fleet of deep-space missions active in 2026.

IN THIS CHAPTER

24.1 The Challenge of Deep Space
24.2 Interplanetary Trajectories
24.3 Gravity Assists
24.4 Low-Thrust Trajectories
24.5 Arrival: Orbit Insertion, Aerobraking and Aerocapture
24.6 Cislunar Space and Libration Points
24.7 Navigating in Deep Space
24.8 Communications, Power and Thermal Design
24.9 Planetary Protection
24.10 From Idea to Arrival: The Mission Life Cycle
24.11 The Deep-Space Fleet in 2026

24.1 The Challenge of Deep Space

Deep-space missions — to the Moon, planets, asteroids, comets, the Sun and beyond — face constraints that Earth-orbiting satellites rarely meet:

- **Energy and time**: escaping the Earth and changing orbit around the Sun require velocity changes of several kilometres per second. Journeys last from days (to the Moon) to more than a decade (to the outer planets).
- **Launch windows** dictated by planetary alignment, which for Mars open only once every 26 months.
- **Distance**: radio signals take up to hours to arrive, so spacecraft must operate autonomously and data rates are limited.
- **Power and temperature**: sunlight falls with the square of the distance from the Sun, and spacecraft experience extremes from the heat near Mercury or the Sun to the cold of the outer Solar System.
- **Environment**: intense radiation at Jupiter, dust near comets, unknown surfaces at landing sites.
- **No second chance**: a missed orbit insertion or a failed landing usually ends the mission.

Missions are commonly classified as **flybys**, **orbiters**, **landers**, **rovers** and **sample-return** missions (Chapters 25–29), and grouped by agencies into cost classes. NASA's **Discovery** programme (cost cap around $500–600 million) and **New Frontiers** programme (around $1 billion) compete missions among scientists, while **flagship** missions such as Cassini, Perseverance and Europa Clipper cost several billion dollars. ESA's Science Programme distinguishes small, medium and large missions; China's **Tianwen** and **Chang'e** series, JAXA's sample-return missions and ISRO's Mars and lunar missions have made deep-space exploration a truly international enterprise. The two Voyager spacecraft, launched in 1977, are still returning data from interstellar space; on 18 November 2026 Voyager 1 reaches one light-day — about 26 billion km — from the Earth.

24.2 Interplanetary Trajectories

24.2.1 Escape and Hyperbolic Excess Speed

Trajectory design uses the **patched-conic** approximation (Chapter 3): near each body, the spacecraft follows a conic section governed by that body alone; between bodies, it follows an ellipse around the Sun. The energy with which a spacecraft leaves the Earth is described by its **hyperbolic excess speed** v_∞ — its speed relative to the Earth after escape. It can also be described by the **characteristic energy** $C_3 = v_\infty^2$. This is the standard way launch-vehicle performance is quoted for interplanetary missions. The velocity change needed to depart from a circular parking orbit of radius r is

$$\Delta v = \sqrt{v_{esc}^2 + v_\infty^2} - v_{circ}, \qquad v_{esc} = \sqrt{\frac{2\mu}{r}},\; v_{circ} = \sqrt{\frac{\mu}{r}}$$

which, by the **Oberth effect**, is much less than v_∞ added to the escape speed separately: a burn deep in a gravity well, at high speed, gains the most energy.

24.2.2 Hohmann Transfers and the Cost of Getting There

The minimum-energy transfer between two circular coplanar orbits is the **Hohmann transfer** (Chapter 3), an ellipse tangent to both orbits (Figure 24.1). Figure 24.2 compares the launch energies and flight times. Venus and Mars are relatively cheap (C3 of about 6–9 km^2/s^2). Mercury is surprisingly expensive — a spacecraft must shed the Earth's orbital speed to fall towards the Sun. The outer planets require C3 values of 80–140 km^2/s^2 and flight times of years to decades. Only a few launch vehicles can deliver useful masses at such energies, which is why outer-planet missions rely on gravity assists.

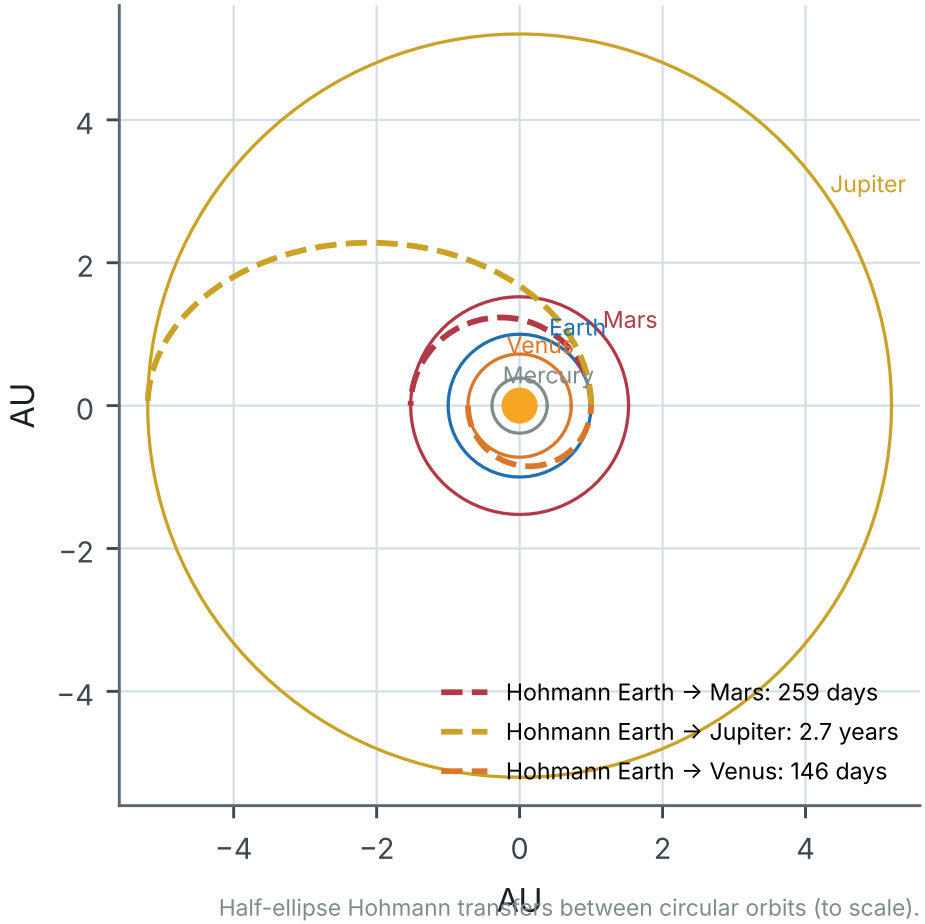
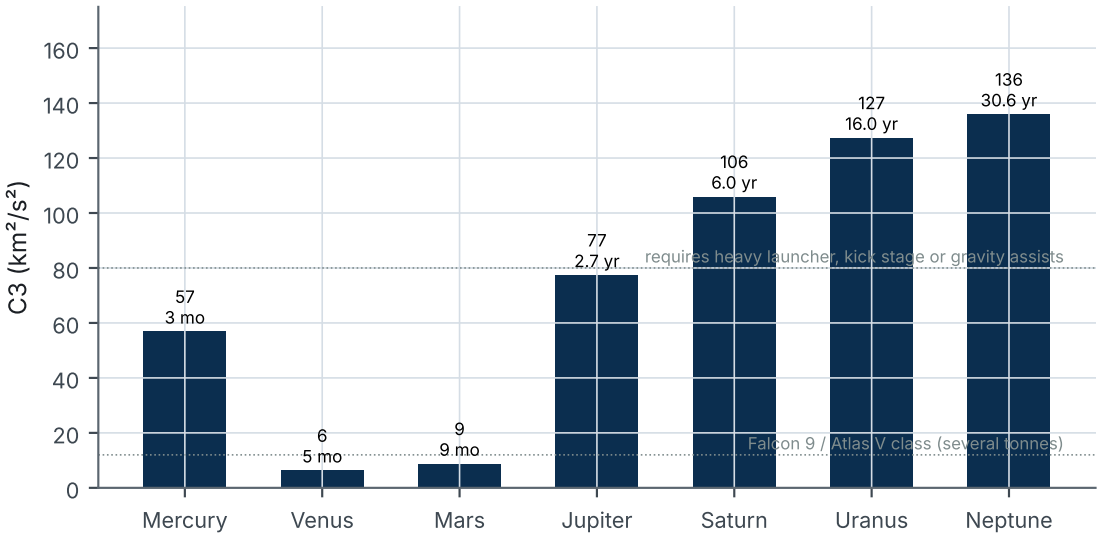


FIGURE 24.1 Hohmann transfer ellipses from the Earth to Venus, Mars and Jupiter, drawn to scale.



Circular coplanar orbits; labels give C3 (km²/s²) and transfer time. Real missions use gravity assists to reach the outer planets with lower C3.

FIGURE 24.2 Launch energy (C3) and flight time for Hohmann transfers from the Earth to each planet.

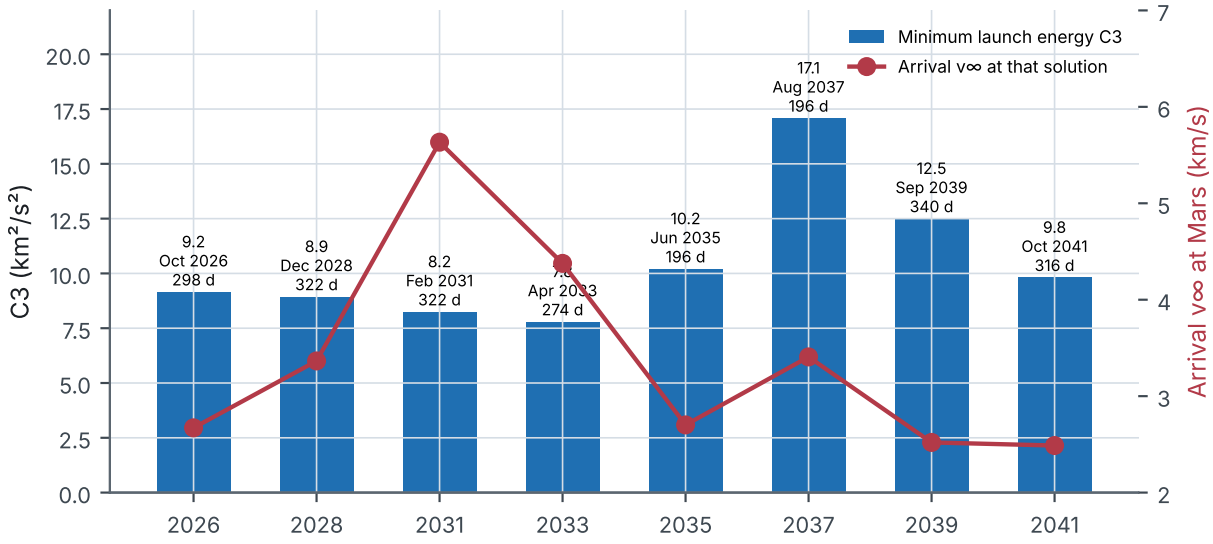
24.2.3 Launch Windows

A transfer works only if the target planet arrives at the end of the transfer at the right moment, so launches must take place when the planets are in the right relative positions. These **launch opportunities** recur at the **synodic period** of the two planets: about 584 days for Venus, 780 days (26 months) for Mars, and about 13 months for Jupiter and the more distant planets (Table 24.1). Each opportunity is a **launch period** of a few weeks, with a daily **launch window** of minutes to hours determined by the launch site and parking orbit.

TABLE 24.1 Synodic periods and transfer characteristics.

Planet	Synodic period with Earth	Hohmann transfer time	Typical C3 (km²/s²)
Mercury	116 days	3.5 months	≈ 57 (direct; real missions use Venus and Mercury flybys)
Venus	584 days (1.6 yr)	4.8 months	≈ 6–10
Mars	780 days (2.1 yr)	8.5 months	≈ 8–17 depending on opportunity
Jupiter	399 days (1.1 yr)	2.7 years	≈ 77–85
Saturn	378 days (1.04 yr)	6.0 years	≈ 105–110

Because planetary orbits are elliptical and inclined, the real transfers are found by solving **Lambert's problem** (Chapter 3) for many combinations of launch and arrival dates. The results are plotted as a "porkchop plot" (Figure 3.16 in Chapter 3). Repeating this computation for each Mars opportunity (Figure 24.3) shows that their quality varies over a cycle of about 15–17 years. This is because Mars's eccentric orbit brings it closer to or farther from the Earth at opposition. The 2033 opportunity requires the least energy in the next decade, while 2037 is among the most demanding. Transfers shorter than half an orbit are called **Type I**, longer ones **Type II**.



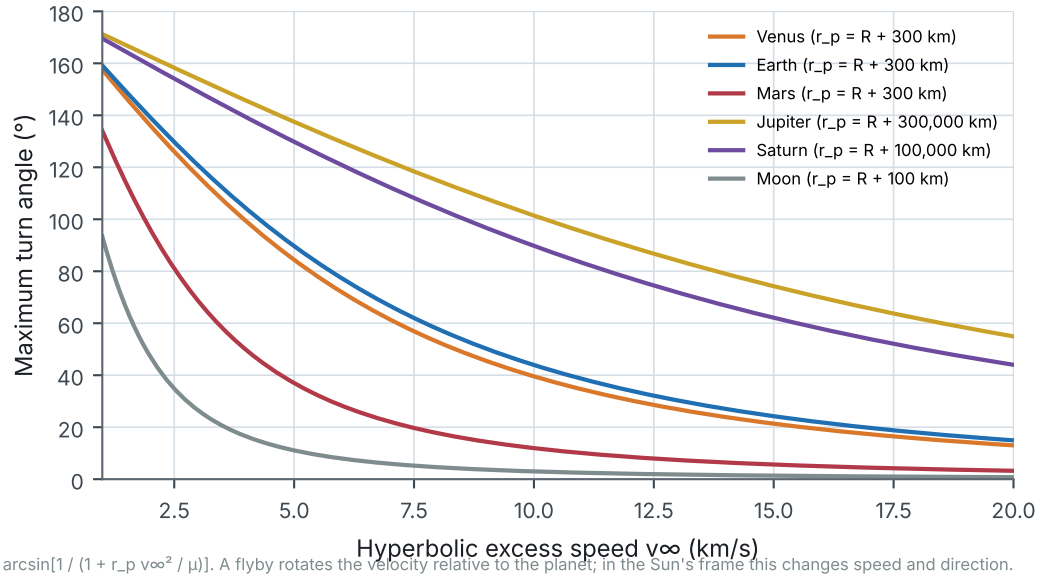
Computed for this manual: single-revolution Lambert solutions minimising C3 in each window (labels: C3, launch month, flight time).

FIGURE 24.3 Minimum launch energy and the corresponding arrival speed for each Mars launch opportunity from 2026 to 2041, computed with a Lambert solver (flight times limited to 130–340 days).

24.3 Gravity Assists

24.3.1 Borrowing Momentum from Planets

A spacecraft flying past a planet enters and leaves the planet's sphere of influence with the same speed relative to the planet. However, its direction is turned through an angle that depends on how close it passes and how fast it is moving (Figure 24.4). Seen from the Sun, adding the planet's orbital velocity to this rotated relative velocity changes the spacecraft's heliocentric speed and direction. This is a **gravity assist**, which can add or remove several kilometres per second without using propellant (Chapter 3, Figure 3.17). Massive planets such as Jupiter can turn a spacecraft by large angles even at high speeds; small bodies like Mars or the Moon give only modest changes.



δ = 2 arcsin[1 / (1 + r_p v∞² / μ)]. A flyby rotates the velocity relative to the planet; in the Sun's frame this changes speed and direction.

FIGURE 24.4 Maximum turn angle of a spacecraft's velocity during a flyby, as a function of the approach speed v∞, for different bodies.

24.3.2 Gravity Assists in Practice

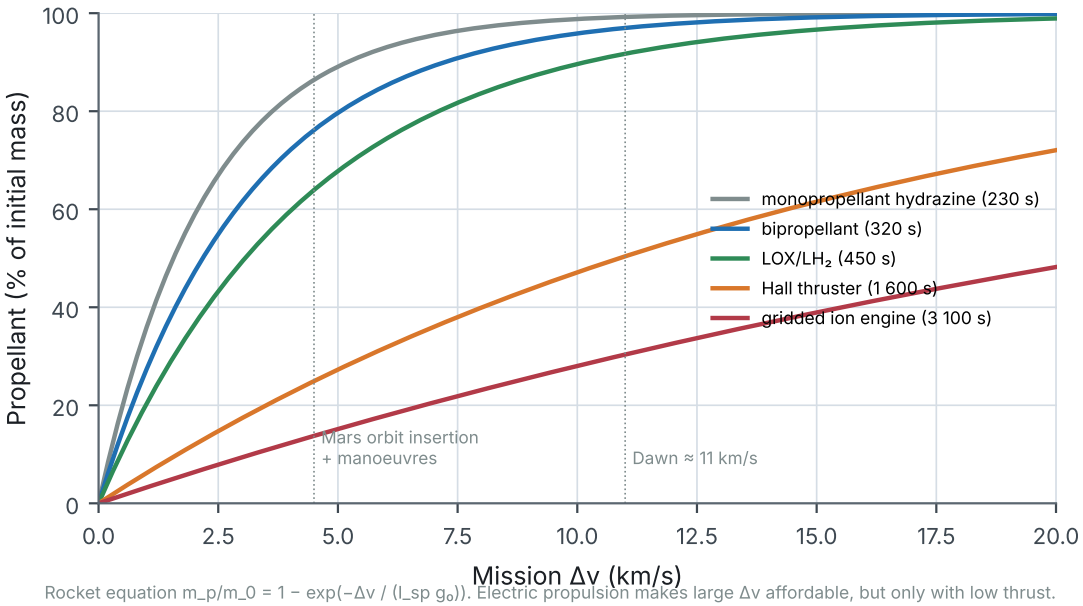
Gravity assists have made the exploration of the Solar System possible:

- **Mariner 10** (1973) used Venus to reach Mercury — the first gravity assist.
- **Voyager 1 and 2** (1977) exploited a rare alignment of the outer planets, recurring only every 175 years, to tour Jupiter, Saturn, Uranus and Neptune.
- **Galileo** (1989) and **Cassini** (1997) used Venus and Earth flybys (VEEGA and VVEJGA trajectories) because their launch vehicles could not send them directly to Jupiter or Saturn.
- **MESSENGER** (2004) used six flybys to slow down enough to orbit Mercury. **BepiColombo** (2018) used nine — one of Earth, two of Venus and six of Mercury — before entering orbit around Mercury in November 2026.
- **Parker Solar Probe** used seven Venus flybys to lower its perihelion to 6.1 million km.
- ESA's **JUICE** (2023) performed the first-ever lunar–Earth double flyby in 2024 and a Venus flyby in 2025. Further Earth flybys follow in 2026 and 2029, before it arrives at Jupiter in 2031. NASA's **Europa Clipper** (2024) flew past Mars in March 2025 and returns to Earth in December 2026 on its way to Jupiter in 2030 (Figure 24.15).

Once in orbit around a giant planet, missions use repeated flybys of its moons to reshape their orbits. Cassini made more than 120 flybys of Titan, and Europa Clipper will make about 49 flybys of Europa. Designing such **tours** is a combinatorial problem solved with tools such as Tisserand graphs and global optimisation.

24.4 Low-Thrust Trajectories

Electric propulsion (Chapter 6) produces only millinewtons to a fraction of a newton of thrust. However, its exhaust velocities are 5–10 times higher than those of chemical rockets. It can therefore provide total velocity changes of 10 km/s or more with modest propellant (Figure 24.5). Instead of short impulsive burns, the engine operates for months or years, spiralling the spacecraft gradually from one orbit to another.



Rocket equation m_p/m_0 = 1 − exp(−Δv / (I_sp g₀)). Electric propulsion makes large Δv affordable, but only with low thrust.

FIGURE 24.5 Propellant mass fraction versus mission Δv for chemical and electric propulsion.

NASA's **Deep Space 1** (1998) demonstrated ion propulsion, and JAXA's **Hayabusa** (2003) used it for the first asteroid sample return. NASA's **Dawn** (2007) used three ion engines to orbit two different bodies — Vesta and Ceres. It achieved a record total Δv of about 11 km/s (Figure 24.6). ESA–JAXA's BepiColombo, NASA's Psyche (launched in 2023 with Hall-effect thrusters, flying past Mars in May 2026 on its way to the metal-rich asteroid Psyche in 2029) and many smaller missions use electric propulsion. Low-thrust trajectories are designed with numerical optimisation methods that balance thrust direction and timing against propellant use and flight time.

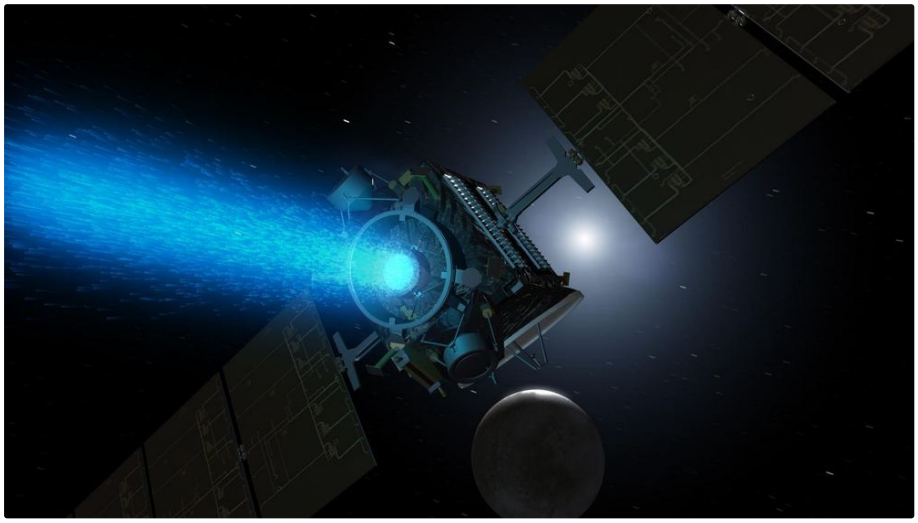


FIGURE 24.6 Artist's impression of NASA's Dawn spacecraft approaching Ceres, its xenon ion engine glowing blue. Dawn's ion engines provided about 11 km/s of velocity change over its mission. Image: NASA/JPL (NASA/JPL-Caltech).

24.5 Arrival: Orbit Insertion, Aerobraking and Aerocapture

A spacecraft arrives at its target on a hyperbolic trajectory and must slow down to be captured. **Orbit insertion** burns of 0.5–1.5 km/s are typical at Mars and Venus, and can exceed 2 km/s at Mercury or for low orbits at the giant planets. They must be executed entirely autonomously at precisely the right moment, often during a communication blackout behind the planet. Failures have ended several missions. Mars Observer (1993) was lost just before insertion. Mars Climate Orbiter (1999) flew too deep into the atmosphere because one team used imperial units and another metric units in navigation software. This is the most famous units error in engineering history.

To save propellant, many Mars and Venus orbiters first enter a large elliptical orbit and then use **aerobraking**. Dipping repeatedly into the upper atmosphere at periapsis, they lose a little energy on each pass until the orbit is circularised. Mars Global Surveyor, Mars Odyssey, Mars Reconnaissance Orbiter, ESA's Trace Gas Orbiter and Venus Express all aerobraked; it takes months and careful monitoring of atmospheric density. **Aerocapture** is a single pass deep through the atmosphere inside a heat shield to go directly from a hyperbolic to an elliptical orbit. It could save even more mass, especially at Uranus and Neptune, but has not yet been flown.

24.6 Cislunar Space and Libration Points

Missions near the Moon and the Sun–Earth Lagrange points use the dynamics of the three-body problem (Chapter 3). The Sun–Earth **L1** and **L2** points, 1.5 million km from the Earth, host solar observatories (SOHO, DSCOVR, Aditya-L1) and astronomical telescopes (Gaia, JWST, Euclid, Roman). These fly in large **halo** or **Lissajous** orbits around the unstable equilibrium points, which require small station-keeping manoeuvres every few weeks.

Around the Moon, new families of orbits have become important (Figure 24.7). The **near-rectilinear halo orbit (NRHO)** is a highly elongated halo orbit around the Earth–Moon L2 point with a period of about 6.5 days. It was selected for the planned Gateway station and tested by the small CAPSTONE spacecraft in 2022. It is stable, almost always in view of the Earth, and accessible from both the Earth and the lunar surface (though Gateway itself was paused in 2026). Artemis I's Orion flew in a **distant retrograde orbit (DRO)**, a very stable orbit around the Moon. **Low-energy (ballistic) transfers** loop out to about 1.5 million km and exploit the Sun's gravity. They take three to four months instead of three days but save hundreds of metres per second of Δv. They were used by Korea's Danuri, CAPSTONE, Japan's SLIM and ispace's Hakuto-R commercial landers.

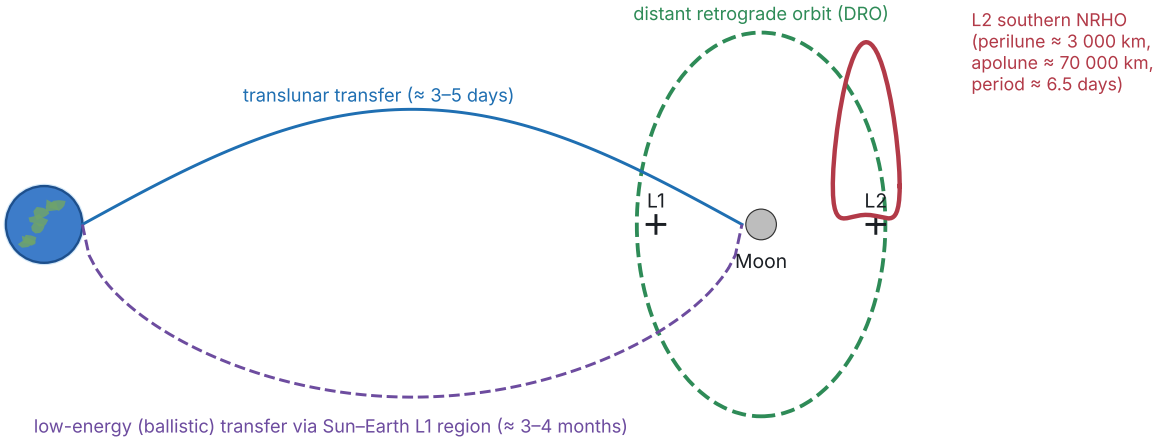


FIGURE 24.7 Orbits and transfers in cislunar space (schematic, not to scale).

24.7 Navigating in Deep Space

24.7.1 Radiometric Tracking

Deep-space navigation is mostly done from the ground. Large antennas of NASA's **Deep Space Network (DSN)** — complexes at Goldstone (California), Madrid and Canberra, spaced about 120° apart in longitude so that every spacecraft is always in view of one — measure:

- **range**, by timing a coded signal sent to the spacecraft and returned by its transponder, to an accuracy of about 1 m;
- **range-rate**, from the Doppler shift of the carrier, to about 0.05 mm/s;

- **angular position**, by **delta differential one-way ranging (ΔDOR)** (Figure 24.8): two widely separated stations observe the spacecraft and then a quasar close to it in the sky. The difference in arrival times gives the spacecraft's direction with an accuracy of about one nanoradian — about 200 m at the distance of Mars.

Navigators combine these measurements with dynamical models in an **orbit determination** filter. The models include the gravity of all relevant bodies, solar radiation pressure, thruster firings and even the thermal recoil of radiation from the spacecraft. Navigators also design **trajectory correction manoeuvres** to keep the spacecraft on course. Errors at a target are expressed in the **B-plane** (Figure 24.9), the plane through the target body perpendicular to the incoming asymptote. Delivering a Mars lander to its entry corridor requires B-plane accuracy of a few kilometres.

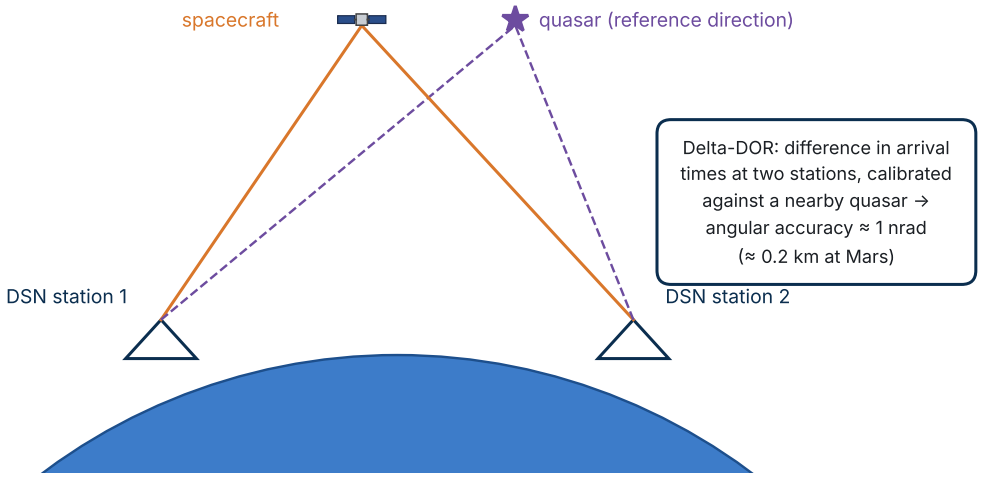


FIGURE 24.8 Delta-DOR: comparing the arrival of a spacecraft's signal at two distant stations with that of a nearby quasar gives a very precise direction.

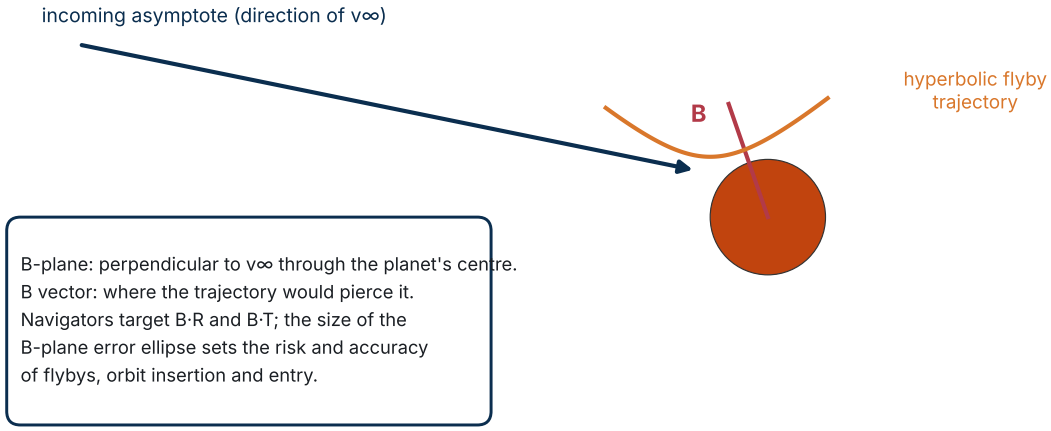


FIGURE 24.9 The B-plane used to target planetary flybys, orbit insertion and atmospheric entry.

24.7.2 Optical and Autonomous Navigation

Near a target, cameras provide **optical navigation**: images of a moon or asteroid against background stars give its direction relative to the spacecraft with high precision. Autonomous navigation systems close the loop on board. NASA's **AutoNav** (Deep Space 1, Deep Impact,

Stardust) and the **SMART Nav** system, which guided DART into Dimorphos in 2022, identified their targets in images. They steered without help from the ground in the final hours. Landing systems use **terrain-relative navigation** (Chapter 27). Experimental techniques include navigation using X-ray pulsars as natural clocks (demonstrated by NASA's SEXTANT on the ISS) and onboard atomic clocks that enable one-way ranging (Chapter 18).

24.8 Communications, Power and Thermal Design

24.8.1 Talking Across the Solar System

Light takes 1.3 seconds to reach the Moon, between 3 and 22 minutes to reach Mars, and almost a day to reach Voyager 1 (Figure 24.10). Spacecraft therefore carry out pre-programmed sequences autonomously and protect themselves with fault-protection software that can put them in a safe mode and wait for instructions. Data rates fall with the square of distance (Chapter 7): a Mars orbiter can return several megabits per second, while Voyager 1 sends about 160 bit/s. The DSN's 70 m antennas (Figure 24.11) and arrays of 34 m antennas, ESA's ESTRACK 35 m deep-space antennas, and Chinese, Indian and Japanese deep-space networks support all missions. NASA's Deep Space Optical Communications experiment on Psyche demonstrated laser data rates of hundreds of megabits per second from tens of millions of kilometres.

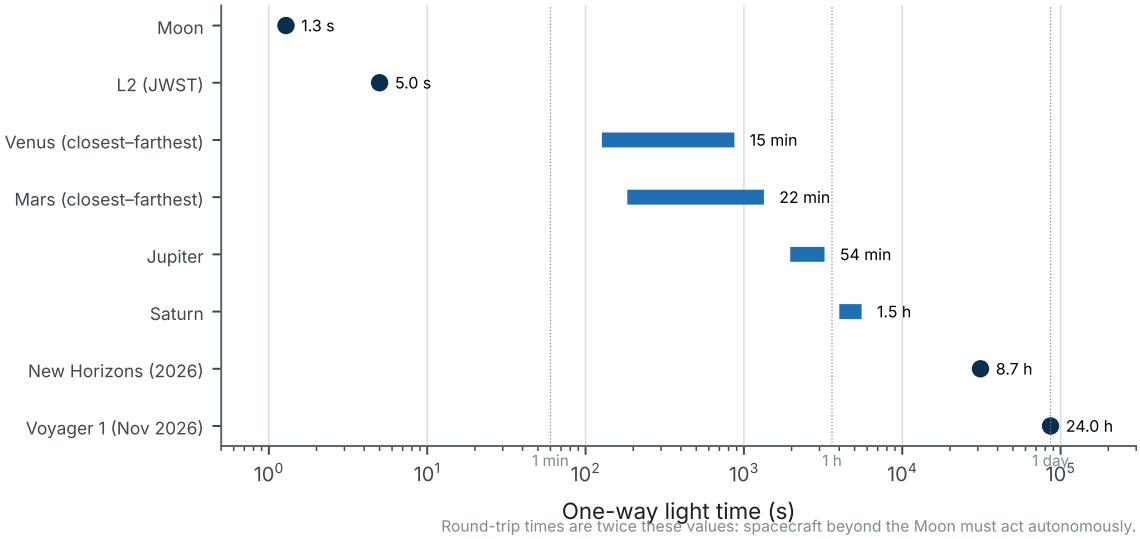


FIGURE 24.10 One-way light time to destinations across the Solar System.

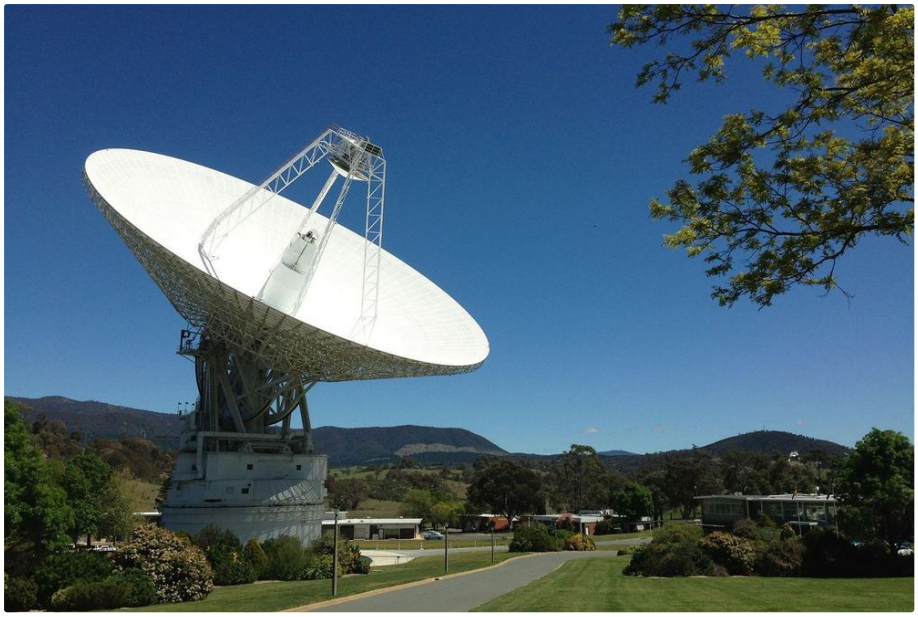


FIGURE 24.11 The 70 m antenna DSS-43 at the Canberra Deep Space Communication Complex in Australia — the only antenna able to send commands to Voyager 2, which is below the Earth's equatorial plane. Image: NASA/JPL (NASA/Canberra Deep Space Communication Complex).

24.8.2 Power

At Jupiter, sunlight is 27 times weaker than at the Earth; at Saturn, 90 times (Figure 24.12). Missions as far as Jupiter can use very large solar arrays combined with efficient cells that work at low intensity and low temperature. Juno's three arrays span about 20 m and produce about 500 W at Jupiter, and Europa Clipper's span 30.5 m. ESA's JUICE carries 85 m² of cells, and NASA's Lucy, with two 7.3 m circular arrays, reaches 5.7 AU. Beyond about Jupiter, missions need **radioisotope power**: radioisotope thermoelectric generators (RTGs) convert the heat of plutonium-238 decay into about 100–300 W of electricity for decades. Voyager, Galileo, Cassini, New Horizons, Curiosity and Perseverance all use RTGs; the supply of plutonium-238 — whose US production restarted in the 2010s — limits how many such missions can fly.

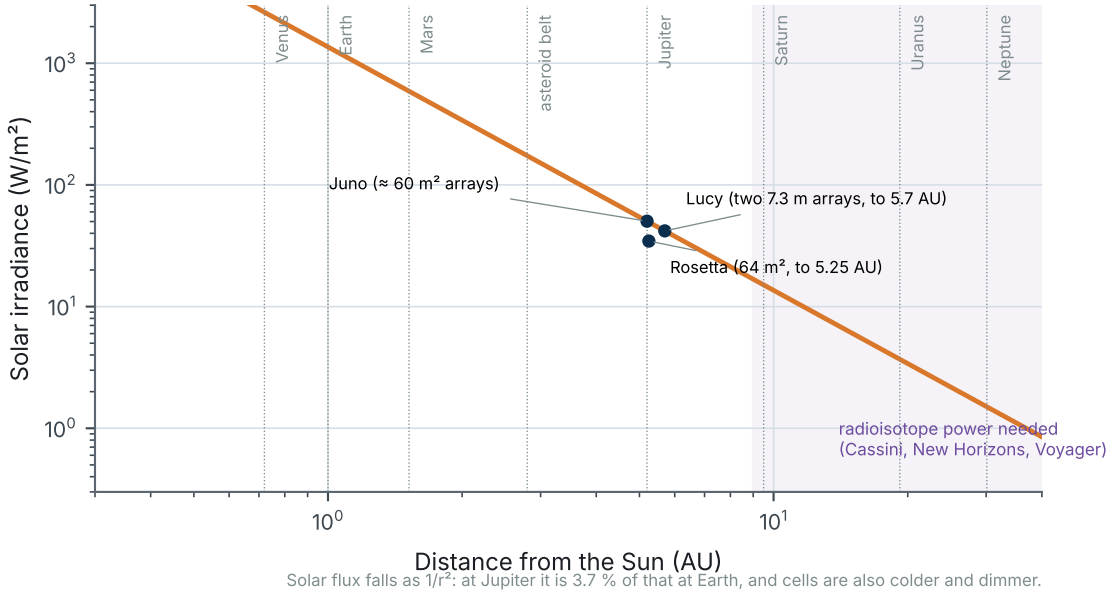


FIGURE 24.12 Solar irradiance as a function of distance from the Sun. Large solar arrays have powered missions out to Jupiter; beyond, radioisotope power is required.

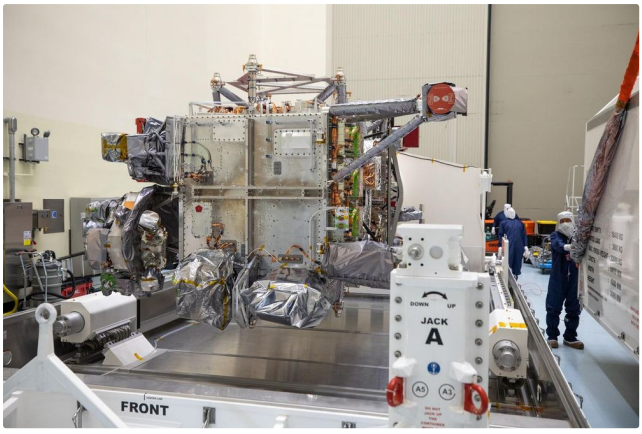


FIGURE 24.13 Europa Clipper — NASA's largest planetary spacecraft, with solar arrays spanning 30.5 m — being unpacked at Kennedy Space Center in May 2024 before its launch in October 2024. Image: NASA/KSC (NASA/Kim Shiflett).

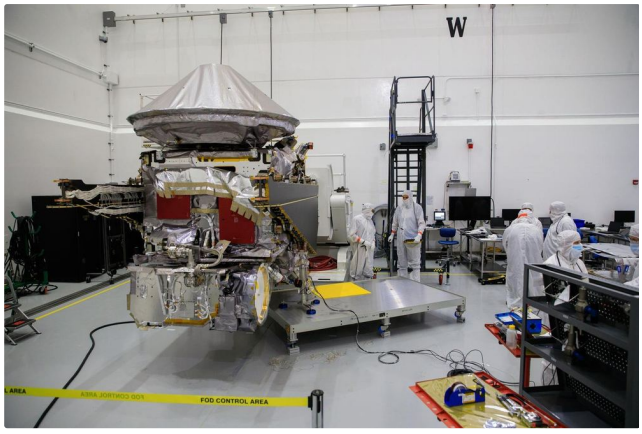


FIGURE 24.14 One of Lucy's two 7.3 m circular solar arrays being deployed in a test in August 2021. Lucy is visiting Trojan asteroids at the distance of Jupiter's orbit using solar power. Image: NASA/KSC (NASA/Glenn Benson).

24.8.3 Thermal Extremes and Radiation

Thermal design must cope with radiation intensities from many times the Earth's value to almost nothing in the outer Solar System. At the hot extreme, Parker Solar Probe's heat shield reaches about 1 400 °C. At the cold extreme, spacecraft rely on radioisotope heater units and electrical heaters to keep propellant and batteries warm. At Jupiter, radiation is so intense that Juno and Europa Clipper protect their electronics inside **radiation vaults** with thick titanium or aluminium walls, and their orbits are designed to limit time in the worst regions.

24.9 Planetary Protection

Spacecraft must not contaminate other worlds with terrestrial organisms that could confuse the search for life or alter habitable environments, and samples returned to Earth must not pose a risk to the biosphere. The **COSPAR planetary protection policy**, implementing Article IX of the Outer Space Treaty, assigns missions to categories (Table 24.2).

TABLE 24.2 COSPAR planetary-protection categories (simplified).

Category	Mission type and target	Requirements
I	Any mission to bodies of no interest for chemical evolution or life (e.g. Mercury, undifferentiated asteroids)	None
II	Missions to bodies of significant interest but with remote chance of contamination (Venus, the Moon, comets, Jupiter, Saturn, Titan)	Documentation and impact-avoidance analysis
III	Flybys and orbiters of bodies with a significant chance of contamination (Mars, Europa, Enceladus)	Limits on impact probability or bioburden
IV	Landers and probes to these bodies	Strict cleaning and sterilisation; IVc for special regions (e.g. Martian ice or brines)

Category	Mission type and target	Requirements
V	Sample return to Earth: “unrestricted” (Moon, asteroid Ryugu) or “restricted” (Mars, Europa)	Restricted: break the chain of contact; containment and biohazard testing

The Viking landers of 1976 were heat-sterilised as complete spacecraft. Today’s Mars landers are cleaned to limits of a few hundred thousand bacterial spores. At the end of their missions near Jupiter and Saturn, Galileo and Cassini were deliberately plunged into the planets to avoid any possibility of crashing on Europa or Enceladus.

24.10 From Idea to Arrival: The Mission Life Cycle

Deep-space missions follow the project life cycle of Chapter 4, but over unusually long timescales. Europa Clipper illustrates this (Figure 24.15). Studies began more than a decade before its launch in October 2024, and its cruise includes Mars (2025) and Earth (2026) gravity assists. Its science phase at Jupiter, beginning in 2030, will last about four years. Its life-cycle cost is about $5 billion.

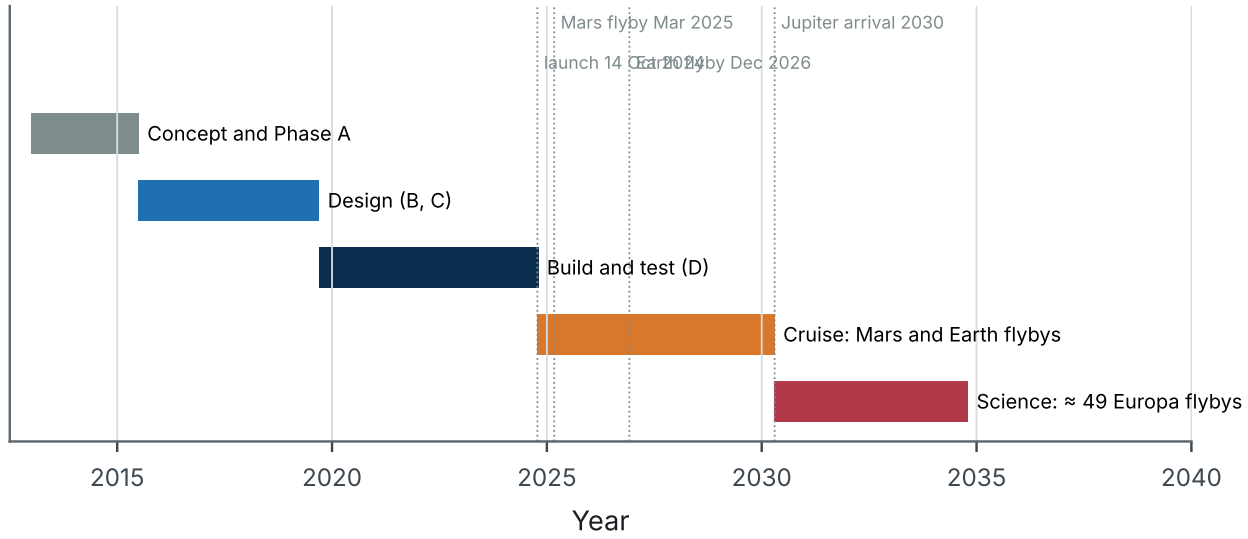


FIGURE 24.15 The life cycle of NASA’s Europa Clipper mission, from concept studies to the end of its prime science phase.

24.11 The Deep-Space Fleet in 2026

TABLE 24.3 Selected deep-space missions in operation or en route in October 2026.

Destination	Active or en-route missions (selected)
Moon	LRO (USA), Chandrayaan-2 orbiter (India), Danuri (Korea), Chang’e-4 and Queqiao relays (China), commercial and Artemis missions (Chapters 25–27)
Sun and heliosphere	Parker Solar Probe, Solar Orbiter, SOHO, IMAP, Aditya-L1, PUNCH
Mercury	BepiColombo (ESA/JAXA), Mercury orbit insertion November 2026

Destination	Active or en-route missions (selected)
Mars	Mars Odyssey, MRO, MAVEN, Mars Express, ExoMars TGO, Tianwen-1 orbiter, Hope; Curiosity and Perseverance rovers; ESCAPADE twin smallsats en route; MMX launching to Phobos in October 2026
Jupiter	Juno (in orbit since 2016; extended operations uncertain after 2025); JUICE (arrival 2031) and Europa Clipper (arrival 2030) en route
Asteroids and comets	Lucy (Trojans), Psyche (arrival 2029), Hera (Didymos–Dimorphos, arrival November 2026), OSIRIS-APEX (Apophis, 2029), Hayabusa2 extended mission (2031), Tianwen-2 (Kamoʻoalewa, sample return ≈ 2027)
Outer Solar System and beyond	New Horizons (Kuiper Belt), Voyager 1 and 2 (interstellar space)

EXAMPLE: THE ΔV BUDGET OF A 2026 MARS ORBITER

From Figure 24.3, the best 2026 opportunity requires C3 ≈ 9.2 km²/s² (v∞ ≈ 3.0 km/s) with launch on about 29 October 2026 and arrival with v∞ ≈ 2.7 km/s.

1. *Departure* from a 300 km parking orbit (r = 6 671 km): v_{circ} = 7.73 km/s and v_{esc} = 10.93 km/s. Then $v_p = \sqrt{10.93^2 + 9.2} = 11.34$ km/s, so Δv ≈ 3.61 km/s. This is normally provided by the launch vehicle's upper stage.
2. *Trajectory corrections* during cruise: typically 20–50 m/s.
3. *Mars orbit insertion* at 250 km altitude (r = 3 640 km, μ = 42 828 km³/s²): the hyperbolic periapsis speed is $\sqrt{2\mu/r + v_\infty^2} = \sqrt{23.53 + 7.13} = 5.54$ km/s. To capture into an elliptical orbit with a period of one Martian day (a = 20 450 km), the periapsis speed must be $\sqrt{\mu(2/r - 1/a)} = 4.63$ km/s, so the insertion burn is about 0.91 km/s.
4. *Aerobraking* then lowers the apoapsis to a low circular science orbit over several months, saving about 1.2 km/s compared with doing the same with the engine.

With a bipropellant engine of 320 s specific impulse, the 0.95 km/s of spacecraft-provided Δv requires propellant equal to about 26 % of the spacecraft's mass at launch.

AI IN DEEP-SPACE MISSION DESIGN

Machine learning and global optimisation are accelerating trajectory design (searching enormous spaces of gravity-assist sequences and low-thrust profiles), autonomous navigation and fault protection, scheduling of DSN antennas, and the analysis of returned data. Future spacecraft will need far greater on-board autonomy to explore distant worlds with hours of communication delay (Chapters 48 and 51).

KEY TAKEAWAYS

- Deep-space missions are constrained by energy, launch windows, distance, power, environment and the need for autonomy.
- Departure energy is expressed as $C3 = v_\infty^2$; Hohmann transfers to Venus and Mars need C3 ≈ 6–10 km^2/s^2, to the outer planets 80–140 km^2/s^2.
- Launch opportunities recur at synodic periods (26 months for Mars); their quality varies with orbital geometry — 2033 is the best Mars window of the next decade.
- Gravity assists and low-thrust electric propulsion make otherwise impossible missions feasible; BepiColombo needed nine flybys to reach Mercury, Dawn achieved 11 km/s of Δv with ion engines.
- Orbit insertion, aerobraking and (future) aerocapture capture spacecraft at their targets; NRHOs, DROs and low-energy transfers are reshaping cislunar operations.
- Navigation relies on DSN range, Doppler and ΔDOR plus optical and autonomous navigation; light time ranges from seconds (Moon) to a day (Voyager 1).
- Solar power works to about Jupiter; beyond, RTGs are needed. Radiation vaults protect electronics at Jupiter.
- Planetary protection categories govern cleanliness and sample return; flagship missions take 15–20 years from concept to science.

REVIEW QUESTIONS

1. List six constraints that make deep-space missions harder than Earth-orbit missions.
2. Calculate the Δv needed to depart from a 200 km circular parking orbit with a C3 of 10 km^2/s^2.
3. Why do Mars launch opportunities recur every 26 months, and why are some windows better than others?
4. Explain how a gravity assist can increase a spacecraft's speed relative to the Sun without violating energy conservation.
5. How did Dawn achieve a total Δv of about 11 km/s, and why would that have been impossible with chemical propulsion?
6. Compare orbit insertion by propulsive burn, aerobraking and aerocapture.
7. Describe range, Doppler and ΔDOR tracking, and the information each provides.
8. Why is solar power practical out to about Jupiter but not beyond? What do missions to the outer Solar System use instead?

25 CHAPTER Planetary Orbiters

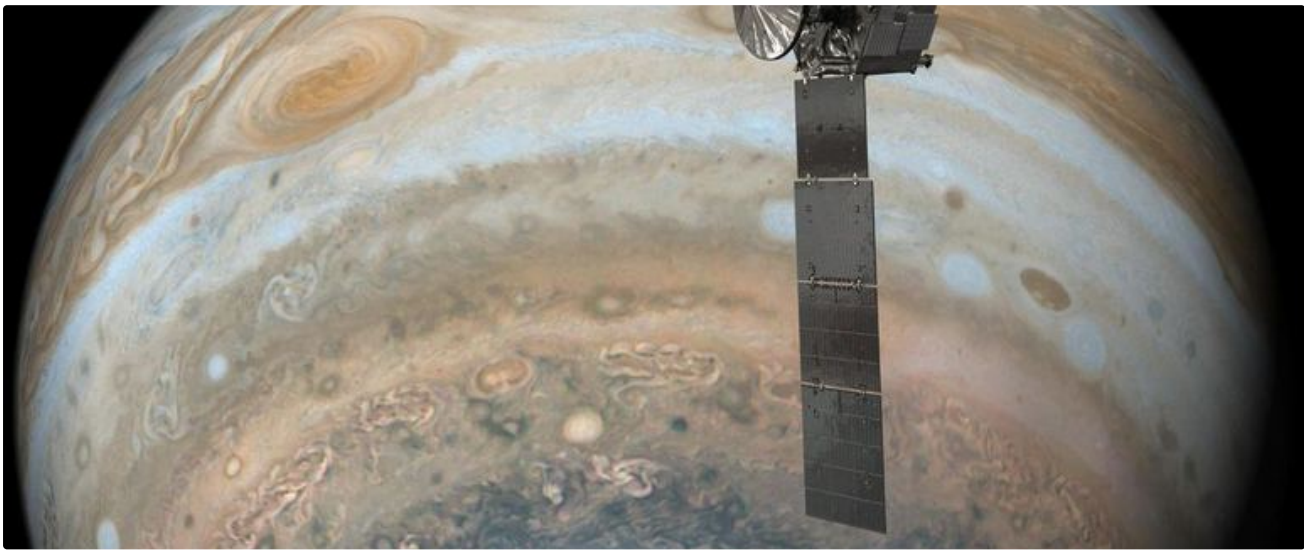


Image: NASA/JPL (NASA/JPL-Caltech)

Orbiters are the workhorses of planetary exploration. Circling a world for years, they map its whole surface, probe beneath it with radar, weigh its interior through its gravity field, sample its atmosphere and magnetic environment, and relay the data of landers and rovers back to Earth. This chapter explains how orbits are designed around bodies as different as the Moon, Mars, Venus, Mercury, the giant planets and small asteroids, describes the instruments that orbiters carry, and reviews the orbiter missions — from Luna 10 and Mariner 9 to Juno, Cassini and BepiColombo — that have transformed our knowledge of the Solar System, including those operating in 2026.

IN THIS CHAPTER

25.1 Why Orbit?
25.2 Designing Orbits Around Other Worlds
25.3 Getting Into Orbit
25.4 What Orbiters Carry
25.5 Orbiters World by World
25.6 Ending an Orbiter Mission

25.1 Why Orbit?

A flyby gives a few hours of close observation; an orbiter can observe a world for years, through changing seasons and lighting, at resolutions limited only by its instruments and its orbit. Orbiters provide:

- **global mapping** of surface composition, topography, geology and temperature;
- **long-term monitoring** of atmospheres, weather, seasonal ice caps and active processes such as dust storms, volcanism and geysers;
- **interior structure**, through precise measurements of the gravity and magnetic fields;

- **reconnaissance** of landing sites and safety;
- **communication relays** for surface missions.

TABLE 25.1 First orbiters of Solar System bodies.

Body	First orbiter	Year	Notes
Moon	Luna 10 (USSR)	1966	First spacecraft to orbit another body
Mars	Mariner 9 (USA)	1971	First to orbit another planet; mapped Mars after a global dust storm
Venus	Venera 9 (USSR)	1975	Orbiter of the first lander mission to return surface images
Jupiter	Galileo (USA)	1995	With an atmospheric entry probe
Asteroid	NEAR Shoemaker (USA)	2000	Orbited and then landed on Eros
Saturn	Cassini (USA/Europe/Italy)	2004	Carried ESA's Huygens Titan probe
Mercury	MESSENGER (USA)	2011	Found water ice in permanently shadowed polar craters
Dwarf planets	Dawn (USA)	2011, 2015	Orbited Vesta and then Ceres
Comet	Rosetta (Europe)	2014	Deployed the Philae lander on 67P/Churyumov–Gerasimenko

25.2 Designing Orbits Around Other Worlds

25.2.1 Gravity Fields and Perturbations

The basic orbital mechanics are the same everywhere (Chapter 3), but each body's mass, size, rotation, shape and gravity field produce different orbits (Figure 25.1). A low orbit around the Moon takes about two hours; around Mars about 1 hour 50 minutes; around Earth about 90 minutes.

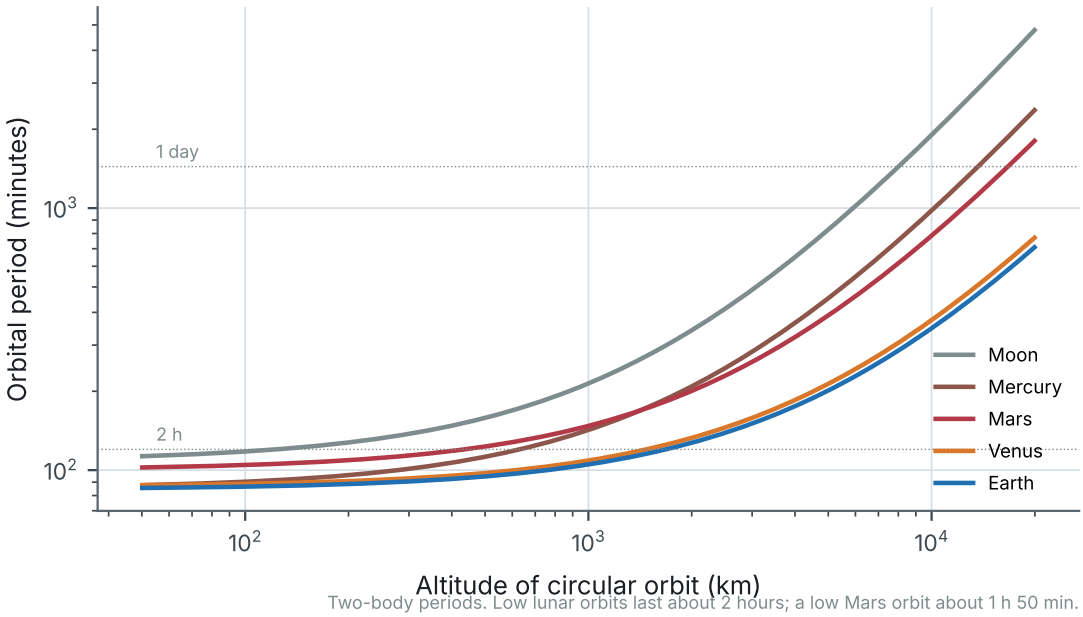


FIGURE 25.1 Orbital period as a function of altitude around the Moon, Mercury, Mars, Venus and the Earth.

Irregularities in the gravity field strongly affect low orbits:

- The **Moon** has concentrations of mass under its large impact basins (**mascons**), discovered by tracking Lunar Orbiter spacecraft in 1968. They perturb low orbits so strongly that most low lunar orbits crash within months unless maintained. Only certain **frozen orbits**, at inclinations of about 27°, 50°, 76° and 86°, are naturally stable for long periods. NASA's **GRAIL** twin spacecraft (2012) measured the lunar gravity field with unprecedented precision by tracking the distance between them, revealing the structure of the lunar crust.
- **Mars** has a large equatorial bulge (J_2 about twice the Earth's), which allows **sun-synchronous orbits** that pass over each point at the same local time (Figure 25.2). Because Mars's year is longer, the required precession is slower, and its sun-synchronous orbits are nearly polar (about 93°). Most Mars orbiters since Mars Global Surveyor use them.
- **Venus** rotates once every 243 days (backwards) and has almost no bulge, so sun-synchronous orbits are impossible. A near-polar orbit lets the planet rotate slowly beneath it. NASA's **Magellan** (1990–1994) mapped 98 % of the surface with radar over three Venusian days (Figure 25.13).
- **Small bodies** — asteroids and comet nuclei — have such weak, irregular gravity that orbits are slow (hours to weeks) and strongly perturbed by solar radiation pressure and outgassing. OSIRIS-REx orbited the 500 m asteroid Bennu at about 1 km — the smallest body ever orbited.

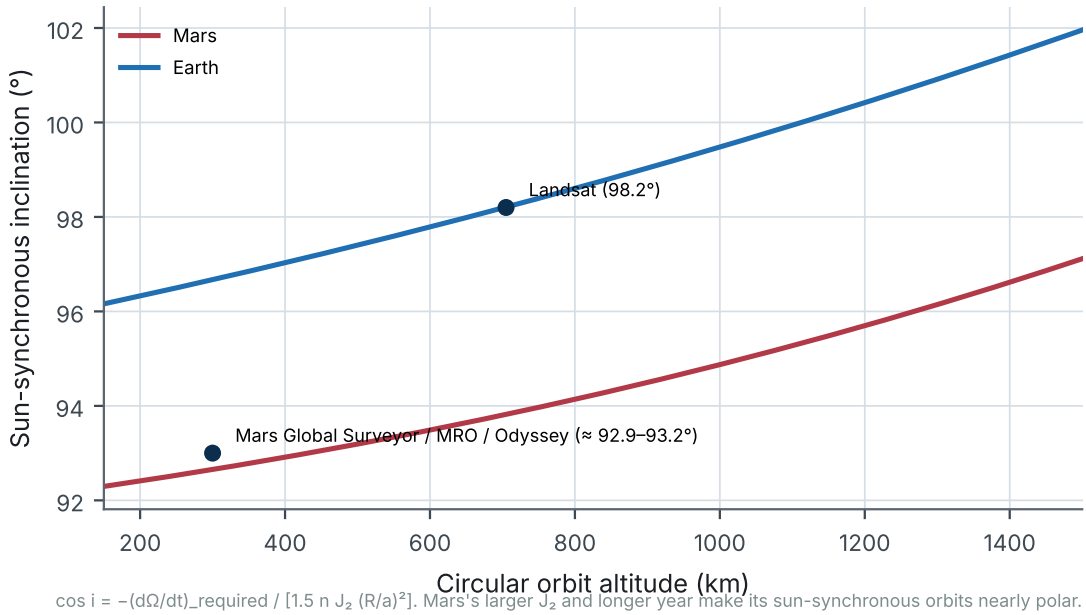


FIGURE 25.2 Sun-synchronous inclination as a function of altitude for Mars and the Earth.

25.2.2 Thermal and Radiation Constraints

Orbits are also shaped by the environment. At **Mercury**, sunlight is up to 10 times more intense than at the Earth, and the dayside surface reaches 430 °C, radiating heat back at the spacecraft. MESSENGER flew a highly elliptical orbit to limit time over the hot surface, shaded behind a ceramic-cloth sunshade. BepiColombo's orbiters use high-temperature multilayer insulation and radiators that view away from the planet. At **Jupiter**, the radiation belts dominate orbit design (Chapter 23): Juno's elongated polar orbit (Figure 25.3) dives beneath the belts at each perijove, minimising the dose per orbit.

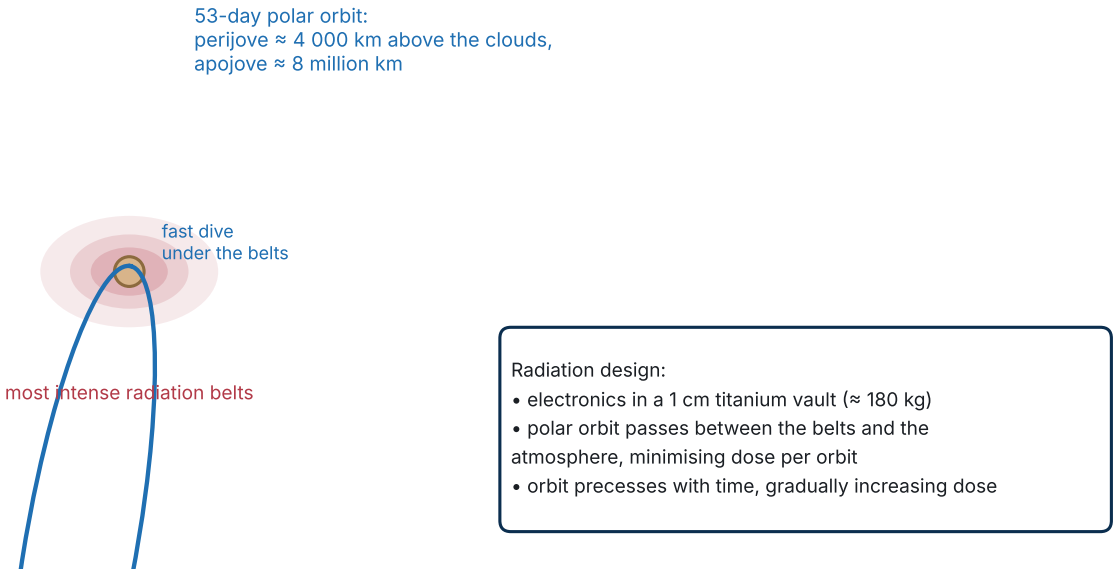


FIGURE 25.3 Juno's 53-day polar orbit passes between Jupiter's atmosphere and its most intense radiation belts at each close approach (schematic).

25.2.3 Moon Tours

At the giant planets, orbiters use repeated flybys of the large moons to change their orbits (Chapter 24). Cassini's 13-year tour of the Saturn system included more than 120 Titan flybys and 23 close Enceladus flybys. It ended with the **Grand Finale** of 2017 — 22 orbits through the 2 000 km gap between the rings and the planet's atmosphere. This was followed by a deliberate plunge into Saturn to protect Enceladus and Titan from contamination (Figure 25.4, Figure 25.5).

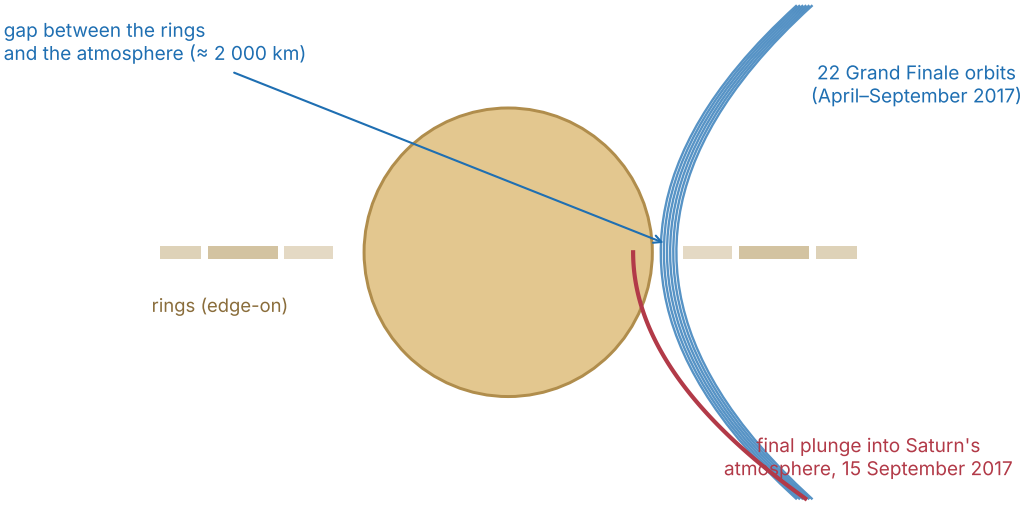


FIGURE 25.4 Cassini's Grand Finale: 22 orbits through the gap between Saturn's rings and atmosphere, followed by the final plunge (schematic).

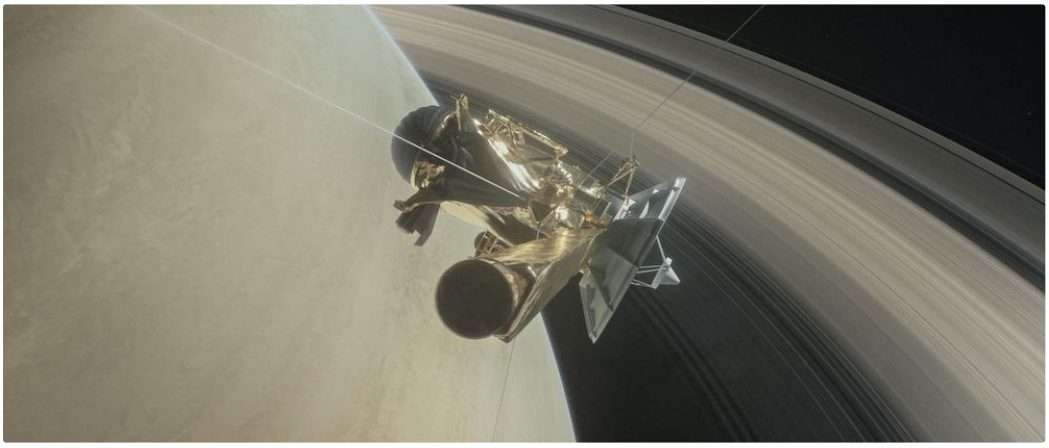


FIGURE 25.5 Artist's impression of Cassini diving between Saturn and its rings during the Grand Finale in 2017. Image: NASA/JPL (NASA/JPL-Caltech).

25.3 Getting Into Orbit

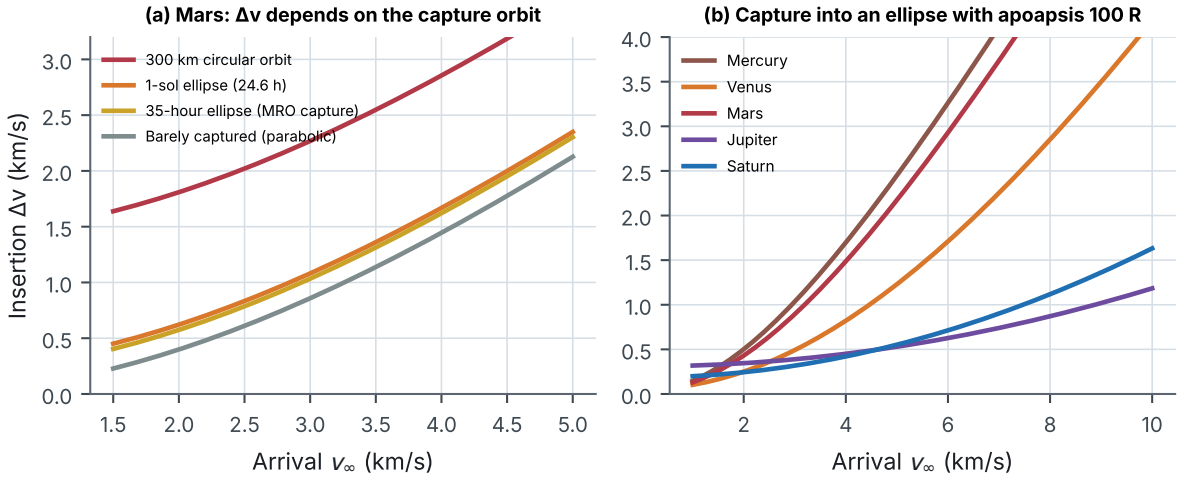
25.3.1 The Orbit-Insertion Burn

A spacecraft arrives at a planet on a hyperbola, moving faster than the local escape speed. To be captured, it must fire its main engine near closest approach and slow down by

$$\Delta v_{\text{insertion}} = \sqrt{v_\infty^2 + \frac{2\mu}{r_p}} - v_{\text{capture}},$$

where v_∞ is the arrival speed relative to the planet (Chapter 24), r_p is the periapsis radius and v_{capture} is the periapsis speed on the desired capture orbit. Two facts dominate the design (Figure 25.6):

- **Capturing into a long ellipse is much cheaper than capturing into a low circular orbit.** At Mars, the difference is more than 1 km/s. Most orbiters therefore first enter a long, elliptical orbit and lower it later.
- **Deep gravity wells make capture cheap.** Because the burn is made where the spacecraft is moving fastest (the Oberth effect), Jupiter and Saturn can be entered for a few hundred metres per second. Juno's Jupiter orbit insertion in 2016 needed about 542 m/s (a 35-minute burn). Cassini's Saturn insertion in 2004 needed about 626 m/s (96 minutes). Mercury, with a shallow well deep inside the Sun's, is the hardest planet to orbit, which is why MESSENGER and BepiColombo needed six or more planetary flybys to reduce their arrival speed.



Periapsis 300 km in (a); 400 km (1.1 R for the giants) in (b). A deep gravity well makes capture cheap: the burn is made where the spacecraft is fastest (the Oberth effect).

FIGURE 25.6 Orbit-insertion Δv. (a) At Mars, capturing into a long ellipse rather than a low circular orbit saves more than 1 km/s. (b) Capture into a long ellipse costs least at the giant planets with their deep gravity wells.

Orbit insertion is a **critical event**: it happens at a fixed time that cannot be postponed, often while the spacecraft is behind the planet and out of contact. It must therefore be fully autonomous, with fault protection that keeps the burn going through most failures. Losses have shaped today's practice. **Mars Observer** fell silent in 1993 just before insertion, probably after a propellant line ruptured while its tanks were being pressurised. **Mars Climate Orbiter** was lost in 1999 because ground software reported thruster impulse in pound-force seconds while the navigation software expected newton-seconds. The error accumulated over the cruise. The spacecraft passed about 57 km above the surface instead of the planned 140–150 km. Japan's **Akatsuki** failed to enter Venus orbit in 2010 when its main engine was damaged. Engineers kept it alive for five years in a solar orbit and inserted it in December 2015 using only its small attitude thrusters. This was one of the great recoveries in spaceflight.

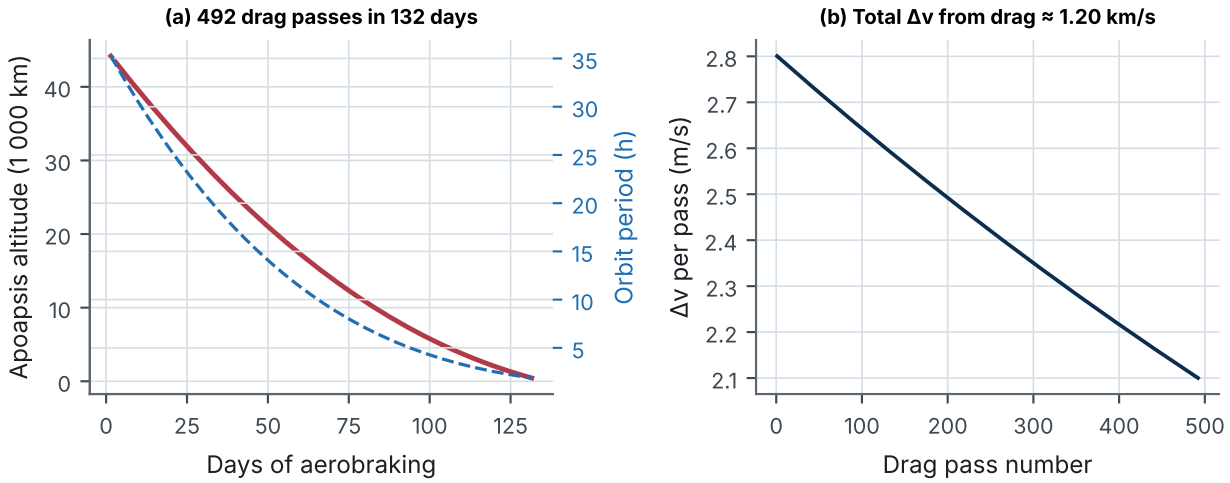
25.3.2 Aerobraking and Aerocapture

After capture, the orbit must be lowered from a long ellipse to the low science orbit. Doing this with the engine at Mars would cost another 1.2 km/s, so orbiters at Mars and Venus use **aerobraking**. The spacecraft lowers its periapsis to about 100–130 km. There, each pass through

the thin upper atmosphere removes a few metres per second of velocity, gradually shrinking the apoapsis (Figure 25.7). The solar arrays act as a drag brake. Ground teams (and increasingly onboard software) manage the **aerobraking corridor**. Small burns at apoapsis raise or lower the periapsis to keep dynamic pressure and heating within limits. This is needed because the density of the upper atmosphere can change by 30 % or more from one orbit to the next during dust storms. Aerobraking has three phases. In the **walk-in**, the periapsis is lowered step by step into the atmosphere. The **main phase** has hundreds of drag passes. In the **walk-out**, the periapsis is raised out of the atmosphere when the target apoapsis is reached.

TABLE 25.2 Aerobraking campaigns.

Mission	Planet	Years	Duration and notes
Magellan	Venus	1993	First aerobraking, used to circularise the orbit for gravity mapping
Mars Global Surveyor	Mars	1997–1999	About 16 months; slowed by a damaged solar-array hinge
Mars Odyssey	Mars	2001–2002	About 3 months
Mars Reconnaissance Orbiter	Mars	2006	426 drag passes in about 5 months; from 35-hour to 2-hour orbit
Venus Express	Venus	2014	Experimental campaign at the end of the mission
ExoMars Trace Gas Orbiter	Mars	2017–2018	About 11 months, interrupted by solar conjunction



Model: periapsis 100 km, density held at 4.5×10^{-8} kg/m³ by corridor control, scale height 10 km, ballistic coefficient 18 kg/m², drag applied impulsively at periapsis.

FIGURE 25.7 Aerobraking at Mars (illustrative model of an MRO-like orbiter). (a) Hundreds of drag passes shrink the orbit from a 35-hour ellipse to the 2-hour science orbit. (b) Each pass removes only a few metres per second; together they provide about 1.2 km/s — the equivalent of hundreds of kilograms of propellant.

Aerocapture goes further: the spacecraft, protected by an aeroshell, makes a single deep pass through the atmosphere to slow from hyperbolic speed directly into orbit, using lift to steer to the right exit conditions. It has never been flown, but studies show it could reduce the mass of orbiters bound for Uranus, Neptune or Titan by a third or more. It is a candidate technology for a future Uranus orbiter (Chapter 42).

EXAMPLE: CAPTURING AT MARS

An orbiter of 2 000 kg arrives at Mars from the 2026 launch window with v_∞ = 2.67 km/s (Chapter 24). Its periapsis is 300 km (radius 3 696 km); its engine has a specific impulse of 320 s.

1. Speed at periapsis on the arrival hyperbola: $\sqrt{2.67^2 + 2\times 42\,828/3\,696}$ = 5.50 km/s.
2. **Option A — direct capture into a 300 km circular orbit** (speed $\sqrt{\mu/r}$ = 3.40 km/s): Δv = 2.10 km/s. By the rocket equation, $1 - e^{-2\,100/(320\times 9.81)}$ = 49 % of the arrival mass — 976 kg — would be propellant.
3. **Option B — capture into a 35-hour ellipse** (semi-major axis 25 825 km, apoapsis altitude about 44 600 km, periapsis speed 4.64 km/s): Δv = 0.87 km/s. This uses 24 % of the mass — 482 kg.
4. Lowering the 35-hour orbit to the 300 km circle with the engine would cost another 1.23 km/s. By aerobraking, it costs only a few tens of metres per second for corridor control and the final periapsis raise.
5. Option B plus aerobraking saves about 450 kg — almost a quarter of the arrival mass — at the cost of about 5 months of intensive operations before the science mission starts.

25.4 What Orbiters Carry

25.4.1 Instruments

Planetary orbiters carry many of the instrument types described in Chapter 12, adapted to their targets (Table 25.3).

TABLE 25.3 Typical planetary orbiter instruments.

Instrument type	Measures	Examples
High-resolution cameras	Morphology, geology, landing-site hazards	HiRISE (MRO, 25–30 cm/pixel), LROC (LRO, 0.5 m), CaSSIS (TGO), HiRIC (Tianwen-1)
Context and stereo cameras	Regional mapping, topography	CTX (MRO, 6 m), HRSC (Mars Express), Kaguya Terrain Camera
Visible–infrared imaging spectrometers	Minerals, ices, hydrated materials	CRISM (MRO), OMEGA (Mars Express), M3 (Chandrayaan-1), MISE (Europa Clipper)
Thermal infrared radiometers	Surface temperature, thermal inertia, rock abundance	THEMIS (Odyssey), Diviner (LRO)
Gamma-ray and neutron spectrometers	Elemental composition, subsurface hydrogen (ice)	Mars Odyssey GRS, Lunar Prospector, MESSENGER
Radar sounders	Subsurface layers, ice, Europa's ice shell	MARSIS (Mars Express), SHARAD (MRO), REASON (Europa Clipper), RIME (JUICE)
Laser and radar altimeters	Topography, shape, tides	MOLA (MGS), LOLA (LRO), BELA (BepiColombo), GALA (JUICE)
Radio science / gravity	Interior structure, atmospheres by radio occultation	GRAIL, Juno, all orbiters' telecom systems
Magnetometers, particles and fields	Magnetic fields, plasma, induced fields from oceans	Galileo, Juno, MAVEN, Europa Clipper
Atmospheric spectrometers	Trace gases, winds, escape	TGO's NOMAD and ACS, MAVEN's IUVS, Juno's MWR

The **High Resolution Imaging Science Experiment (HiRISE)** on NASA's Mars Reconnaissance Orbiter, with a 0.5 m telescope, resolves objects about a metre across from 300 km. It has photographed landers descending under their parachutes (Chapter 27 shows Perseverance on its way down), tracked the movement of dunes and avalanches, and certified the safety of every NASA landing site since 2008.

Radar sounders operating at a few megahertz penetrate hundreds of metres to kilometres into ice and dry rock (Figure 25.8). On Mars, they have mapped the layered polar caps, buried glaciers at mid-latitudes and the thickness of ice deposits — essential information for future human missions seeking water.

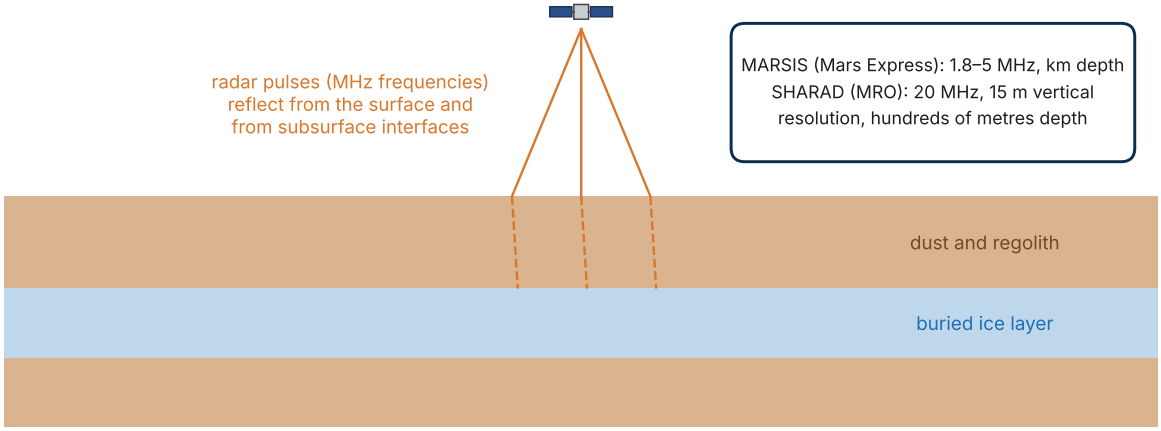


FIGURE 25.8 Principle of orbital radar sounding of subsurface layers.

25.4.2 Relays for Surface Missions

Landers and rovers have small antennas and limited power, and they can talk directly to the Earth only at low data rates. Orbiters overhead act as **relays** (Figure 25.9). A rover transmits to an orbiter at UHF during a pass of about ten minutes, at up to about 2 Mbit/s. The orbiter then forwards the data to the Deep Space Network over its high-gain antenna. Most of the data from NASA's Mars rovers have been relayed this way, mainly by Mars Odyssey and MRO, with MAVEN and ESA's Trace Gas Orbiter. Chinese missions use Tianwen-1's orbiter and, on the Moon's far side, the Queqiao relay satellites. Ageing relay orbiters — Odyssey was launched in 2001 and MRO in 2005 — are a growing concern for future Mars surface missions.

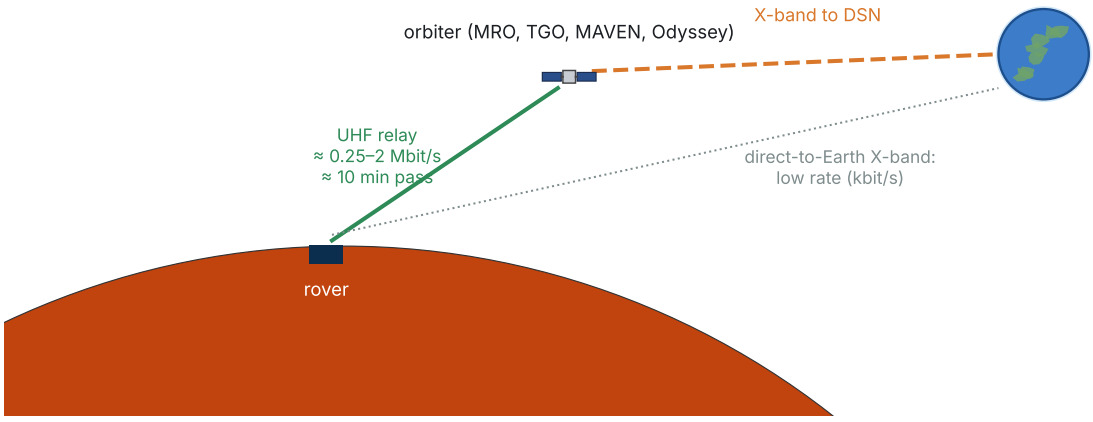


FIGURE 25.9 Relay of data from a Mars rover via an orbiter.

25.5 Orbiters World by World

25.5.1 The Moon

The Moon has been orbited by more spacecraft than any other body. After Luna 10 (1966), NASA's five **Lunar Orbiters** (1966–67) photographed potential Apollo landing sites and 99 % of the surface. A long gap followed until **Clementine** (1994) and **Lunar Prospector** (1998), whose neutron spectrometer detected excess hydrogen at the poles — possibly water ice in permanently shadowed craters. The 2000s brought ESA's ion-propelled **SMART-1**, Japan's **Kaguya**, China's **Chang'e-1** and **Chang'e-2**, and India's **Chandrayaan-1**, whose NASA-supplied Moon Mineralogy Mapper detected hydroxyl and water molecules on the sunlit surface in 2009.

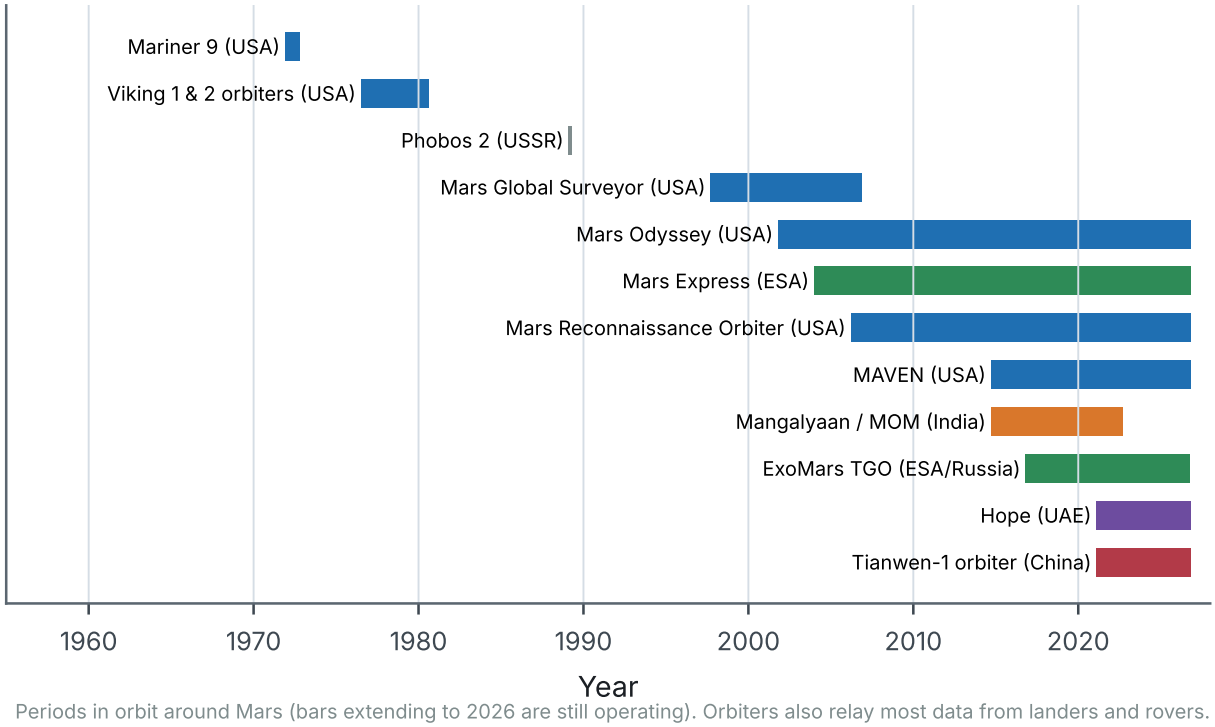
NASA's **Lunar Reconnaissance Orbiter (LRO)**, launched in 2009 and still operating, has mapped the Moon in unprecedented detail. Its cameras resolve the Apollo landers and astronauts' tracks, and its laser altimeter has produced a global topographic map. Its Diviner radiometer found the coldest places measured in the Solar System — about 25 K in polar craters (Figure 25.10). Its companion **LCROSS** deliberately crashed a rocket stage into the crater Cabeus in 2009 and detected water in the ejecta. Today, India's **Chandrayaan-2** orbiter, Korea's **Danuri** and China's Queqiao relays orbit the Moon, preparing for the Artemis and Chinese crewed landings (Chapters 27, 31). Danuri carries NASA's ShadowCam, which images inside shadowed craters by faint reflected light.

FIGURE 25.10 Earthrise as seen by NASA's Lunar Reconnaissance Orbiter in 2015, with the crater Compton on the Moon's far side in the foreground. Image: NASA/GSFC.

25.5.2 Mars

Mars has had a continuous orbital presence since 1997 (Figure 25.11). **Mariner 9** (1971) revealed giant volcanoes, canyons and dry river valleys; the **Viking orbiters** (1976) mapped the planet; **Mars Global Surveyor** (1997–2006) measured its topography with a laser altimeter (MOLA) and discovered remnant crustal magnetism. **Mars Odyssey** (2001) found hydrogen-rich ground — ice — beneath the surface at high latitudes. ESA's **Mars Express** (2003) and NASA's **Mars**

Reconnaissance Orbiter (2006) discovered clays and sulphates formed in water, revealing a warmer, wetter early Mars. They also imaged active processes from avalanches to seasonal carbon-dioxide jets. NASA's **MAVEN** (2014) showed how the solar wind has stripped Mars's atmosphere. ESA–Roscosmos's **Trace Gas Orbiter** (2016) placed strict limits on atmospheric methane, and India's **Mangalyaan** (2014–2022) was the first Asian Mars orbiter. The UAE's **Hope** (2021) studies the atmosphere from a high orbit, and China's **Tianwen-1** (2021) delivered the Zhurong rover. NASA's twin **ESCAPADE** smallsats, launched in 2025, will study the Martian magnetosphere from 2027.



Periods in orbit around Mars (bars extending to 2026 are still operating). Orbiters also relay most data from landers and rovers.

FIGURE 25.11 Orbiters at Mars since 1971. Six orbiters from four agencies and the UAE were operating in 2026.

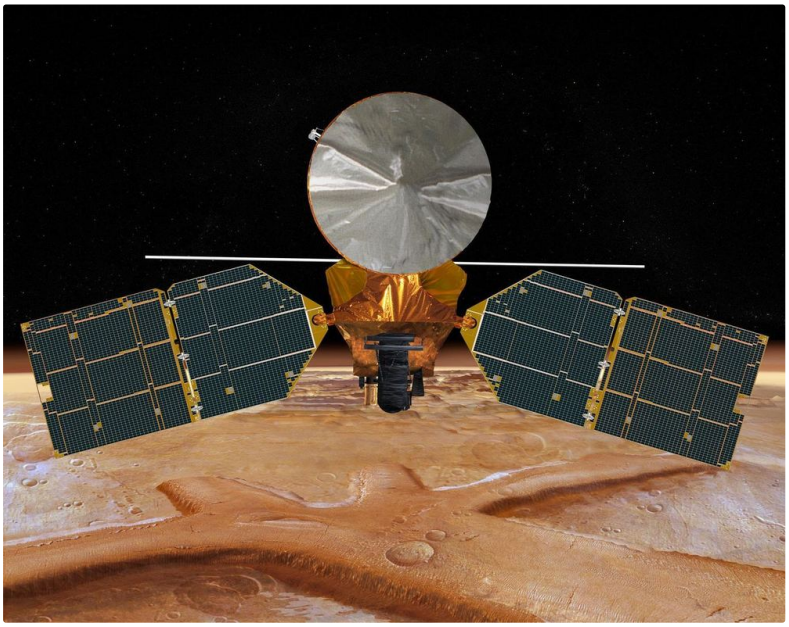


FIGURE 25.12 Artist's concept of NASA's Mars Reconnaissance Orbiter, with its 3 m high-gain antenna and large solar arrays. Since 2006 it has returned more data than all other Mars missions combined. Image: NASA/JPL (NASA/JPL).

25.5.3 Venus

Venus's thick clouds hide its surface from optical cameras, so orbiters use radar. The Soviet **Venera 15 and 16** (1983) and NASA's **Magellan** (1990–1994) mapped it, revealing a young volcanic surface (Figure 25.13). ESA's **Venus Express** (2006–2014) and Japan's **Akatsuki** (2015–2024) studied its atmosphere, including the "super-rotation" in which the cloud tops circle the planet in four days. Three new orbiters were planned — NASA's **VERITAS**, ESA's **EnVision** and ISRO's **Shukrayaan** — together with NASA's DAVINCI descent probe, although US budget pressures in 2025–26 made the NASA missions' future uncertain.

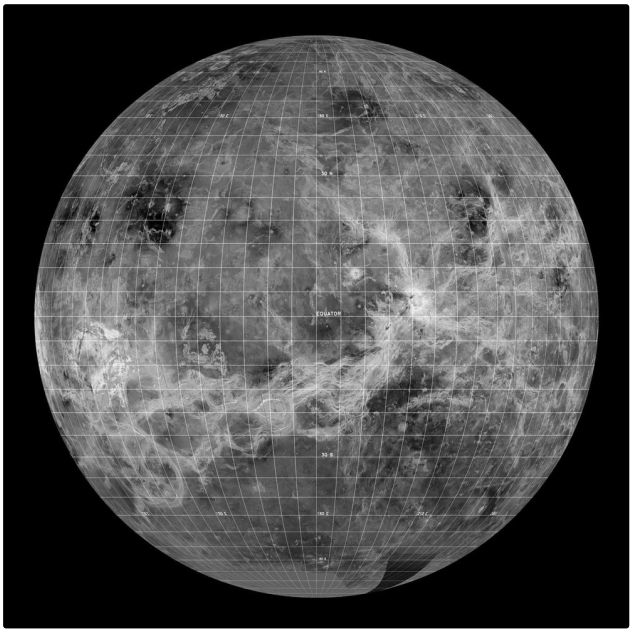


FIGURE 25.13 A global radar view of Venus from Magellan data, centred on 180° longitude. Venus's surface is dominated by volcanic plains and highlands. Image: NASA/JPL (NASA/JPL).

25.5.4 Mercury

NASA's **MESSENGER** (2011–2015) was the first Mercury orbiter. It found that Mercury's surface is unexpectedly rich in volatile elements such as sulphur and potassium, and that its magnetic field is offset to the north. It also found that permanently shadowed craters near its poles contain water ice — on the planet closest to the Sun (Figure 25.14). The ESA–JAXA **BepiColombo** mission, launched in 2018, comprises two orbiters: ESA's Mercury Planetary Orbiter and JAXA's Mio, carried by a transfer module with electric propulsion. After a thruster power problem in 2024 delayed its arrival by about a year, it enters orbit around Mercury in November 2026.

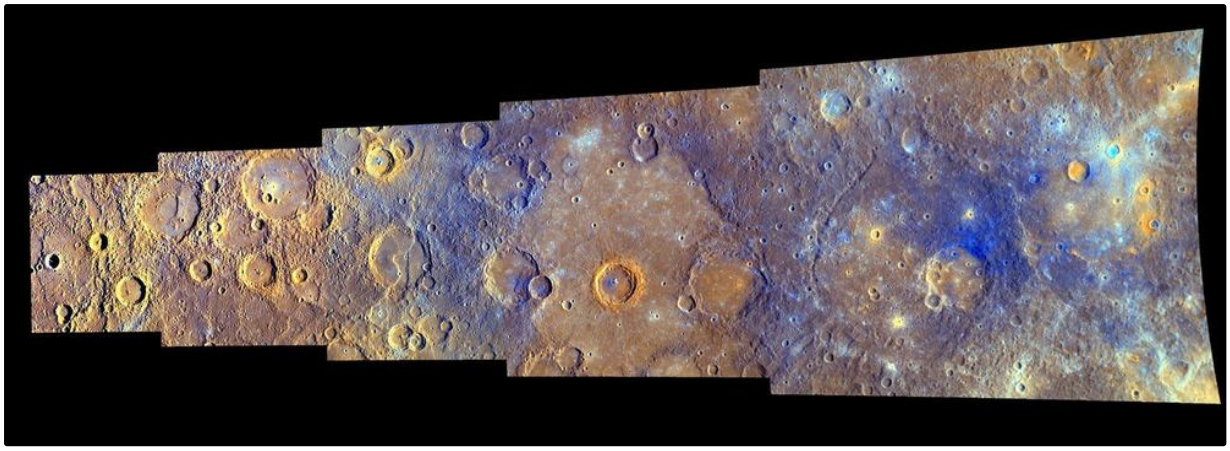


FIGURE 25.14 MESSENGER colour mosaic of Mercury's surface. Enhanced colours reveal variations in composition between volcanic plains and older terrain. Image: NASA/JPL.

25.5.5 Jupiter

Galileo (1995–2003) dropped a probe into Jupiter's atmosphere and found evidence from magnetic-field measurements that Europa, Ganymede and Callisto have subsurface oceans. **Juno**, in orbit since 2016, studies Jupiter's interior from a polar orbit (Figure 25.3). It found that the planet's core is "dilute" — mixed with the surrounding envelope. It also found that its zonal winds extend about 3 000 km deep, and that its poles are covered by clusters of giant cyclones. Its JunoCam has produced spectacular images (Figure 25.15). Two new missions are on their way. NASA's **Europa Clipper** will make about 49 close flybys of Europa from 2030 to assess its habitability. ESA's **JUICE** will tour the Galilean moons from 2031 before entering orbit around Ganymede in 2034 — the first spacecraft to orbit a moon of another planet.

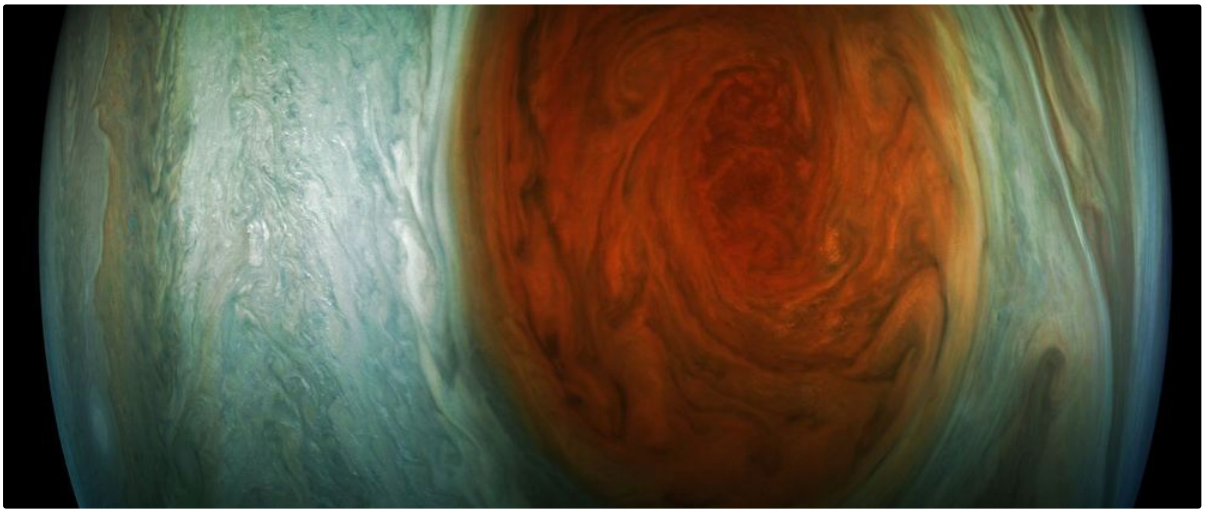


FIGURE 25.15 Jupiter's Great Red Spot imaged by JunoCam on Juno in July 2017 (enhanced colour). The storm, larger than the Earth, extends about 300 km below the clouds. Image: NASA/JPL.

25.5.6 Saturn

Cassini–Huygens (2004–2017), a collaboration of NASA, ESA and the Italian Space Agency, was one of the most successful planetary missions. It delivered the Huygens probe to the surface of Titan (Chapter 27) and found hydrocarbon lakes and seas near Titan's poles. In 2005 it discovered

water-vapour and ice plumes erupting from the south pole of Enceladus (Figure 25.16). The plumes are fed by a subsurface ocean that contains salts, silica and organic molecules, making Enceladus one of the most promising places to search for life.

FIGURE 25.16 Plumes of ice and water vapour rising from the south polar region of Enceladus, back-lit by the Sun, imaged by Cassini. Image: NASA/JPL (NASA/JPL-Caltech/Space Science Institute).

25.5.7 Small Bodies

Orbiting small bodies demands precise navigation in very weak gravity. NASA's **NEAR Shoemaker** orbited Eros for a year and landed on it in 2001. **Dawn** orbited Vesta (2011–12) and Ceres (2015–18), finding bright salt deposits on Ceres. ESA's **Rosetta** accompanied comet 67P/Churyumov–Gerasimenko for two years (2014–2016) as it passed perihelion. The sample-return missions Hayabusa2 and OSIRIS-REx also mapped their asteroids from orbit before collecting samples (Chapter 29). ESA's **Hera** will arrive at the binary asteroid Didymos–Dimorphos in November 2026, a month earlier than first planned. It will measure the effects of NASA's DART impact of 2022. NASA's **Psyche** will orbit the metal-rich asteroid Psyche from 2029.

TABLE 25.4 Planetary orbiters operating or arriving in 2026 (selected).

Mission	Agency	Target	Arrival	Status (Oct 2026)
LRO	NASA	Moon	2009	Operating
Chandrayaan-2 orbiter	ISRO	Moon	2019	Operating
Danuri (KPLO)	KARI	Moon	2022	Operating
Mars Odyssey, MRO, MAVEN	NASA	Mars	2001, 2006, 2014	Operating (relay and science)
Mars Express, ExoMars TGO	ESA (TGO with Roscosmos until 2022)	Mars	2003, 2016	Operating
Hope	UAE	Mars	2021	Operating
Tianwen-1 orbiter	CNSA	Mars	2021	Operating
Juno	NASA	Jupiter	2016	Extended operations uncertain after 2025

Mission	Agency	Target	Arrival	Status (Oct 2026)
BepiColombo (MPO, Mio)	ESA, JAXA	Mercury	Nov 2026	Orbit insertion
Hera	ESA	Didymos–Dimorphos	Nov 2026	Final approach
Europa Clipper, JUICE	NASA, ESA	Jupiter system	2030, 2031	En route
Psyche	NASA	Asteroid Psyche	2029	En route

25.6 Ending an Orbiter Mission

Orbiters usually end when they run out of propellant for attitude control, when a critical component fails, or when funding stops. How they end is governed by **planetary protection** (Chapter 24): an orbiter must not contaminate a world that might host life, or that future missions will search for life.

- **Disposal into the planet.** Galileo was deliberately plunged into Jupiter in 2003 and Cassini into Saturn in 2017, so that they could never crash into Europa, Enceladus or Titan. Juno's mission plan also ends with a plunge into Jupiter.
- **Long-lived parking orbits.** Mars orbiters that have not been sterilised must stay in orbit for at least 20 years after launch (with 99 % probability) and 50 years (with 95 % probability). When Dawn ran out of hydrazine in 2018, it was left in an orbit around Ceres that is expected to remain stable for decades.
- **Controlled impacts.** Where contamination is not a concern, orbiters often end with an impact that is itself an experiment. LCROSS, GRAIL's twin spacecraft (2012) and LADEE (2014) impacted the Moon, and MESSENGER crashed into Mercury in 2015. Rosetta made a slow descent onto comet 67P in 2016, taking images until the final seconds.
- **Unplanned endings.** Mars Global Surveyor was lost in 2006 after a software error pointed a solar array wrongly and exposed its battery to the Sun. India's Mangalyaan fell silent in 2022, probably after a long eclipse drained its battery. Contact with Akatsuki was lost in 2024.

EXAMPLE: CHOOSING A MARS MAPPING ORBIT

A new Mars orbiter must image the surface at the same local time each day (to keep shadows consistent) from an altitude of about 300 km.

1. Mars's $J_2 = 1.96 \times 10^{-3}$, radius 3 396 km and $\mu = 42\,828$ km^3/s^2 give, at 300 km, a mean motion $n = \sqrt{\mu/a^3} = 9.2 \times 10^{-4}$ rad/s.
2. The orbit plane must precess 360° in one Martian year (687 days): $\dot{\Omega} = 1.06 \times 10^{-7}$ rad/s.
3. From $\dot{\Omega} = -\tfrac{3}{2} n J_2 (R/a)^2 \cos i$, $\cos i = -0.052$, so $i \approx 93.0^\circ$ (Figure 25.2).
4. The ground track shifts westward by about 28° of longitude per orbit (Mars rotates 360° in 24.6 hours while the orbit takes about 1.9 hours). A camera with a 6 km swath therefore needs thousands of orbits — several Earth years — to map the planet completely, as MRO's Context Camera did.
5. Planetary protection (Category III) requires the orbit to remain above the atmosphere for at least 20 years after launch, which a 300 km orbit easily satisfies if the spacecraft is clean or later raised.

AI FOR PLANETARY ORBITERS

Planetary scientists use machine learning to detect craters, landslides, dust devils and new impacts in millions of orbital images, to map minerals from hyperspectral data, and to prioritise downlink of the most interesting observations. On board, autonomous instruments can detect transient events — such as plumes, storms or impacts — and decide what to observe and return (Chapter 51).

KEY TAKEAWAYS

- Orbiters provide global, long-duration mapping, interior measurements and relays for surface missions.
- Orbit design depends on each body. Lunar mascons allow only a few frozen low orbits, and Mars's bulge allows nearly polar sun-synchronous orbits. Venus and Mercury pose thermal challenges, Jupiter's radiation shapes Juno's orbit, and small bodies have weak, irregular gravity.
- Instruments include high-resolution cameras (HiRISE, about 30 cm/pixel), imaging spectrometers, radar sounders, altimeters, gravity science and particle and field sensors.
- Key discoveries include water ice at the lunar and Mercurian poles, Mars's watery past and atmospheric loss, Venus's volcanic surface, Jupiter's dilute core, and Enceladus's ocean plumes.
- In 2026, orbiters operate at the Moon, Mars and Jupiter; BepiColombo and Hera arrive at Mercury and Didymos, and Europa Clipper and JUICE are en route to Jupiter.

REVIEW QUESTIONS

1. Give four kinds of science that an orbiter can do better than a flyby or a lander.
2. Why are only a few low lunar orbits stable for long periods? What are frozen orbits?
3. A spacecraft arrives at Mars with v_∞ = 2.65 km/s. Calculate the Δv to enter a circular orbit 400 km above the surface in one burn at periapsis.
4. Why do many Mars orbiters first capture into a long elliptical orbit and then aerobrake? What are the risks of aerobraking?
5. Explain why a radiation vault is needed at Jupiter and how Juno's orbit was designed to limit radiation dose.
6. How do Mars orbiters act as relays for surface missions, and why does this increase the data returned by rovers?
7. Describe three major discoveries made by planetary orbiters.
8. Why are orbiters around Jupiter and Saturn deliberately crashed into the planet at the end of their missions?

CHAPTER

26 Space Probes and Flyby Missions

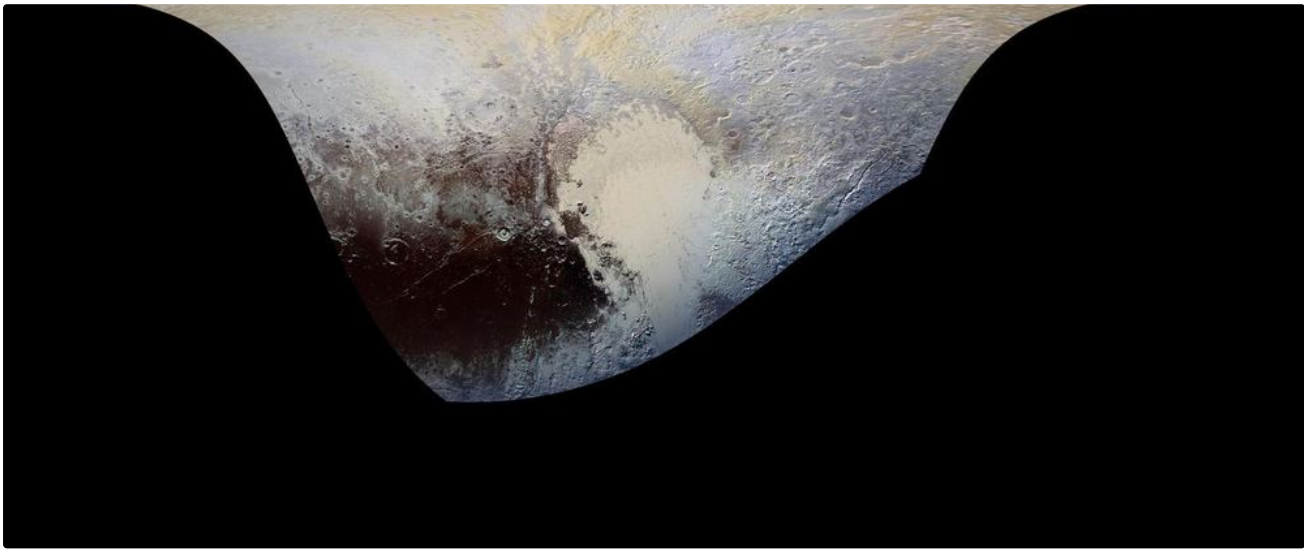


Image: NASA/JPL

A flyby gives a few hours of close-up science after years of travel, yet flybys revealed most of the Solar System to us — the volcanoes of Io, the rings of Uranus, the geysers of Triton, the nitrogen glaciers of Pluto. This chapter explains the different kinds of space probes — flyby spacecraft, atmospheric entry probes, impactors and interstellar travellers — and the engineering behind them, from encounter navigation and imaging at 14 km/s to radioisotope power and data links across 25 billion kilometres. It follows the Pioneers, Voyagers and New Horizons to the edge of the heliosphere, describes how DART moved an asteroid, and reviews the probes active in 2026.

IN THIS CHAPTER

26.1 Kinds of Probes
26.2 Sixty Years of Flybys
26.3 The Grand Tour: Pioneer and Voyager
26.4 Engineering a Flyby
26.5 Power and Communications Far from the Sun
26.6 Atmospheric Entry Probes
26.7 Impactors and Planetary Defence
26.8 Comets, Asteroids and Interstellar Visitors
26.9 Into Interstellar Space
26.10 Probes in 2026

A **space probe** is an uncrewed spacecraft that leaves Earth orbit to explore the Solar System. In a broad sense every planetary mission is a probe, but this chapter concentrates on missions that do not stay in orbit or land. These are spacecraft that **fly past** their targets, **plunge into** atmospheres, **crash into** surfaces on purpose, or travel **out of the Solar System**. Orbiters are described in Chapter 25, landers in Chapter 27, rovers in Chapter 28 and sample-return missions in Chapter 29.

26.1 Kinds of Probes

TABLE 26.1 The main kinds of probes discussed in this chapter.

Type	What it does	Advantages	Limitations	Examples
Flyby	Passes a target once, observing for hours to days	Cheapest way to reach a new world; can visit several targets	Brief, one-sided view; high speed blurs images	Mariner 4, Voyager 1 and 2, New Horizons, Lucy
Atmospheric entry probe	Enters an atmosphere, measuring it during descent	Direct measurement of composition, temperature, winds	Short life (minutes to hours); extreme entry heating	Pioneer Venus, Galileo probe, Huygens
Impactor	Strikes a surface deliberately	Excavates subsurface material; tests asteroid deflection	Single event; must be observed by another spacecraft or telescope	Deep Impact, LCROSS, DART
Heliospheric and interstellar	Travels outward to sample the solar wind and interstellar medium	The only direct measurements of the Sun's outer boundary	Power and communications limits; decades-long missions	Pioneer 10 and 11, Voyager 1 and 2
Close-solar	Dives towards the Sun	In-situ measurements of the corona	Intense heat; needs many gravity assists	Helios, Parker Solar Probe, Solar Orbiter (Chapter 22)

Flybys are the natural first step in exploring a world. A flyby needs no braking propellant and can be combined with gravity assists to reach several targets in one mission (Chapter 24). They are followed, where budgets allow, by orbiters for global mapping and landers for surface measurements. For the most distant worlds — Uranus, Neptune, Pluto and the Kuiper Belt — flybys remain our only close-up views.

26.2 Sixty Years of Flybys

The first spacecraft to escape Earth was the Soviet **Luna 1** in January 1959. It missed the Moon by about 6 000 km and became the first artificial object in orbit around the Sun. The same year **Luna 3** photographed the Moon's far side. The planetary era began with NASA's **Mariner 2**, which flew past Venus in December 1962 and showed that its surface was hot enough to melt lead. **Mariner 4** returned the first close-up pictures of another planet — 21 images of a cratered, Moon-like Mars — in July 1965 (Figure 26.1).

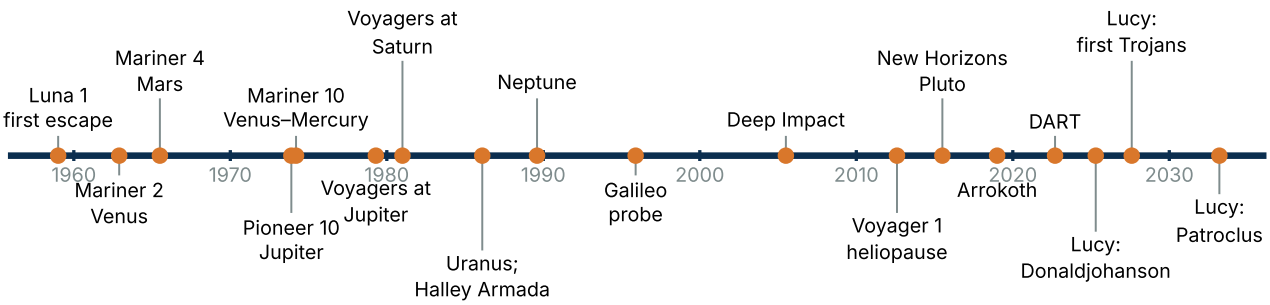


FIGURE 26.1 Milestones in flyby and probe exploration, 1959–2033.

Mariner 10 (1973–75) was the first spacecraft to use a gravity assist: a Venus flyby bent its path towards Mercury, which it passed three times. **Pioneer 10** became the first spacecraft to cross the asteroid belt and fly past Jupiter (December 1973); **Pioneer 11** followed it past Jupiter and then made the first Saturn flyby in 1979. In 1986 an international fleet — the **Halley Armada** of ESA's Giotto, the Soviet Vega 1 and 2, and Japan's Suisei and Sakigake — met Halley's Comet. Giotto passed 596 km from the nucleus and returned the first images of a comet's core.

TABLE 26.2 Firsts in flyby and probe exploration.

Year	Spacecraft	Achievement
1959	Luna 1 (USSR)	First spacecraft to escape Earth and orbit the Sun
1959	Luna 3 (USSR)	First images of the Moon's far side
1962	Mariner 2 (USA)	First successful planetary flyby (Venus)
1965	Mariner 4 (USA)	First close-up images of Mars
1967	Venera 4 (USSR)	First data from inside another planet's atmosphere (Venus)
1973	Pioneer 10 (USA)	First crossing of the asteroid belt; first Jupiter flyby
1974	Mariner 10 (USA)	First gravity assist; first Mercury flyby
1979	Pioneer 11 (USA)	First Saturn flyby
1985	ICE (USA)	First comet encounter (Giacobini–Zinner)
1986	Giotto (ESA)	First close images of a comet nucleus (Halley)
1986, 1989	Voyager 2 (USA)	Only flybys of Uranus and Neptune
1991, 1993	Galileo (USA)	First asteroid flybys (Gaspra, Ida) and first asteroid moon (Dactyl)
1995	Galileo probe (USA)	First entry probe into a giant planet
2005	Deep Impact (USA)	First impact experiment on a comet
2012	Voyager 1 (USA)	First spacecraft in interstellar space
2015	New Horizons (USA)	First Pluto flyby
2019	New Horizons (USA)	Most distant object explored (Arrokoth, 6.6 billion km)
2022	DART (USA)	First change of an asteroid's orbit by a human-made object

26.3 The Grand Tour: Pioneer and Voyager

26.3.1 Pioneer 10 and 11

The two Pioneers were simple, spin-stabilised spacecraft of about 260 kg, powered by four SNAP-19 radioisotope generators and communicating through a 2.74 m dish (Figure 26.2). Their main task was to find out whether a spacecraft could survive the asteroid belt and Jupiter's radiation belts. They could — just: Pioneer 10's instruments recorded radiation doses close to their limits. Both spacecraft carry a gold-anodised plaque showing a man and a woman and the Sun's location relative to 14 pulsars. Pioneer 11 fell silent in 1995 and Pioneer 10 in 2003, at a distance of 80 AU.



FIGURE 26.2 Pioneer 10 being prepared for launch in 1972. The 2.74 m antenna dominates the spacecraft; the radioisotope generators are on booms to keep their radiation away from the instruments. Image: NASA/ARC.

26.3.2 Voyager 1 and 2

In the late 1970s the four giant planets lined up so that a single spacecraft could visit all of them, using each one's gravity to reach the next. This alignment occurs about once every 175 years. NASA's two **Voyagers**, launched in August and September 1977, made use of it (Figure 26.3). Voyager 1 flew past Jupiter (March 1979) and Saturn (November 1980); it was then deliberately sent close to Saturn's moon Titan, which bent its path northward out of the ecliptic. Voyager 2 visited Jupiter (July 1979), Saturn (August 1981), Uranus (January 1986) and Neptune (August 1989) — still the only spacecraft to have seen the two ice giants close up.

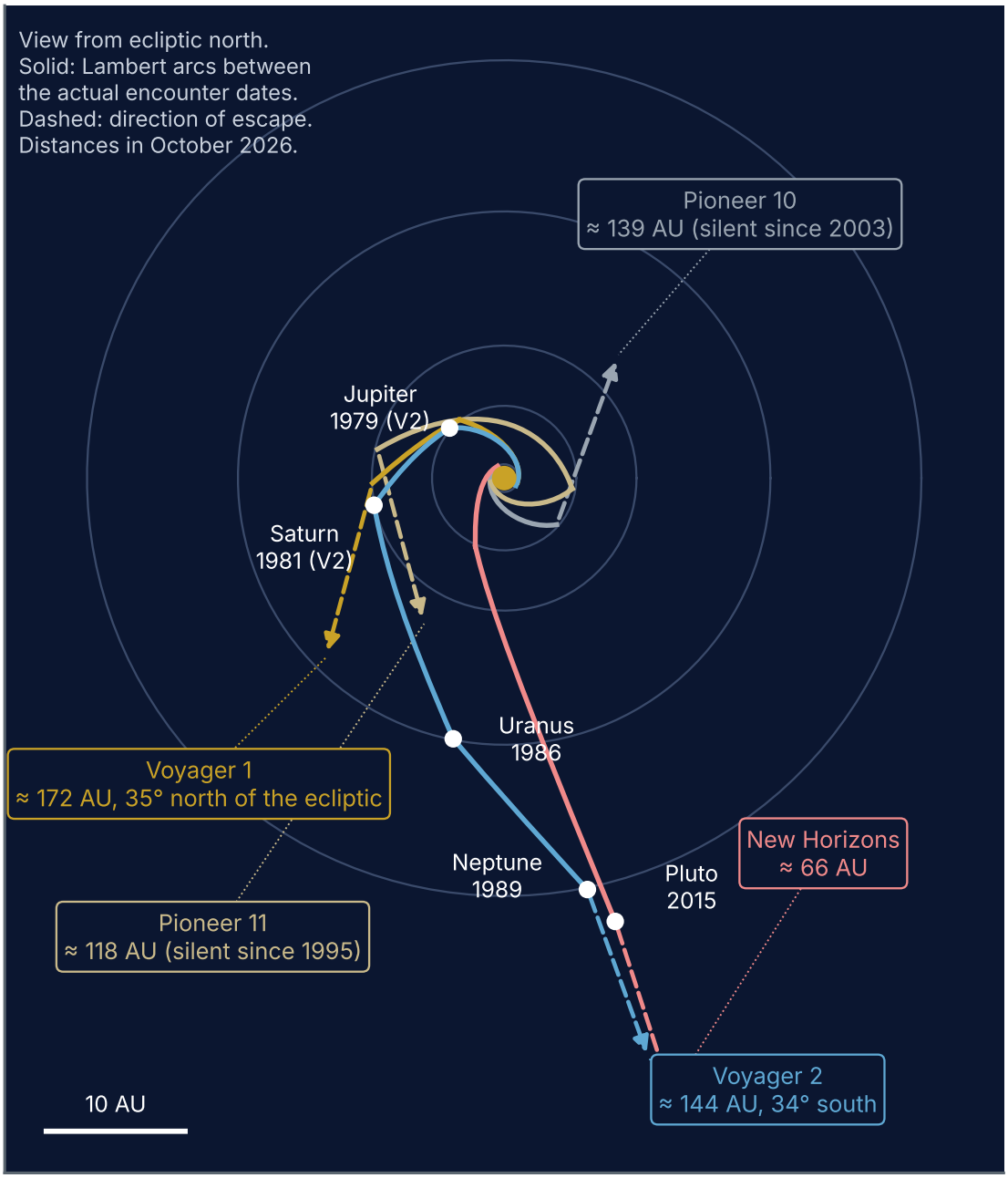


FIGURE 26.3 The trajectories of the five spacecraft that are leaving the Solar System, computed from the planets' positions on the actual encounter dates.

Each Voyager weighed 825 kg at launch and carried ten science instruments, a 3.66 m high-gain antenna and three radioisotope generators producing 470 W. It also had a scan platform that pointed the cameras and spectrometers without turning the whole spacecraft (Figure 26.4). Their three onboard computers had a combined memory of about 69 kilobytes, and data were recorded on an eight-track digital tape recorder.

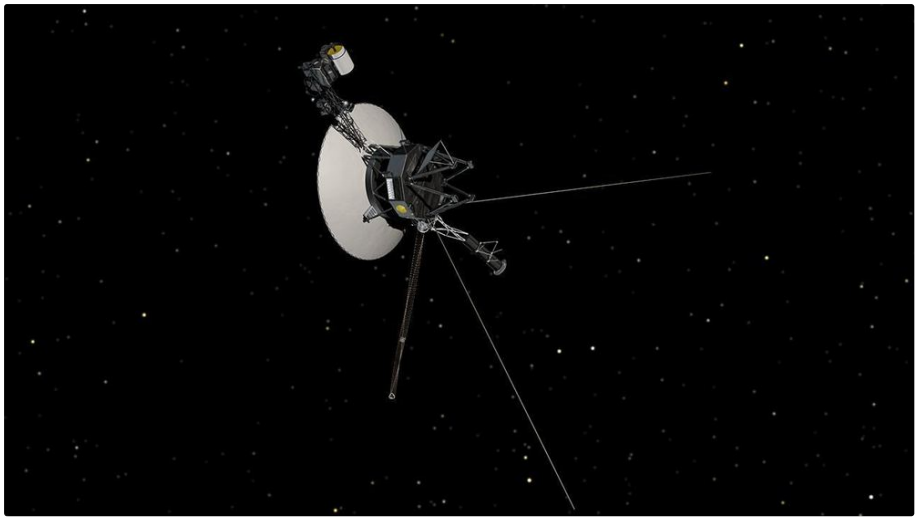


FIGURE 26.4 Artist's concept of a Voyager spacecraft. The 3.66 m antenna points to Earth; the magnetometer boom (13 m) and the RTG boom extend to the sides. Image: NASA/JPL (NASA/JPL-Caltech).

The Voyagers transformed planetary science:

- **Jupiter:** active volcanoes on Io — the first seen beyond Earth — a thin ring, and the cracked, icy surface of Europa, hinting at an ocean beneath.
- **Saturn:** thousands of ringlets and the "spokes" in the rings, shepherd moons that confine narrow rings, and a thick nitrogen atmosphere on Titan.
- **Uranus:** a magnetic field tilted by 59° to the rotation axis, ten new moons and the bizarre, patchwork surface of Miranda.
- **Neptune:** the Great Dark Spot, the fastest winds in the Solar System (about 2 000 km/h), and nitrogen geysers on its large moon Triton. Triton orbits backwards and is probably a captured Kuiper Belt object (Figure 26.5).

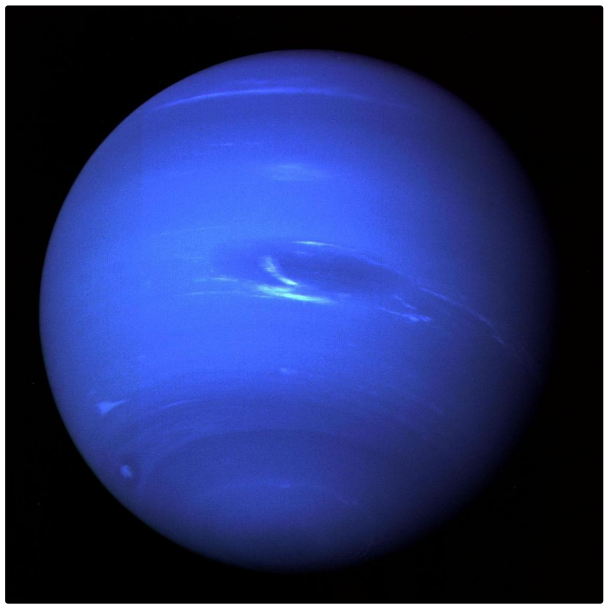


FIGURE 26.5 Neptune imaged by Voyager 2 in August 1989, showing the Great Dark Spot (centre) and bright methane-ice clouds. Image: NASA/JPL (Voyager 2).

On 14 February 1990, at a distance of 6 billion km, Voyager 1 turned its cameras back towards the Sun and photographed the planets for a "family portrait". In one frame Earth appears as a pale blue dot less than a pixel across (Figure 26.6). The cameras were then switched off to save power

and memory; they have not been used since.



FIGURE 26.6 The "Pale Blue Dot": Earth (in the band of scattered sunlight, right of centre) seen by Voyager 1 from 6 billion km in 1990, reprocessed in 2020. Image: NASA/JPL (NASA/JPL-Caltech).

Each Voyager carries a **Golden Record** in case either spacecraft is ever found by another civilisation. It is a gold-plated copper disc with sounds and images of Earth, greetings in 55 languages and 90 minutes of music.

26.4 Engineering a Flyby

26.4.1 Arriving at the Right Place at the Right Time

A flyby is aimed in the **B-plane**, the plane through the target's centre perpendicular to the incoming asymptote of the hyperbolic trajectory (Chapter 24). The aim point in this plane determines the closest-approach distance and the bending of the trajectory; the arrival time determines which side of the target is lit and which moons can be observed. Errors grow as the spacecraft approaches: a small velocity error made weeks before encounter can move the aim point by thousands of kilometres. Navigators therefore combine radio tracking (Doppler, ranging and delta-DOR) with **optical navigation** — images of the target against the stars taken by the spacecraft's own cameras. They also make a series of trajectory correction manoeuvres in the final weeks. For the Pluto flyby, the uncertainty in Pluto's own position, known only from 85 years of telescopic observations, was larger than that of the spacecraft. Optical navigation reduced it to a few hundred kilometres before the encounter. New Horizons arrived within about 90 seconds of the planned time.

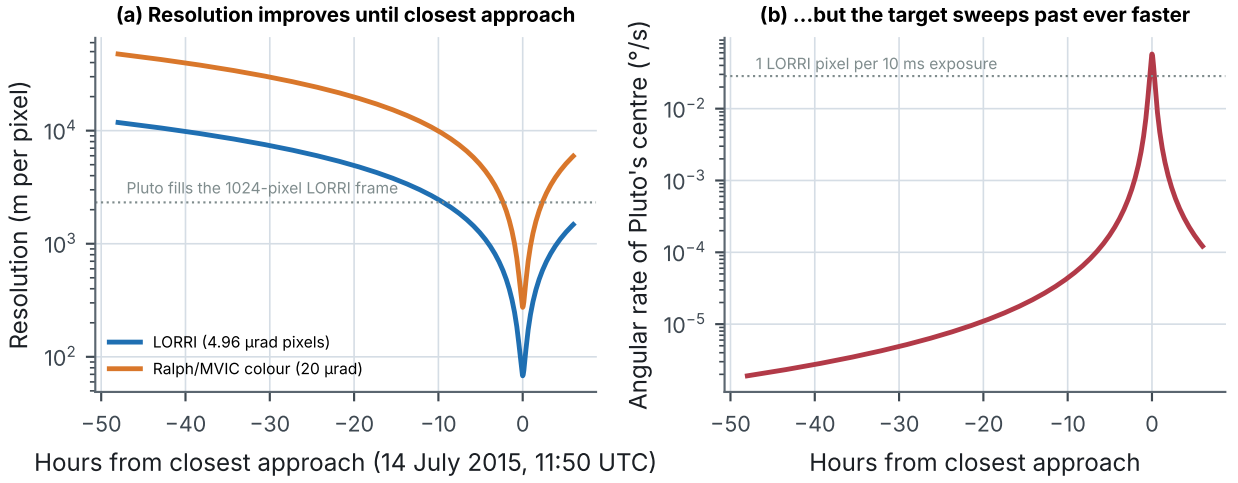
26.4.2 Sequences and Fault Protection

The most important hours of a flyby must run without help from the ground: at Pluto, a radio signal took 4.5 hours each way. The encounter is therefore executed by a **command sequence** loaded days in advance, which points the spacecraft and instruments, sets exposure times and records the data. **Fault protection** software monitors the spacecraft and, if a problem occurs,

puts it into a safe state — but during an encounter, entering safe mode could mean losing the unrepeatable observations. Flyby sequences therefore include special "critical sequence" fault protection that keeps the observations going through most faults. Ten days before the Pluto flyby, on 4 July 2015, New Horizons went into safe mode. Its main computer was overloaded while compressing data and loading the encounter sequence at the same time. The team recovered the spacecraft in three days and lost only minor science.

26.4.3 Imaging at High Speed

A flyby spacecraft sees its target grow from a dot to a world in a few days, and then sweep past in hours. The resolution — the size of a camera pixel projected on the target — improves in proportion to the distance, but the target's angular rate rises sharply near closest approach (Figure 26.7). New Horizons passed Pluto at 13.78 km/s. At closest approach, 12 500 km above the surface, Pluto's centre moved across the sky at 0.058° per second. A feature would cross one pixel of the LORRI telescopic camera in about 5 milliseconds. Exposures had to be short, or the spacecraft had to turn to follow the target, which it did with its thrusters.



Flyby speed 13.78 km/s; closest approach 12 500 km above Pluto's surface. Data were stored on board and returned over the next 15 months.

FIGURE 26.7 New Horizons at Pluto. (a) The resolution of the LORRI camera and the colour imager improved from kilometres to about 70 m per pixel at closest approach. (b) Pluto's angular rate rose more than ten-thousandfold in the last two days.

Voyager mounted its cameras on a **scan platform** that could point independently of the spacecraft, so the high-gain antenna could stay pointed at Earth. New Horizons, to save mass and complexity, had no scan platform: the whole spacecraft turned to point its instruments, so it could not transmit while observing. It recorded about 6.25 gigabytes during the Pluto encounter on two solid-state recorders and then spent 15 months sending it to Earth at 1–4 kilobits per second.

EXAMPLE: IMAGE SMEAR DURING A FLYBY

At closest approach to Pluto, New Horizons was 13 690 km from Pluto's centre, moving at 13.78 km/s.

1. Angular rate of the target as seen from the spacecraft: $\omega = v/d$ = 13.78 / 13 690 = 1.0×10^{-3} rad/s (0.058°/s).
2. LORRI's pixels subtend 4.96 µrad. The time for the image to move by one pixel is 4.96×10^{-6} / $1.0 \times 10^{-3} \approx 5$ ms.
3. Sunlight at 33 AU is 1/1 100 as bright as at Earth, so long exposures are tempting. But an exposure of 50 ms would smear the image over 10 pixels unless the spacecraft rotated to track the target.
4. A day earlier, at 1.2 million km, Pluto was almost straight ahead. Its angular rate, $v\,b/r^2$ with b = 13 690 km, was only about 10^{-7} rad/s and smear was negligible — but each pixel covered 6 km.

The design of a flyby encounter is therefore a trade between distance (resolution), speed of tracking (smear) and the time needed to observe each target.

26.4.4 Pluto and Arrokoth: New Horizons

New Horizons was launched on 19 January 2006 on an Atlas V with an additional solid-fuel upper stage, leaving Earth faster than any spacecraft before it (about 16.3 km/s relative to Earth). A Jupiter flyby in February 2007 shortened its journey by three years. Most of the 9.5-year cruise was spent in hibernation, with the spacecraft spinning slowly and sending a weekly beacon tone.

At Pluto on 14 July 2015, it found a geologically active world. **Sputnik Planitia** is a 1 000 km plain of nitrogen ice slowly convecting in cells tens of kilometres across. It also found mountains of water ice several kilometres high, a blue, layered haze and evidence of a possible ocean beneath the ice. Its large moon **Charon** has a vast canyon system and a reddish polar cap made of material from Pluto's escaping atmosphere (Figure 26.8).



FIGURE 26.8 Charon in enhanced colour from New Horizons. The reddish polar region (top) is coloured by organic material formed from methane that escaped from Pluto. Image: NASA/JPL.

On 1 January 2019, New Horizons flew 3 500 km past **Arrokoth**, a 36 km Kuiper Belt object 6.6 billion km from the Sun — the most distant object ever explored. Arrokoth turned out to be a "contact binary" of two flattened lobes that merged gently at walking speed, supporting the

theory that planetesimals formed by the gentle collapse of clouds of pebbles. New Horizons continues to study the Kuiper Belt, the heliosphere and the faint cosmic optical background, which can only be measured from far beyond the zodiacal dust of the inner Solar System.

26.5 Power and Communications Far from the Sun

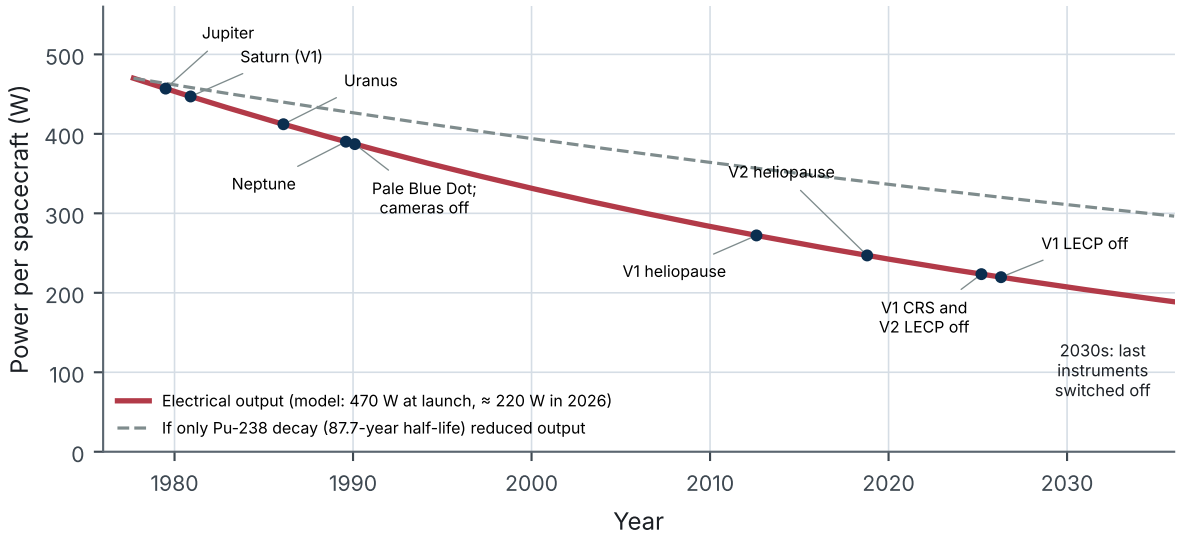
26.5.1 Radioisotope Power

Sunlight weakens with the square of the distance: at Jupiter it is 1/27 as strong as at Earth, at Saturn 1/90 and at Pluto about 1/1 100. Solar arrays work as far as Jupiter — Juno, JUICE, Europa Clipper and Lucy all use very large arrays. Beyond that, every probe has relied on **radioisotope thermoelectric generators** (RTGs), which convert heat from the decay of plutonium-238 into electricity (Chapter 5).

TABLE 26.3 Radioisotope thermoelectric generators used on US deep-space probes.

Generator	Electrical power at launch	Mass	Missions
SNAP-19	≈ 40 W	14 kg	Pioneer 10 and 11 (four each), Viking landers
MHW-RTG	≈ 158 W	38 kg	Voyager 1 and 2 (three each), LES-8 and 9
GPHS-RTG	≈ 290 W	56 kg	Ulysses, Galileo, Cassini, New Horizons
MMRTG	≈ 110 W	45 kg	Curiosity, Perseverance, Dragonfly
Next-Generation RTG	≈ 250 W (design)	—	In development for missions of the 2030s

An RTG's output declines for two reasons: the plutonium decays with a half-life of 87.7 years, and the thermocouples degrade as they age. For the Voyagers, the combined effect is a loss of about 4 W per year (Figure 26.9). From 470 W at launch, each Voyager had about 220 W in 2026. To keep the spacecraft operating, engineers have switched off the cameras (1990), the scan platforms, heaters and, one by one, the science instruments. Voyager 1's cosmic ray instrument was switched off in February 2025 and its low-energy charged particle instrument in April 2026. Voyager 2's low-energy charged particle instrument was switched off in March 2025. In 2026 Voyager 1 retained only its magnetometer and plasma wave instruments. The team plans to keep at least one instrument running on each spacecraft into the 2030s.



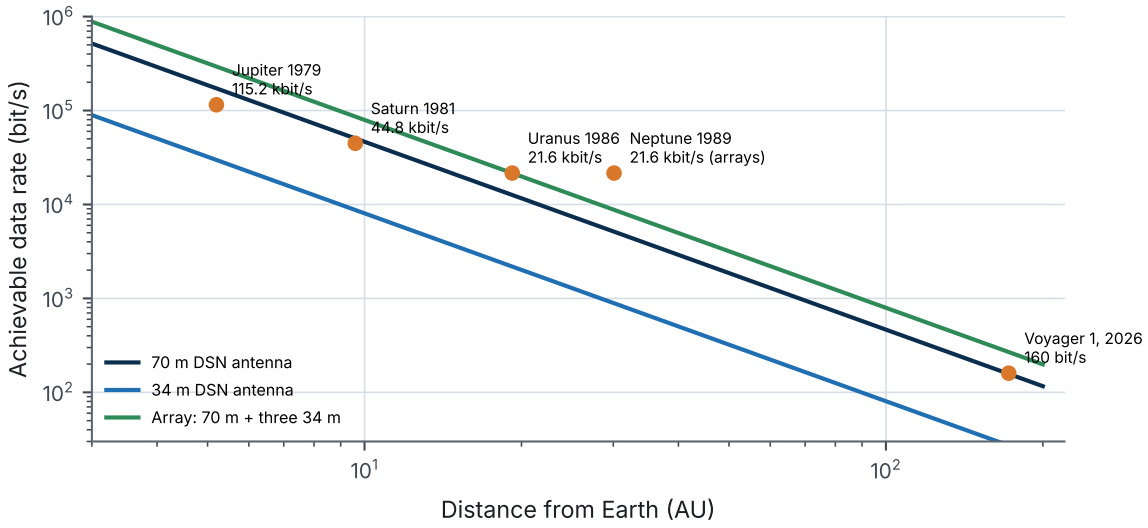
Three MHW-RTGs per spacecraft. Output also falls as the thermocouples age; the decline is now about 4 W per year.

FIGURE 26.9 Electrical power available on each Voyager. Ageing thermocouples make the decline faster than plutonium decay alone; instruments are switched off as power falls.

Keeping 49-year-old spacecraft alive has required remarkable engineering. In November 2023, Voyager 1 began sending meaningless data. Engineers traced the fault to one failed memory chip in its flight data system. In April 2024, they moved the affected code to other parts of the memory — a software patch sent across 24 billion km. In 2025 they revived Voyager 1's primary roll thrusters, which had been considered dead since 2004, as a backup to its degrading secondary thrusters, whose fuel lines are slowly clogging with silicon dioxide.

26.5.2 Talking across the Solar System

Radio signals weaken with the square of the distance as well, so a probe's data rate falls as it travels outward (Figure 26.10). At Jupiter in 1979, each Voyager returned up to 115 kilobits per second. At Neptune, ten years later, it achieved 21.6 kbit/s only because NASA had enlarged its 64 m antennas to 70 m and combined them electronically with other radio telescopes. These included the Very Large Array and Australia's Parkes dish. In 2026, Voyager 1 transmits at 160 bits per second; its signal, from a 21 W transmitter, delivers only about 3×10^{-19} W to a 70 m antenna on Earth.



Model: Voyager X-band link — 21.3 W, 3.66 m antenna, coded Eb/N0 ≈ 2.5 dB plus 3 dB margin.
At Jupiter and Saturn the DSN dishes were 64 m; for Uranus and Neptune they were enlarged and arrayed with other radio telescopes.

FIGURE 26.10 Data rate versus distance for a Voyager-class X-band link. The orange points are rates actually achieved.

Because Voyager 2 is heading south, far below the ecliptic, it can be seen only from the southern hemisphere. Only the 70 m antenna at Canberra (DSS-43) has an S-band transmitter able to command it. When that antenna was upgraded between March and October 2020, Voyager 2 could receive no commands. In August 2023 an incorrect command turned Voyager 2's antenna 2° away from Earth. The team regained contact by "shouting" — transmitting at high power from Canberra in the hope that some of the signal would reach the spacecraft, which it did.

EXAMPLE: VOYAGER 1'S LINK BUDGET IN 2026

Voyager 1 is 172 AU (2.57×10^{13} m) from Earth, transmitting 21.3 W at 8.415 GHz (wavelength 3.56 cm) through a 3.66 m antenna to a 70 m Deep Space Network antenna.

1. Transmit power: $10 \log_{10} 21.3 = 13.3$ dBW.
2. Antenna gains $G = \eta(\pi D/\lambda)^2$: transmit ($\eta = 0.55$) 47.6 dBi; receive ($\eta = 0.60$) 73.6 dBi.
3. Free-space loss $(4\pi d/\lambda)^2 = 319.2$ dB.
4. Received power: $13.3 + 47.6 + 73.6 - 319.2 = -184.7$ dBW, or 3.4×10^{-19} W.
5. Noise density for a 22 K receiving system: $kT = -228.6 + 13.4 = -215.2$ dBW/Hz. Carrier-to-noise density: 30.5 dB-Hz.
6. At 160 bit/s (22.0 dB-Hz), the energy per bit relative to noise is $E_b/N_0 = 8.5$ dB. Voyager's concatenated coding needs about 2.5 dB, leaving about 6 dB of margin for pointing losses, weather and the noise of the sky.
7. The one-way light time is 23.8 hours: a command sent on Monday morning is answered on Wednesday morning. On 18 November 2026, Voyager 1 becomes the first spacecraft one light-day from Earth.

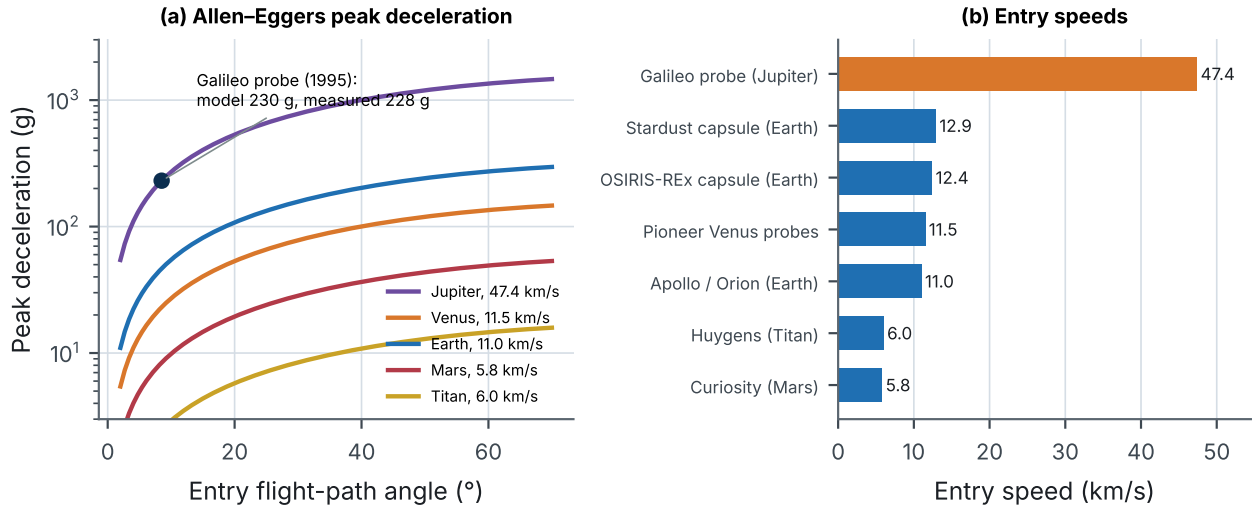
26.6 Atmospheric Entry Probes

An entry probe is released from a carrier spacecraft weeks before arrival, coasts to the planet and enters its atmosphere at high speed, protected by a heat shield. Once it has slowed, it deploys a parachute and descends, measuring temperature, pressure, composition, clouds and winds and

relaying its data to the carrier spacecraft overhead.

The Soviet **Venera 4** returned the first data from inside the atmosphere of Venus in 1967 before being crushed by the pressure. NASA's **Pioneer Venus Multiprobe** dropped one large and three small probes into Venus in December 1978; one of them survived impact and transmitted from the surface for 67 minutes. In 1985 the Soviet–French **Vega** missions placed two balloons in the Venus atmosphere at 54 km altitude, where they floated for two days, tracked by radio telescopes around the world.

The Galileo probe remains the only probe to have entered a giant planet. Released five months before arrival, it entered Jupiter's atmosphere on 7 December 1995 at 47.4 km/s — the fastest atmospheric entry ever — and decelerated at up to 228 *g*. Its carbon-phenolic heat shield, 152 kg of the probe's 339 kg, lost about 80 kg to ablation. The probe transmitted for 58 minutes and descended about 150 km below the cloud tops, to a pressure of 22 bar. It found less water than expected, because it had entered a dry "hot spot". It also found winds of over 600 km/h and heavy noble gases more enriched than models predicted, which changed theories of how Jupiter formed. ESA's **Huygens** probe, which descended through Titan's atmosphere and landed on its surface in 2005, is described in Chapter 27.



Peak deceleration a = v² sin γ / (2 e H) for a ballistic entry into an exponential atmosphere of scale height H; it does not depend on the vehicle's mass or shape.

FIGURE 26.11 Entry physics. (a) Peak deceleration depends only on entry speed, flight-path angle and the atmosphere's scale height; the Galileo probe's measured 228 g matches this simple model. (b) The Galileo probe entered at four times the speed of a lunar return capsule.

The heating rate at the stagnation point of an entry vehicle rises roughly as $\rho^{1/2} v^3$. The Galileo probe's entry was therefore far more severe than any Earth return. Much of its heating came as radiation from the extremely hot gas of the shock layer. For entries into Earth's atmosphere, see Chapters 31 and 29.

EXAMPLE: THE GALILEO PROBE'S DECELERATION

For a ballistic entry at speed v_e and flight-path angle γ into an exponential atmosphere of scale height H, the peak deceleration (Allen and Eggers, 1958) is

$$a_{\max} = \frac{v_e^2 \sin\gamma}{2\,e\,H}.$$

For the Galileo probe, v_e = 47.4 km/s, γ = 8.5° and $H \approx 27$ km in Jupiter's upper atmosphere:

$a_{\max} = (47\,400)^2 \times 0.148 / (2 \times 2.718 \times 27\,000) = 2\,260$ m/s$^2 \approx 230\,g$,

within 1 % of the measured 228 *g*. Steeper entries give higher decelerations but shorter, less intense heat pulses; shallower entries risk skipping out of the atmosphere. Probe designers choose γ to keep both deceleration and total heat load within limits.

New entry probes are planned but not yet secure. NASA's **DAVINCI** would descend through the atmosphere of Venus in the 2030s, and the US planetary science decadal survey (2022) chose a **Uranus Orbiter and Probe** as the highest-priority new flagship mission. In 2026, budget pressures left the timing of both uncertain.

26.7 Impactors and Planetary Defence

26.7.1 Deep Impact

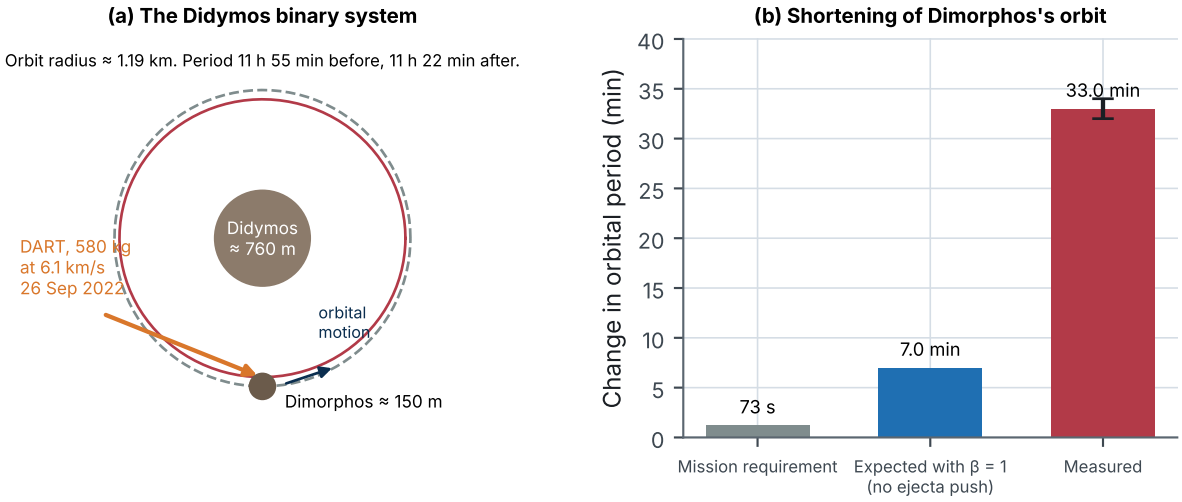
Some questions can be answered only by hitting a target. NASA's **Deep Impact** released a 372 kg copper-rich impactor into the path of comet Tempel 1, which struck the nucleus at 10.2 km/s on 4 July 2005 (Figure 26.12). In its final two hours the impactor steered itself using **AutoNav**, an autonomous navigation system that tracked the brightest part of the nucleus. The flyby spacecraft, 500 km away, observed the plume of ejecta, which revealed that the nucleus is covered in fine dust as loose as talcum powder. In 2011, NASA's Stardust-NExT spacecraft flew past Tempel 1 and imaged the 150 m crater.

FIGURE 26.12 The flash and ejecta of Deep Impact's collision with comet Tempel 1 on 4 July 2005, imaged by the flyby spacecraft 67 seconds after impact. Image: NASA/JPL (NASA/JPL-Caltech/UMD).

26.7.2 DART: Moving an Asteroid

Asteroid impacts are rare but potentially catastrophic. An object 140 m across, striking land, could devastate a whole metropolitan region. The 10 km asteroid that ended the age of the dinosaurs released billions of times the energy of the Hiroshima bomb. The simplest way to deflect an asteroid that is discovered years before impact is a **kinetic impactor** — a spacecraft that strikes it to change its velocity slightly, so that years later it misses Earth.

NASA's **Double Asteroid Redirection Test (DART)** tested this idea on **Dimorphos**, a 150 m moon of the asteroid Didymos (Figure 26.13, Figure 26.14). A binary asteroid was chosen because a change in the moon's 12-hour orbit around its primary could be measured from Earth within weeks. A change in an asteroid's orbit around the Sun, by contrast, would take years to detect. DART struck Dimorphos at 6.1 km/s on 26 September 2022, steering itself during the last four hours with an autonomous navigation system (SMART Nav) that identified Dimorphos only an hour before impact. The Italian cubesat **LICIACube**, released 15 days earlier, flew past three minutes after the impact and imaged the ejecta plume.



Ejecta thrown back from the impact site multiplied the push: momentum enhancement β ≈ 3.6 (2.2–4.9, depending on Dimorphos's density, which Hera will measure).

FIGURE 26.13 DART's impact on Dimorphos. (a) The Didymos system. (b) The orbit period shortened by 33 minutes, far more than DART's momentum alone could explain.

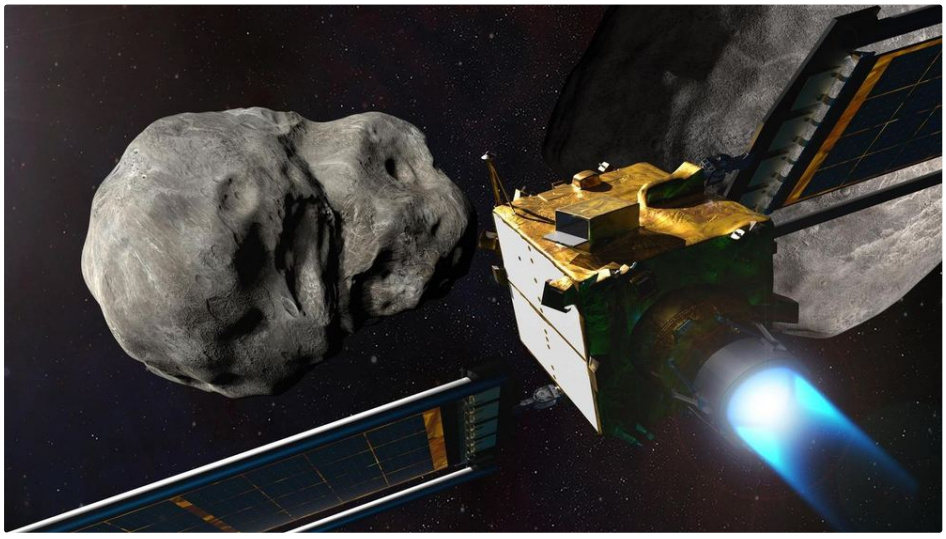


FIGURE 26.14 Artist's illustration of DART approaching the binary asteroid Didymos (left) and its moon Dimorphos. Image: NASA/JPL (NASA/Johns Hopkins APL/Steve Gribben).

Telescope observations showed that Dimorphos's orbital period had shortened by 33.0 ± 1.0 minutes, from 11 h 55 min — more than 25 times the mission's minimum success criterion of 73 seconds. More than a million kilograms of rock were thrown off, forming a tail thousands of kilometres long; the recoil from this ejecta pushed Dimorphos several times harder than the spacecraft did. The ratio of the total momentum change to the momentum DART delivered, the **momentum enhancement factor β**, is about 3.6 (between 2.2 and 4.9, depending on Dimorphos's unknown density). In 2026, stellar occultation and radar measurements showed that the impact had also changed the orbit of the whole Didymos system around the Sun, by about 12 micrometres per second.

EXAMPLE: HOW HARD DID DART PUSH?

Dimorphos orbits 1.19 km from the centre of Didymos with a period T = 11.92 h, so its orbital speed is $v = 2\pi \times 1\,190$ m / 42 900 s = 0.174 m/s.

1. For a nearly circular orbit, a small along-track velocity change Δv changes the period by $\Delta T/T \approx 3\,\Delta v/v$.
2. The measured $\Delta T = -33$ min gives $\Delta T/T = -0.046$, so $\Delta v \approx 0.046 \times 0.174 / 3 = 2.7$ mm/s.
3. Dimorphos has a volume of about 1.8×10^6 m^3. For a density of 2 400 kg/m^3, its mass is 4.3×10^9 kg. DART's momentum (580 kg × 6 145 m/s) could change its velocity by at most $3.6 \times 10^6 / 4.3 \times 10^9 = 0.83$ mm/s.
4. The observed 2.7 mm/s is more than three times larger: the ejecta, thrown backwards out of the crater, provided most of the push.

Because the result depends on Dimorphos's mass, ESA's Hera mission (arriving in November 2026) will measure it, along with the shape of the crater, so that future deflection missions can be designed with confidence.

26.7.3 Finding the Threats First

A deflection mission needs years of warning, so planetary defence begins with discovering near-Earth objects (NEOs). US law set the goal of finding 90 % of NEOs larger than 140 m; by 2026 well under half of them had been found. Ground surveys such as Catalina, Pan-STARRS, ATLAS and the Vera C. Rubin Observatory, which began its survey in 2025, will be joined by NASA's **NEO Surveyor**, an infrared telescope to be stationed at the Sun–Earth L1 point, with launch targeted for September 2027. The value of these surveys was shown by asteroid **2024 YR4**, about 60 m across, discovered in December 2024. In February 2025 its calculated chance of striking Earth in 2032 briefly reached 3 %, the highest ever recorded for an object of its size. Further observations then ruled out an Earth impact (a small chance of hitting the Moon remains).

On 13 April 2029, the 340 m asteroid **Apophis** will pass about 32 000 km above Earth's surface — closer than geostationary satellites — and be visible to the naked eye. It will be studied by NASA's OSIRIS-APEX (the extended OSIRIS-REx spacecraft, Chapter 29) and by ESA's **Ramses**. Ramses was approved in 2025 for launch in April 2028 on a Japanese H3 rocket, and will observe how Earth's gravity shakes and reshapes the asteroid. China plans its own impactor-and-observer test later in the decade.

26.8 Comets, Asteroids and Interstellar Visitors

Flybys of small bodies have revealed their bewildering variety. Galileo's flyby of **Ida** in 1993 discovered **Dactyl**, the first known asteroid moon. **Deep Space 1** (1998–2001) tested ion propulsion and autonomous navigation and flew past comet Borrelly. **Stardust** collected dust from comet Wild 2 (Chapter 29). Rosetta flew past the asteroids Šteins and Lutetia on its way to its comet.

NASA's **Lucy**, launched in October 2021, is touring the **Trojan asteroids** — primitive bodies sharing Jupiter's orbit at its L4 and L5 Lagrange points (Chapter 3). On its way it flew past the main-belt asteroid **Dinkinesh** in November 2023, discovering that it has a moon, Selam, which is itself a contact binary, and past **Donaldjohanson** in April 2025. Lucy will fly past the Trojans Eurybates and its moon Queta (August 2027), Polymele (September 2027), Leucus (April 2028) and Orus (November 2028), and finally the binary pair Patroclus–Menoetius in March 2033.

Three objects from outside the Solar System have been discovered passing through it. The first was **'Oumuamua** (2017), an elongated body that accelerated slightly as it left, and the second was the comet **2I/Borisov** (2019). The third, **3I/ATLAS**, was discovered on 1 July 2025. It passed perihelion at 1.36 AU at the end of October 2025, moving at about 58 km/s relative to the Sun. When 3I/ATLAS was hidden behind the Sun as seen from Earth, spacecraft across the Solar System observed it. Mars orbiters observed it as it passed Mars in early October 2025, and the solar observatories SOHO and STEREO did so near perihelion. Later came ESA's JUICE and NASA's SPHEREx, which found a coma rich in carbon dioxide and a strong increase in water after perihelion. ESA's **Comet Interceptor**, to be launched in 2028 or 2029, will wait at the Sun–Earth L2 point for a newly discovered comet — or interstellar object — and then intercept it with three spacecraft.

26.9 Into Interstellar Space

The Sun blows a supersonic wind of charged particles, the solar wind, that carves a bubble in the interstellar medium — the **heliosphere** (Chapter 22). About 90 AU from the Sun the wind passes through a **termination shock**, where it slows abruptly; beyond it, in the **heliosheath**, the slowed, heated wind is deflected towards a long tail. The **heliopause** is the boundary where the solar wind meets the interstellar plasma (Figure 26.15).

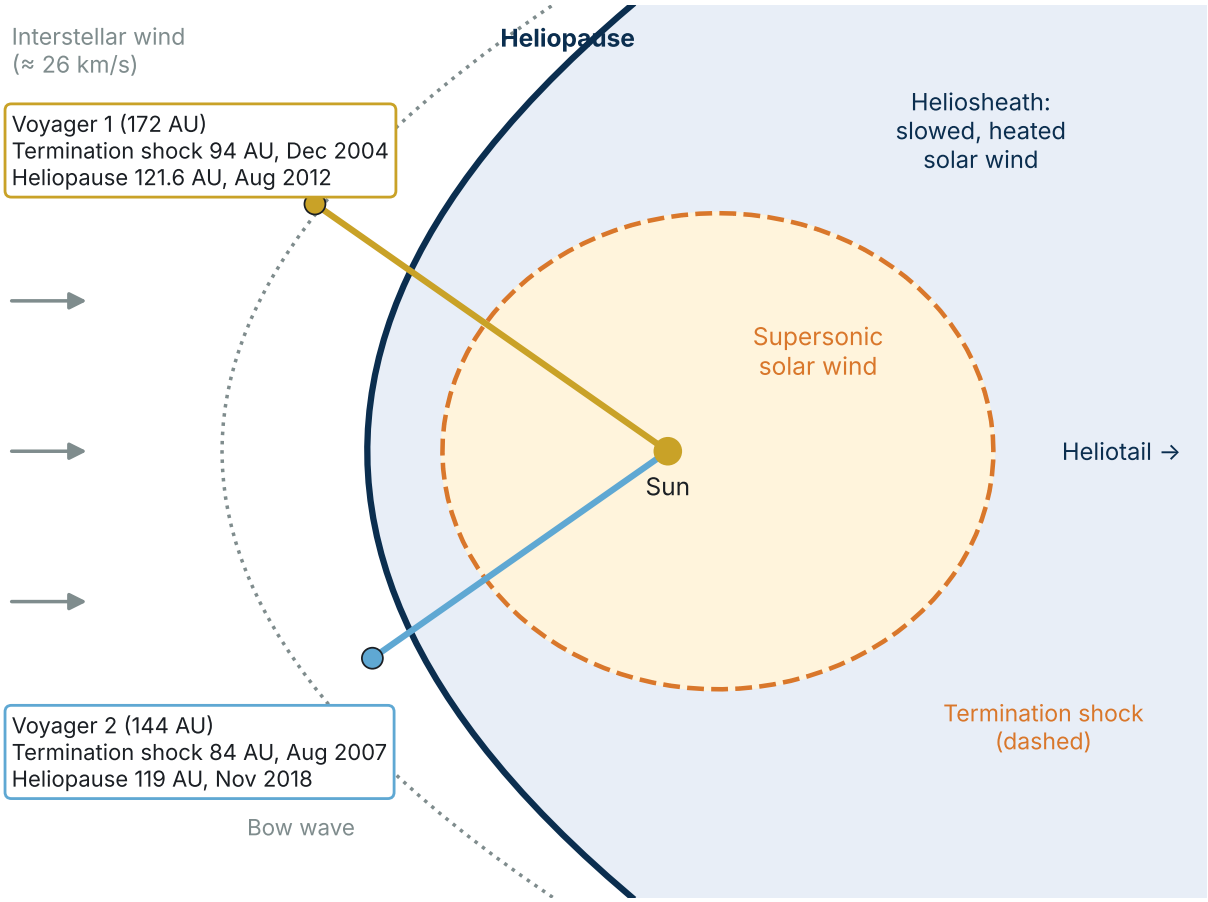


FIGURE 26.15 The heliosphere and the Voyagers' crossings of its boundaries (schematic, not to scale).

Voyager 1 crossed the termination shock in December 2004 at 94 AU and the heliopause on 25 August 2012 at 121.6 AU, becoming the first human-made object in interstellar space. Voyager 2 followed in November 2018 at 119 AU. The crossings were identified by a sudden fall in solar-wind particles and a jump in galactic cosmic rays. Because Voyager 1's plasma instrument had failed decades earlier, they were also identified by its plasma wave instrument. This detected the plasma's natural oscillation frequency and revealed a density about 40 times higher than in the heliosheath. Surprisingly, the direction of the magnetic field barely changed across the heliopause. The Voyagers now measure the interstellar medium directly, including the faint "hum" of plasma waves, and show that the Sun's influence extends well beyond the heliopause.

The **Pioneer anomaly** — a tiny unexplained sunward acceleration of Pioneers 10 and 11, about 8.7×10^{-10} m/s^2 — was once proposed as evidence of new physics. In 2012, a careful analysis of recovered tracking data and a thermal model of the spacecraft explained the anomaly. It was caused by the uneven emission of heat from the RTGs and electronics, which acts as a minute thrust.

No spacecraft is yet on its way to explore the interstellar medium deliberately. Concept studies of an **Interstellar Probe** assume a launch in the 2030s with a super-heavy launcher and a close gravity assist from Jupiter. They aim at speeds of about 7 AU per year — twice the Voyagers' — to reach several hundred AU within a working lifetime. Light-sail concepts that could reach other stars are discussed in Chapter 42.

26.10 Probes in 2026

TABLE 26.4 Probes and flyby missions in 2026.

Spacecraft	Status in October 2026
Voyager 1	172 AU; one light-day from Earth on 18 November 2026; magnetometer and plasma wave instruments operating
Voyager 2	About 144 AU, heading south; magnetometer, plasma wave and cosmic ray instruments operating early in 2026
New Horizons	About 66 AU; Kuiper Belt and heliosphere observations; extended mission until it leaves the Kuiper Belt (≈ 2028–29)
Lucy	Cruising to the L4 Trojans; Eurybates flyby August 2027
Psyche	Mars flyby May 2026; arrival at asteroid Psyche August 2029
Europa Clipper	Earth flyby December 2026; Jupiter arrival April 2030
JUICE	Earth flyby on 28 September 2026 added 3.5 km/s; one more Earth flyby (January 2029); Jupiter in July 2031
Hera	Final approach; arrival at Didymos in November 2026
BepiColombo	Mercury orbit insertion November 2026 (Chapter 25)
Parker Solar Probe	Closest solar passes at 6.1 million km and 192 km/s (Chapter 22)
In development	NEO Surveyor (2027), Ramses (2028), Comet Interceptor (2028–29), Dragonfly (2028, Chapter 28), Tianwen-4 to Jupiter and Uranus (≈ 2029–30)

AUTONOMY AND AI ON FLYBY MISSIONS

Flybys and impacts happen too fast and too far away for ground control, so they were among the first uses of onboard autonomy. **AutoNav**, developed for Deep Space 1 (1999–2001), navigated by images of asteroids against the stars and steered Deep Impact's impactor into Tempel 1. DART's **SMART Nav** found Dimorphos in its camera images an hour before impact and guided the spacecraft into the centre of the 150 m target. Lucy uses an autonomous **terminal tracking** system that keeps an asteroid centred in its instruments' fields of view during the fastest part of each flyby. Machine-learning tools are now used on the ground to plan encounter sequences, to detect new moons, rings and plumes in flyby images, and to search survey data for near-Earth asteroids and interstellar objects (Chapter 51).

KEY TAKEAWAYS

- Probes include flyby spacecraft, atmospheric entry probes, impactors and heliospheric explorers. Flybys are the cheapest way to reach a new world and remain our only close-up views of Uranus, Neptune, Pluto and the Kuiper Belt.
- Flyby engineering requires precise targeting in the B-plane, autonomous encounter sequences protected against faults, and imaging strategies that trade resolution against smear.
- Beyond Jupiter, probes rely on RTGs, whose output declines by a few watts each year; data rates fall with the square of distance, and Voyager 1 now returns 160 bit/s from 172 AU.
- Entry probes face extreme heating and deceleration — up to 228 g for the Galileo probe at Jupiter — that can be estimated from simple physics.
- DART showed that a kinetic impactor can change an asteroid's orbit: Dimorphos's period shortened by 33 minutes, with ejecta multiplying the push about 3.6 times. Hera arrives in November 2026 to measure the result.
- The Voyagers crossed the heliopause in 2012 and 2018 and are now sampling interstellar space; Voyager 1 reaches one light-day from Earth in November 2026.

REVIEW QUESTIONS

1. Why are flybys the cheapest way to reach a new world, and what are their scientific limitations?
2. What is B-plane targeting, and why must a flyby be navigated so precisely?
3. Voyager 1 returns 160 bit/s from about 172 AU. With the same antennas and transmitter, what rate would it have had at 86 AU?
4. Plutonium-238 has a half-life of 87.7 years. What fraction of its heat output remains after 49 years? Why has the electrical output of the Voyager RTGs fallen faster than this?
5. Why did the Galileo probe experience a peak deceleration of about 228 g? What does this imply for the design of entry probes?
6. What did DART demonstrate, and what is the momentum-enhancement factor β? Why did β exceed 1?
7. How do flyby sequences protect against faults when the spacecraft is too far away for real-time ground control?
8. What are the heliopause and interstellar space, and what have the Voyagers measured there?

CHAPTER

27 Landers and Entry, Descent & Landing

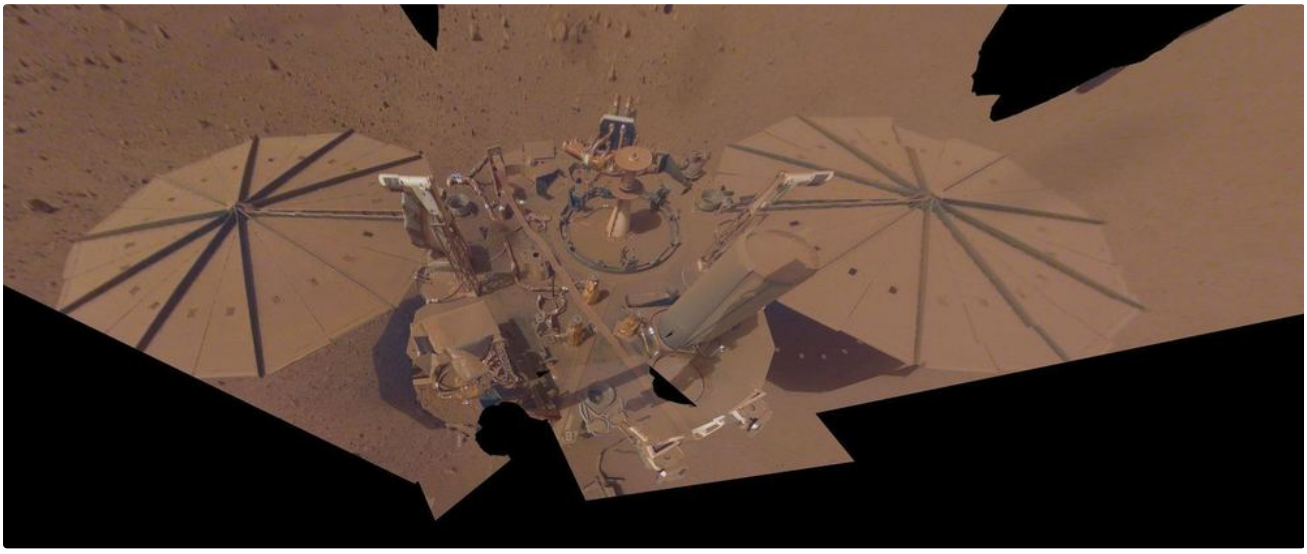


Image: NASA/JPL (NASA/JPL-Caltech)

Landing is the most dangerous minute of any planetary mission. A lander must shed kilometres per second of speed, find a safe spot it has never seen, and touch down gently enough to survive — all on its own, minutes or hours of light-time from Earth. This chapter explains how this is done on airless worlds like the Moon, on Mars with its treacherously thin atmosphere, in the crushing heat of Venus, under the haze of Titan and in the feeble gravity of comets and asteroids. It describes powered descent, heat shields, parachutes, airbags and sky cranes, terrain-relative navigation and hazard avoidance, the record of lunar landings from Luna 9 to the commercial landers of 2024–25, and what landers have discovered on the surface.

IN THIS CHAPTER

27.1 Why Land?
27.2 The Landing Problem
27.3 Landing on the Moon
27.4 Entry, Descent and Landing at Mars
27.5 Venus, Titan and Small Bodies
27.6 Surviving on the Surface

Entry, descent and landing (EDL) is the sequence of events that takes a spacecraft from its approach trajectory or orbit to a safe rest on a planetary surface. It is the phase in which most planetary missions have failed. Of the 34 robotic attempts to land softly on the Moon up to 2025, 12 crashed and 4 more landed damaged or on their sides. Only a little over half of all attempts to land on Mars have succeeded. This chapter covers stationary **landers**; rovers, helicopters and balloons are described in Chapter 28, and the return of samples to Earth in Chapter 29.

27.1 Why Land?

Orbiters map whole worlds, but some measurements can only be made by touching the surface (Table 27.1). A lander can analyse the chemistry and mineralogy of rocks and soil in contact, listen for quakes with a seismometer and measure the flow of heat from the interior. It can also record the weather day after day, search for organic molecules and signs of life, and collect samples for return to Earth. Landers also prove technologies needed for later crewed missions and for building permanent bases.

TABLE 27.1 Science that requires a lander.

Measurement	Instrument	Examples
Interior structure	Seismometer	Apollo passive seismic experiments (Moon, 1969–77); InSight SEIS (Mars, 2018–22)
Heat flow from the interior	Heat-flow probe	Apollo 15 and 17; InSight HP³; Blue Ghost LISTER (2025)
Surface chemistry and mineralogy	X-ray, gamma-ray and laser spectrometers, microscopes	Venera 13 (Venus); Phoenix and Perseverance (Mars); Chandrayaan-3 (Moon)
Volatiles and organics	Gas chromatograph–mass spectrometer, ovens	Viking (Mars); Phoenix; Philae (comet 67P)
Atmosphere and weather	Pressure, temperature, wind and dust sensors	Viking, InSight, Huygens (Titan)
Ground truth for orbital data	Cameras, spectrometers	Every lander

27.2 The Landing Problem

A spacecraft arriving at a planet carries an enormous amount of kinetic energy per kilogram. Coming down from a low lunar orbit at 1.7 km/s, each kilogram carries 1.4 MJ — as much as 30 g of TNT. Entering the Martian atmosphere at 5.4 km/s, it carries 15 MJ; returning to Earth from the Moon at 11 km/s, 60 MJ. All of this energy must be removed before touchdown, together with the potential energy of the altitude. There are only three ways to do it:

- **Aerodynamic drag** — heat shields and parachutes convert kinetic energy into heat in the atmosphere. This is free in terms of propellant but requires an atmosphere and protection from heating.
- **Propulsion** — rocket engines fired against the direction of motion. This works anywhere but is expensive: every metre per second costs propellant, and slowing down against gravity adds **gravity losses**.
- **Impact attenuation** — airbags, crushable structures or shock-absorbing legs absorb the last few metres per second at touchdown.

Every lander combines these in proportions set by its destination (Table 27.2).

TABLE 27.2 Landing on different worlds.

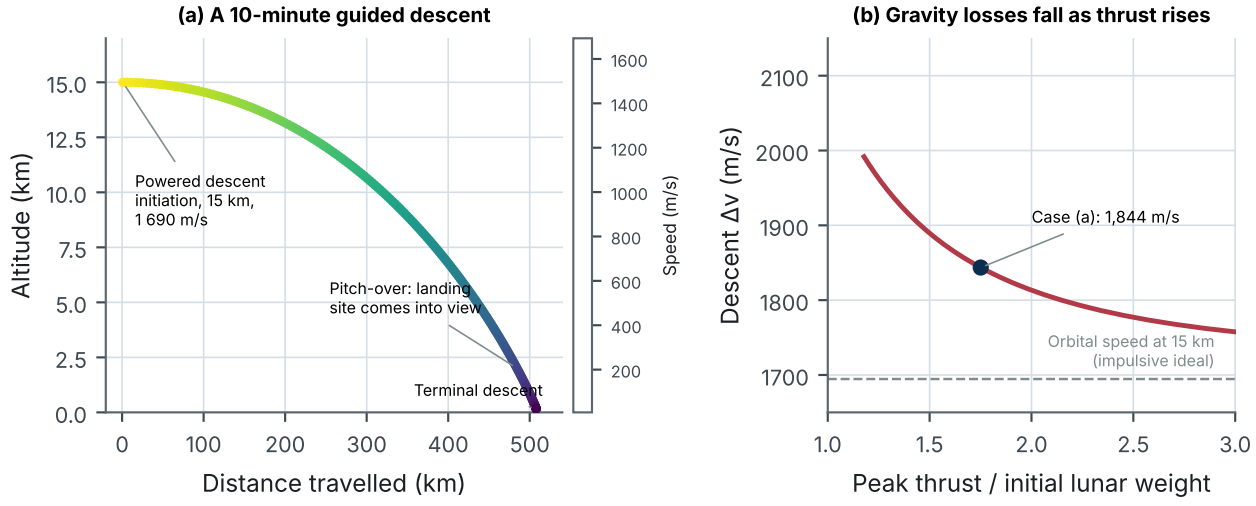
Destination	Surface gravity	Atmosphere	How landers slow down
Moon, Mercury	1.6 / 3.7 m/s²	None	Entirely propulsive, from orbital speed

Destination	Surface gravity	Atmosphere	How landers slow down
Mars	3.7 m/s²	Thin (≈ 0.6 % of Earth's surface pressure)	Heat shield removes ≈ 90 % of speed; parachute; propulsion or airbags for the last 80–100 m/s
Venus	8.9 m/s²	Very dense (92 bar at the surface)	Heat shield, parachute high up, then free fall: the dense air limits the landing speed to a few metres per second
Titan	1.35 m/s²	Dense and extended (1.5 bar)	Heat shield and parachutes all the way to the surface
Asteroids, comets	10^{-5}–10^{-3} m/s²	None	Small thrusters; the challenge is to avoid bouncing off

27.3 Landing on the Moon

27.3.1 Powered Descent

With no atmosphere, a lunar lander must cancel its entire orbital velocity with its engines. The Apollo Lunar Module (LM) set the pattern still used today. It first lowered its orbit to a periapsis of about 15 km, 480 km before the landing site. At **powered descent initiation**, its throttleable descent engine fired against the direction of motion. During the **braking phase**, lasting about eight minutes, the engine removed most of the 1 690 m/s orbital speed while the lander gradually lost height. At the **pitch-over**, about 2 km up, the LM rotated to a more upright attitude so the crew — or, today, the cameras — could see the landing site. In the **approach phase** the site was checked and, if necessary, changed; and in the **terminal descent** the lander came straight down at a metre per second or so (Figure 27.1).



Simulation: Apollo-class lander (15.1 t, Isp 311 s) from a 15 × 110 km orbit to 150 m altitude above the site, using an optimal-guidance law of the kind flown by Apollo.

FIGURE 27.1 Lunar powered descent. (a) A simulated Apollo-class descent from 15 km altitude. (b) Descent Δv versus the peak thrust-to-weight ratio: a weak engine must fight gravity for longer and wastes more propellant.

If the lander could stop instantly, it would need a Δv equal to its orbital speed. In reality it must also hold itself up against gravity during the descent, and the total Δv is 1.8–2.1 km/s, including margins for hovering and diverting to a safer site. The higher the thrust-to-weight ratio, the

shorter the burn and the smaller these **gravity losses** — but bigger engines are heavier and must be throttled deeply to hover.

EXAMPLE: PROPELLANT FOR A LUNAR DESCENT

An Apollo-class lander of 15 100 kg begins powered descent at 15 km. Its engine has a specific impulse of 311 s (v_e = 3 050 m/s).

1. The guided descent of Figure 27.1 needs Δv = 1 844 m/s to reach 150 m above the site. Adding about 200 m/s for the final descent, hovering and redesignation margin gives about 2 040 m/s.
2. Rocket equation: $m_p = m_0\,(1 - e^{-\Delta v / v_e})$ = 15 100 × (1 − e^{−2 040/3 050}) = 7 360 kg.
3. Nearly half the lander's mass is descent propellant. The real LM descent stage carried about 8 200 kg, the rest being reserve. Apollo 11 touched down with less than a minute of usable propellant left before the abort limit, after Neil Armstrong flew beyond a boulder field.

27.3.2 Navigation and Hazard Avoidance

A lander must know its altitude and velocity precisely. Apollo used a **landing radar** that measured altitude and, by the Doppler shift of four beams, velocity. Modern landers add **laser rangefinders** and **Doppler lidar** — NASA's Navigation Doppler Lidar measures velocity to a few centimetres per second. They also add cameras for **terrain-relative navigation (TRN)**. TRN matches images of craters and other landmarks with an onboard map to determine position to tens of metres.

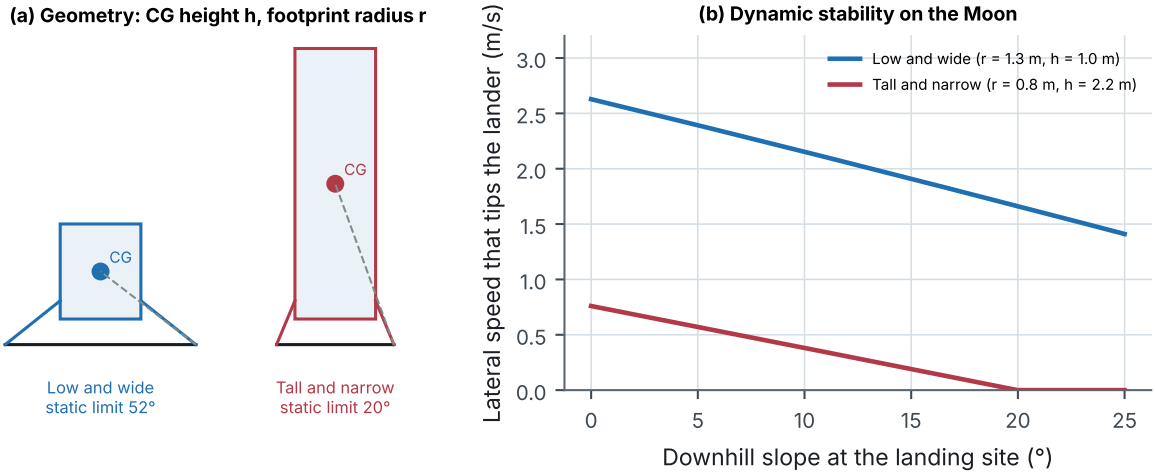
The final hazard is the landing site itself: a boulder or a slope of more than 10–15° can tip a lander over. **Hazard detection and avoidance** systems image the terrain below with lidar or cameras during the last few hundred metres, build a map of slopes and rocks, and steer to a safe spot. China's Chang'e 3 lander hovered 100 m above the Moon in 2013 to choose its landing point, and every Chang'e lander since has done the same. Japan's **SLIM** (Smart Lander for Investigating Moon) demonstrated vision-based "pinpoint" landing in January 2024. It touched down about 55 m from its target, within its 100 m goal. It did so even though one of its two main engines lost a nozzle about 50 m above the surface and the lander came to rest on its nose.

27.3.3 Touchdown

Landing legs must absorb the remaining vertical speed, typically 1–3 m/s, and keep the lander upright on slopes and rocks. Apollo's legs contained **crushable aluminium honeycomb** cartridges that absorbed energy by deforming permanently (Figure 27.2). A lander tips over when its centre of gravity (CG) passes outside the line joining its downhill feet. On level ground, the static limit is the angle whose tangent is the footprint radius divided by the CG height. But a sideways velocity at touchdown makes the lander rotate about its downhill feet. A tall, narrow lander can be tipped by less than 1 m/s of lateral speed on a modest slope (Figure 27.3). Both Intuitive Machines landers to date (2024 and 2025) — tall vehicles 4.3 m high — ended on their sides.



FIGURE 27.2 Apollo 11's Buzz Aldrin beside a leg of the Lunar Module Eagle in July 1969. The footpad and the crushable struts above it absorbed the landing shock. Image: NASA/JSC.



Lander pivots about its downhill feet; it tips if the kinetic energy of rotation exceeds the work needed to raise the CG over the pivot. Radius of gyration 0.8 m; lunar gravity.

FIGURE 27.3 Tip-over stability. (a) A low, wide lander tolerates much steeper slopes than a tall, narrow one. (b) The lateral speed at touchdown that would tip each lander over, as a function of slope.

The engine exhaust also interacts with the surface. Plumes blast away dust and rocks at high speed, erode shallow craters and can damage nearby hardware. When Apollo 12 landed about 160 m from the robotic Surveyor 3 lander in 1969, the astronauts found Surveyor's surfaces sandblasted by the LM's plume (Figure 27.4). Plume–surface interaction is a major concern for the large Artemis landers and for future lunar bases.

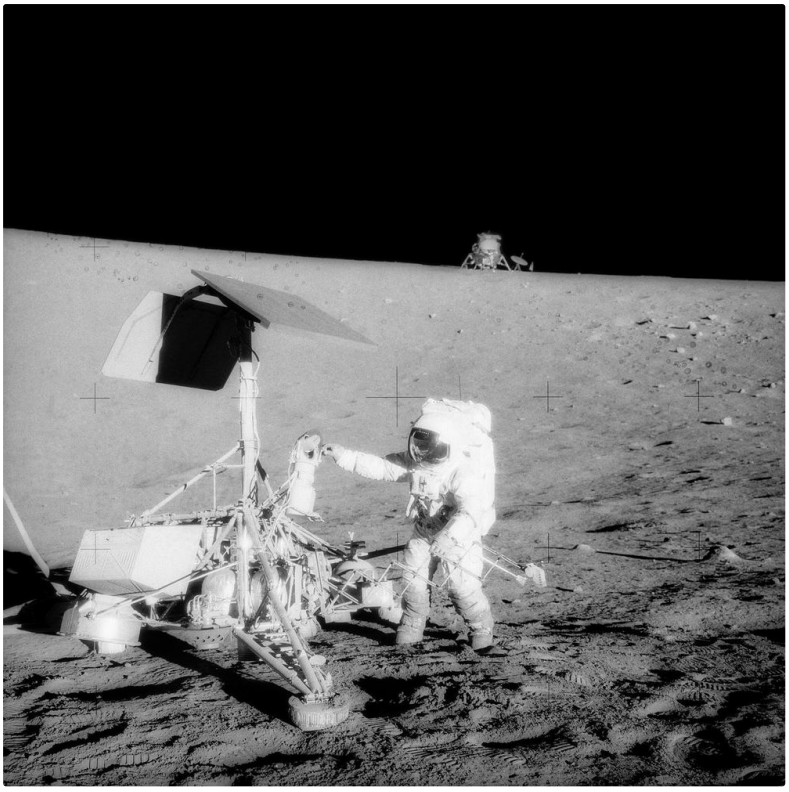
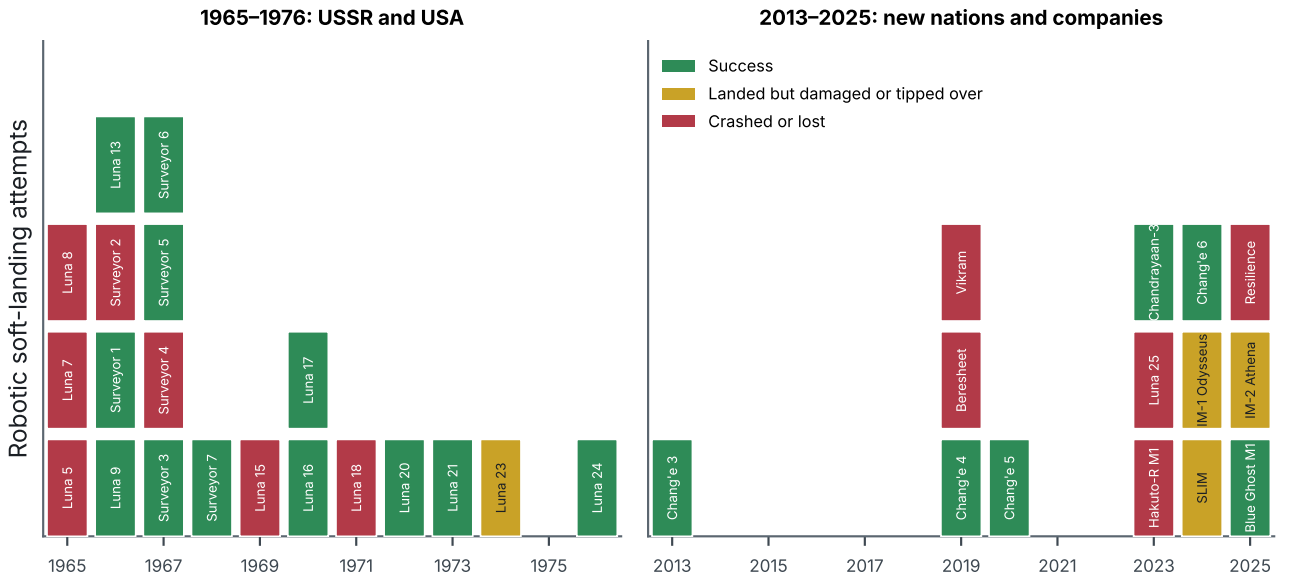


FIGURE 27.4 Apollo 12's Pete Conrad inspects Surveyor 3, which had landed 31 months earlier; the Lunar Module Intrepid stands on the horizon. Parts brought back to Earth showed sandblasting by the LM's plume. Image: NASA/JSC.

27.3.4 Sixty Years of Lunar Landings

The Soviet **Luna 9** made the first soft landing on another world in February 1966, using an inflated airbag-like ball ejected just before impact. NASA's **Surveyor 1** followed four months later with a fully propulsive landing. Between 1966 and 1976, five of seven Surveyors and seven Soviet Luna landers succeeded, including three sample-return missions and two Lunokhod rovers, and six Apollo crews landed. Then, for 37 years, no spacecraft landed on the Moon — until China's **Chang'e 3** in 2013 (Figure 27.5).



Excludes the six crewed Apollo landings (1969–72, all successful) and missions lost before reaching the Moon. No lunar landing was attempted in 2026 up to October.

FIGURE 27.5 Robotic lunar soft-landing attempts by year and outcome.

The 2010s and 2020s opened the Moon to new nations and to private companies. China has landed four times without failure, including the first landings on the far side (Chang'e 4, 2019) and the first far-side sample return (Chang'e 6, 2024). India's **Chandrayaan-3** became the first lander near the south pole region, at 69° S, in August 2023. Through NASA's **Commercial Lunar Payload Services (CLPS)**, companies now deliver NASA instruments under fixed-price contracts. Firefly's **Blue Ghost Mission 1** became the first fully successful commercial landing on 2 March 2025 in Mare Crisium. It operated for a full lunar day of 14 Earth days.

The failures are equally instructive (Table 27.3). Most were caused not by hardware breaking but by **navigation and software**: sensors that were not enabled, data rejected as faulty, or unexpected terrain that confused altitude estimates.

TABLE 27.3 Lessons from recent lunar landing failures and anomalies.

Mission	Date	What happened	Lesson
Beresheet (SpaceIL)	Apr 2019	An inertial measurement unit fault led to a computer reset that shut down the main engine during descent	Fault responses must not remove thrust during descent
Vikram (ISRO)	Sep 2019	Higher-than-expected thrust in the braking phase; the guidance could not recover	Robust guidance margins; Chandrayaan-3 widened them and succeeded
Hakuto-R M1 (ispace)	Apr 2023	Passing over a crater rim, the altitude estimate jumped; software rejected correct radar data, and the lander ran out of propellant 5 km up	Test navigation with realistic terrain
Luna 25 (Roscosmos)	Aug 2023	An orbit-lowering burn lasted 127 s instead of 84 s; the lander crashed	Independent burn-cutoff protection
SLIM (JAXA)	Jan 2024	Nozzle lost 50 m up; landed on its nose but operated	Precision landing works; design for off-nominal touchdown

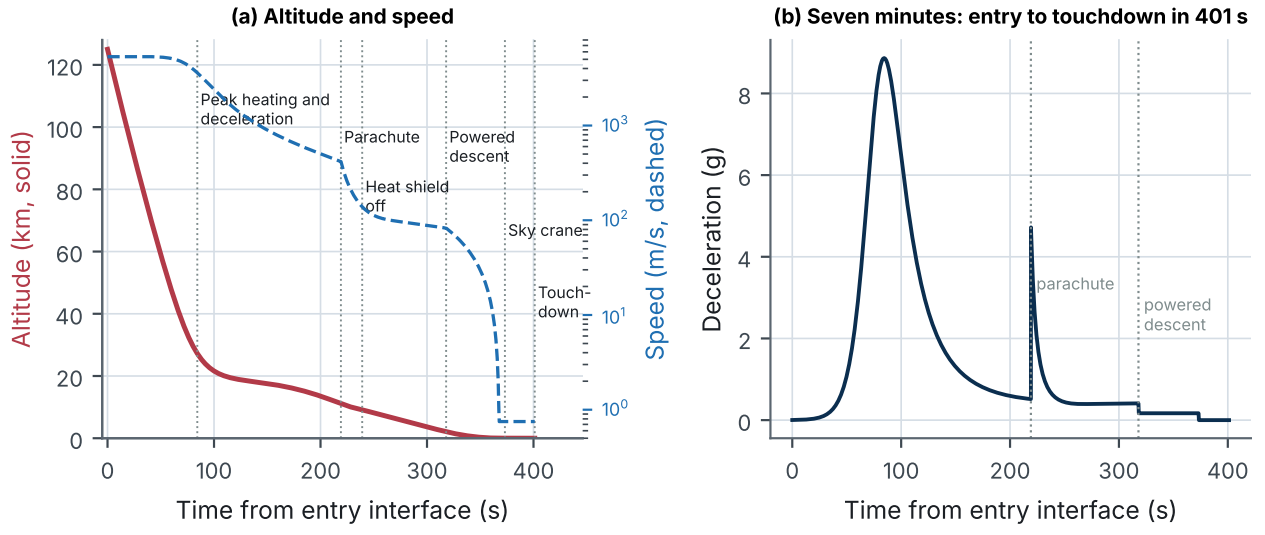
Mission	Date	What happened	Lesson
IM-1 Odysseus (Intuitive Machines)	Feb 2024	Laser rangefinders unusable because a safety switch was not enabled before launch; the team switched to a NASA lidar experiment; a leg caught at touchdown and the lander tipped	Pre-launch checks; end-to-end sensor testing
IM-2 Athena (Intuitive Machines)	Mar 2025	Altimeter data degraded over dark polar terrain; landed on its side in a crater	Sensors for low-Sun polar lighting
Resilience (ispace)	Jun 2025	Laser rangefinder measurements came late; the lander could not slow down in time	Sensor qualification for the descent environment

In 2026, no lunar landing had been attempted by October. Astrobotic's Griffin-1 was due to launch no earlier than November. Blue Origin's Blue Moon MK1, Intuitive Machines' IM-3, Firefly's far-side Blue Ghost Mission 2 and China's Chang'e 7 (which missed its August 2026 launch window) were all expected in 2027.

27.4 Entry, Descent and Landing at Mars

27.4.1 The Thin-Atmosphere Problem

Mars is the hardest place to land in the inner Solar System. Its atmosphere is thick enough to heat an entering spacecraft to well over 1 000 °C. But it is also very thin — less than 1 % of Earth's surface pressure. A heat shield and parachute alone therefore cannot slow a spacecraft to a safe landing speed. Under the same parachute, a spacecraft descends about five times faster on Mars than it would on Earth, and the parachute must open at supersonic speed. Each Mars lander therefore uses a chain of devices, each taking over when the previous one has done all it can: the famous **"seven minutes of terror"** (Figure 27.6).



Simulation of a Perseverance-class entry: 3.4 t, 4.5 m aeroshell, lift-to-drag 0.24, entry at 5.3 km/s and −15.5°, exponential atmosphere (scale height 11.1 km).

FIGURE 27.6 Entry, descent and landing at Mars, simulated for a Perseverance-class spacecraft. (a) Altitude and speed with the main events. (b) Deceleration: the heat shield removes about 90 % of the speed in the first two minutes.

27.4.2 The EDL Sequence

The sequence flown by NASA's Perseverance rover on 18 February 2021 illustrates the state of the art:

1. **Cruise-stage separation and entry.** Ten minutes before entry, the cruise stage that carried the spacecraft from Earth is jettisoned. The 4.5 m aeroshell enters the atmosphere at about 125 km altitude at 5.4 km/s.
2. **Guided entry.** The capsule flies at an angle of attack that gives it a little lift (lift-to-drag ratio about 0.24). By rolling to point this lift up, down or sideways with small thrusters, the onboard guidance corrects for variations in the atmosphere and steers towards the target, like the Apollo capsules returning to Earth. Peak heating and a deceleration of about 10 g occur about 80 seconds after entry; the heat shield's surface reaches about 1 300 °C.
3. **Parachute.** About four minutes after entry, at 9–11 km altitude and close to Mach 1.7, a 21.5 m disk-gap-band parachute is fired out by a mortar. Perseverance used a **range trigger**, opening the parachute when its position estimate showed it was the right distance from the target, rather than at a fixed speed.
4. **Heat-shield separation.** Twenty seconds later the heat shield is released, exposing the radar and cameras.
5. **Terrain-relative navigation.** From about 4 km altitude, Perseverance's **Lander Vision System** photographed the ground, matched the images with an onboard map made from orbital images, and computed its position to within about 40 m. It then chose the safest reachable spot from a pre-loaded hazard map.
6. **Back-shell separation and powered descent.** At about 2.1 km altitude and 90 m/s, the descent stage with the rover beneath it drops away from the back shell and parachute, and eight throttleable engines take over (Figure 27.7).
7. **Sky crane.** About 20 m above the surface, the descent stage lowers the rover on three nylon bridles 6.4 m long. When the rover's wheels touch down at 0.75 m/s, the bridles are cut and the descent stage flies away to crash at a safe distance (Figure 27.9).

The whole sequence is autonomous: with 11 minutes of light-time, the landing was already over when the first signal of entry reached Earth.



FIGURE 27.7 Perseverance's powered descent vehicle (descent stage, top, with the rover beneath) being stacked at Kennedy Space Center in 2020. Image: NASA/KSC (NASA JPL/Christian Mangano).

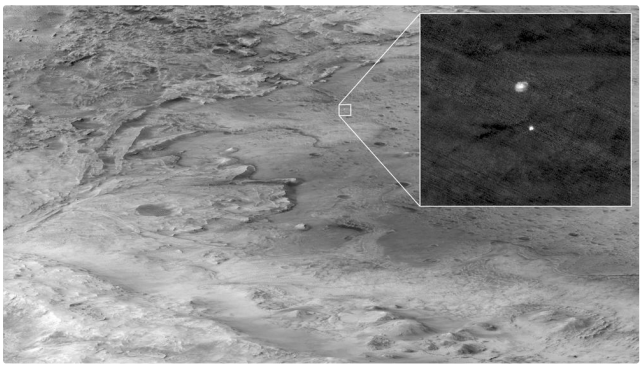


FIGURE 27.8 Perseverance under its parachute, photographed from orbit by MRO's HiRISE camera during the descent on 18 February 2021. Image: NASA/JPL (NASA/JPL-Caltech/University of Arizona).

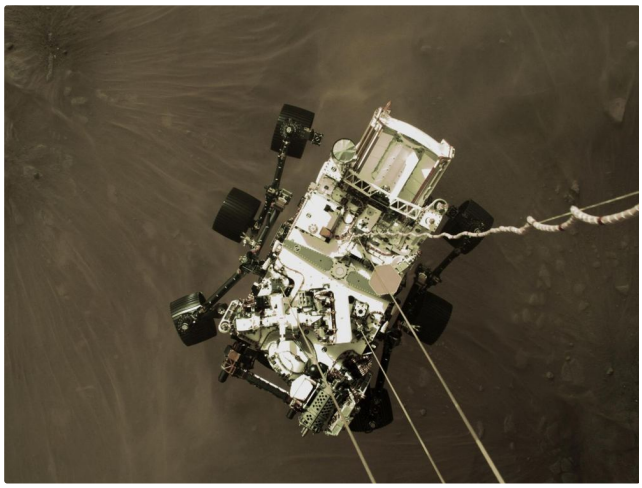


FIGURE 27.9 The sky crane manoeuvre, seen from the descent stage: Perseverance hangs on its bridles a few metres above the surface. Image: NASA/JPL (NASA/JPL-Caltech).

EXAMPLE: WHY A MARS PARACHUTE IS NOT ENOUGH

A 3 000 kg spacecraft hangs under a 21.5 m parachute with drag coefficient 0.6 (drag area $C_D A$ = 218 m²). At terminal velocity, drag equals weight: $v_t = \sqrt{2mg/(\rho C_D A)}$.

- **Mars**, 2 km above the surface: ρ ≈ 0.017 kg/m³, g = 3.71 m/s²: $v_t = \sqrt{2\times3\,000\times3.71/(0.017\times218)}$ ≈ **78 m/s** (280 km/h).
- **Earth**, at the same altitude: ρ ≈ 1.0 kg/m³, g = 9.81 m/s²: $v_t \approx$ **16 m/s**.

Even a parachute of 50 m diameter would leave the spacecraft falling at about 30 m/s on Mars. The final 80–100 m/s must be removed by rockets, or the impact must be absorbed by airbags. Mars Pathfinder and the Mars Exploration Rovers used airbags, bouncing up to 15 m high before rolling to a stop.

27.4.3 Landing Architectures

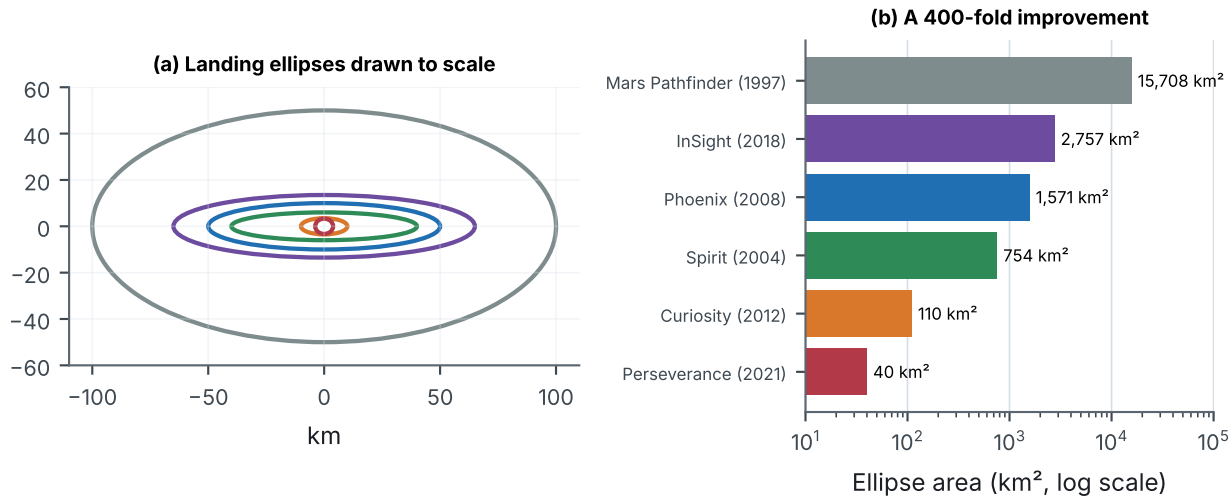
TABLE 27.4 Successful and attempted Mars landing systems (selected).

Mission	Year	Final landing system	Notes
Viking 1 and 2 (USA)	1976	Throttled rockets and three legs	Biology experiments; operated until 1980 and 1982
Mars Pathfinder (USA)	1997	Airbags	Carried the first Mars rover, Sojourner
Spirit and Opportunity (USA)	2004	Airbags	Opportunity operated for 14 years
Phoenix (USA)	2008	Pulsed rockets and legs	Arctic plains, 68° N; dug into ground ice
Curiosity (USA)	2012	Sky crane	First guided lifting entry at Mars; 899 kg rover
Schiaparelli (ESA)	2016	Crushable structure	Failed: navigation error, see below
InSight (USA)	2018	Pulsed rockets and legs	Seismometer and heat-flow probe
Perseverance (USA)	2021	Sky crane	First terrain-relative navigation at Mars; 1 025 kg rover
Tianwen-1 / Zhurong (China)	2021	Rockets and legs, with hover and hazard avoidance	First successful landing on Mars at the first attempt by a new nation

Airbags are simple and tolerant of rocks and slopes but limited to landed masses of a few hundred kilograms. Legged landers, as on Phoenix and InSight, can carry heavier instruments but need flat, rock-free sites. The **sky crane**, introduced for the 899 kg Curiosity, keeps the rocket engines high above the surface, avoiding dust and craters, and puts the rover directly on its wheels, ready to drive. Landing heavier payloads — such as the tonnes of cargo needed for human missions — will require new techniques, such as inflatable heat shields and **supersonic retropropulsion**. In retropropulsion, rocket engines are fired into the oncoming supersonic flow. SpaceX's Falcon 9 first stages use this technique when returning through Earth's upper atmosphere, and Starship is designed to land on Mars the same way.

27.4.4 Shrinking the Landing Ellipse

Because the atmosphere and the spacecraft's approach are never known exactly, a lander can only be targeted to land somewhere within a **landing ellipse**, and every point inside it must be safe. Better approach navigation, guided entry, the range trigger and terrain-relative navigation have shrunk Mars landing ellipses about 400 times in area in 25 years (Figure 27.10). This has opened scientifically rich but rugged sites such as Perseverance's Jezero crater, an ancient river delta.



Guided lifting entry (Curiosity) and terrain-relative navigation (Perseverance) shrank the ellipses; Japan's SLIM landed within 100 m of its lunar target in 2024.

FIGURE 27.10 Mars landing ellipses. Smaller ellipses allow landings in complex, scientifically interesting terrain.

27.4.5 Failures at Mars

The Soviet **Mars 3** made the first soft landing on Mars in December 1971 but fell silent after about 20 seconds, possibly because of a dust storm. NASA's **Mars Polar Lander** was lost in 1999, almost certainly because the jolt of its legs deploying was misread by software as touchdown, shutting off the engines 40 m above the ground. Britain's **Beagle 2** landed in 2003 but was not heard from; in 2015 HiRISE images showed it on the surface with its solar panels only partly unfolded, blocking its antenna. ESA's **Schiaparelli** test lander crashed in October 2016 when its inertial measurement unit briefly saturated during parachute descent. The navigation software then calculated a negative altitude and released the parachute. It fired the braking thrusters for only three seconds while the lander was still 3.7 km up.

27.4.6 What Mars Landers Found

The two **Viking** landers of 1976 returned the first images from the surface (Figure 27.11) and ran three biology experiments. Their results — chemically active soil but no organic molecules — are still debated. The later discovery of perchlorates in the soil by **Phoenix** suggests that heating during the analysis destroyed any organics. Phoenix scraped through the soil at 68° N and found water ice a few centimetres down (Figure 27.12).



FIGURE 27.11 The first photograph taken by Viking Lander 2 at Utopia Planitia in September 1976, showing rocks and part of the lander's footpad (lower right). Image: NASA/JPL (NASA/JPL).



FIGURE 27.12 Phoenix's "Dodo-Goldilocks" trench in June 2008. Lumps of bright material in the lower left (top image) had vanished four days later (bottom), proving that they were water ice that sublimated. Image: NASA/JPL (NASA/JPL-Caltech/University of Arizona/Texas A&M University).

NASA's **InSight** (2018–2022) placed a seismometer on the Martian surface with its robotic arm (Figure 27.13) and recorded 1 319 marsquakes. These included one of magnitude 4.7 in May 2022 and the shaking from meteorite impacts whose fresh craters were then found by orbiters. Seismic waves passing through the planet revealed its internal structure (Figure 27.14). It has a crust 24–72 km thick and a large liquid iron-rich core surrounded by a layer of molten rock. According to a study published in 2025, it may also have a solid inner core. InSight's heat-flow probe, a self-hammering "mole", could not penetrate the unexpectedly cohesive soil. Dust slowly covered the solar panels (see the chapter-opening image) and the mission ended in December 2022.

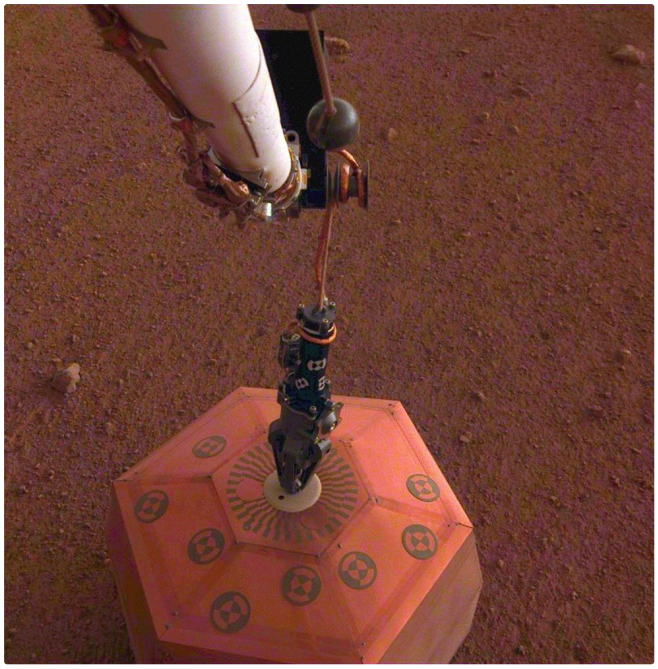


FIGURE 27.13 InSight's robotic arm placing the SEIS seismometer on the surface of Mars in December 2018. Image: NASA/JPL (NASA/JPL-Caltech).

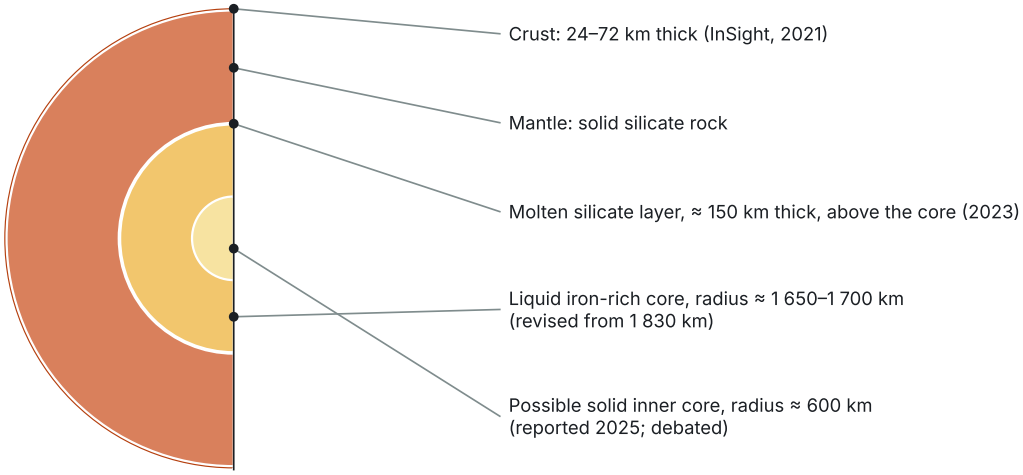


FIGURE 27.14 The interior of Mars revealed by InSight's seismometer.

27.5 Venus, Titan and Small Bodies

27.5.1 Venus: Landing in an Oven

The surface of Venus is at 465 °C and 92 bar — the pressure 900 m deep in Earth's oceans — under clouds of sulphuric acid. The Soviet Union nevertheless landed ten spacecraft there between 1970 and 1985. **Venera 7** (1970) transmitted for 23 minutes from the surface, the first data ever received from the surface of another planet. **Venera 9** (1975) returned the first pictures. **Venera 13** (1982) returned colour panoramas, drilled a sample and analysed it with an X-ray fluorescence spectrometer, surviving for 127 minutes.

Venus landers are built like deep-sea submersibles: a spherical titanium pressure vessel, thick insulation, and a heat sink of lithium nitrate or similar phase-change material, with the electronics pre-chilled before entry. They are doomed from the moment they land, and the race is to return data before the interior overheats. Because the atmosphere is so dense, the parachute is released at about 50 km altitude. The lander falls the rest of the way with only a drag plate, reaching the surface at about 8 m/s. New electronics based on silicon carbide, which NASA operated for 60 days in a Venus simulation chamber in 2021, may one day allow long-lived Venus landers.

27.5.2 Titan: Huygens

ESA's **Huygens** probe, carried to Saturn by Cassini, entered Titan's atmosphere on 14 January 2005. Titan's dense, extended atmosphere and low gravity made the landing gentle. Huygens descended under parachutes for 2 hours 27 minutes, imaging river channels and shorelines below. It landed at about 4.5 m/s on damp ground strewn with rounded pebbles of water ice (Figure 27.15). It recorded a surface temperature of −179 °C and a pressure of 1.5 bar. It transmitted from the surface for 72 minutes until Cassini set below the horizon. Its landing is still the most distant ever made. NASA's Dragonfly rotorcraft will return to Titan in 2034 (Chapter 28).

FIGURE 27.15 The surface of Titan photographed by Huygens after landing in 2005. The "pebbles" in the foreground, 10–15 cm across, are made of water ice. Image: NASA/JPL (NASA/JPL/ESA/University of Arizona).

27.5.3 Comets and Asteroids: Avoiding the Bounce

Landing on a small body is the opposite problem: gravity is so weak that a lander risks bouncing back into space. The escape speed from the 900 m asteroid Ryugu is about 0.37 m/s, a slow walking pace. NASA's **NEAR Shoemaker** orbiter, not designed as a lander, touched down on Eros at the end of its mission in 2001 and kept operating for two weeks. ESA's **Philae** landed on comet 67P/Churyumov–Gerasimenko on 12 November 2014, but its harpoons failed to fire and its hold-down thruster did not work. It bounced about a kilometre high and came to rest two hours later in the shadow of a cliff. There it operated on its batteries for about 64 hours before falling silent.

JAXA's Hayabusa2 deployed the first hopping rovers, **MINERVA-II-1A and 1B**, and the German–French **MASCOT** lander on Ryugu in 2018; sample-collecting "touch-and-go" landings are described in Chapter 29.

EXAMPLE: HOW SLOWLY MUST YOU LAND ON A COMET?

Comet 67P has a mass of about 1.0×10^{13} kg and an effective radius of about 2 km.

1. Escape speed: $v_{esc} = \sqrt{2GM/r} = \sqrt{2 \times 6.67 \times 10^{-11} \times 10^{13} / 2\,000}$ = 0.82 m/s.
2. Philae touched down at about 1 m/s. Without anchoring, any rebound faster than 0.82 m/s would have escaped the comet entirely. Philae's damping legs absorbed most of the energy, and it rebounded at about 0.4 m/s. This was slow enough to fall back, but only after a two-hour flight.
3. Surface gravity is about GM/r^2 = 1.7 × 10^{-4} m/s², so the 100 kg Philae weighed only about 0.02 N — the weight of a 2 g object on Earth. This is why it carried harpoons, ice screws and a thruster to press it down: without them, almost any disturbance could launch it back into space.

27.6 Surviving on the Surface

Landing is only the beginning. A lander must then survive the local environment:

- **Night and cold.** A lunar night lasts 14 Earth days, and the surface cools to about −170 °C. Solar-powered landers usually die at sunset unless they carry radioisotope heater units, as China's Chang'e 3 and 4 landers do; SLIM, not designed to survive, revived after three lunar nights before failing in 2024. Martian nights reach −90 °C, and winter is the end of most solar-powered polar landers, as it was for Phoenix.
- **Dust.** Lunar dust is sharp and electrostatically charged, sticking to radiators, seals and optics; Martian dust settles on solar panels, steadily reducing power, as it ended InSight's mission.
- **Communications.** A lander on the Moon's near side can talk directly to Earth, but one on the far side, which never faces Earth, needs a relay satellite such as China's Queqiao and Queqiao-2. Mars landers relay most of their data through orbiters (Chapter 25).

AI AND AUTONOMY FOR LANDING

Landing is the most autonomous phase of any mission, and modern landers rely on computer vision. Terrain-relative navigation systems detect craters or match image patches with onboard maps. Hazard-detection systems turn lidar or stereo images into maps of slope and roughness and select a safe spot within seconds. SLIM's vision-based navigation demonstrated 100 m landing accuracy. On the ground, deep-learning models trained on orbital images detect boulders and craters to map hazards at candidate landing sites — for the Artemis south-pole sites, they have catalogued millions of boulders from LRO images. Future landers will combine these into fully autonomous "land anywhere" systems (Chapters 48 and 51).

KEY TAKEAWAYS

- Landing removes huge kinetic energy — 1.4 MJ/kg from lunar orbit, 15 MJ/kg at Mars entry — using drag, propulsion and impact attenuation in proportions set by the destination.
- Airless bodies require fully propulsive descent; gravity losses add 10–20 % to the Δv, and the final phase depends on accurate altimetry, terrain-relative navigation and hazard avoidance.
- Tall, narrow landers are vulnerable to tipping over; most recent lunar failures and anomalies were caused by navigation sensors and software rather than hardware breakdowns.
- Mars's thin atmosphere requires a chain of heat shield, supersonic parachute and propulsive or airbag landing; guided entry and terrain-relative navigation have shrunk landing ellipses 400-fold.
- Venus landers survive for hours at 465 °C and 92 bar; Huygens landed on Titan under parachutes; small-body landers must avoid bouncing off in micro-gravity.
- Landers have revealed Mars's interior, water ice at its high latitudes and the surfaces of Venus and Titan; in 2026 a new wave of commercial and national lunar landers was preparing for 2027.

REVIEW QUESTIONS

1. Calculate the specific kinetic energy that must be removed when landing from lunar orbit (1.68 km/s) and at Mars entry (5.5 km/s).
2. A lunar lander needs a Δv of 2.0 km/s with engines of 320 s specific impulse. What fraction of its initial mass must be propellant?
3. Why do airless bodies require fully propulsive descent, and why do gravity losses add 10–20 % to the Δv?
4. Why have recent commercial lunar landers tipped over? What design and navigation lessons follow?
5. Describe the Mars EDL sequence from entry to touchdown for a sky-crane landing.
6. How have guided entry and terrain-relative navigation reduced the size of Mars landing ellipses?
7. How did the Venera landers survive on the surface of Venus, and for how long?
8. Why is landing on a comet or small asteroid difficult? What happened to Philae?

CHAPTER

28 Planetary Rovers and Aerial Explorers

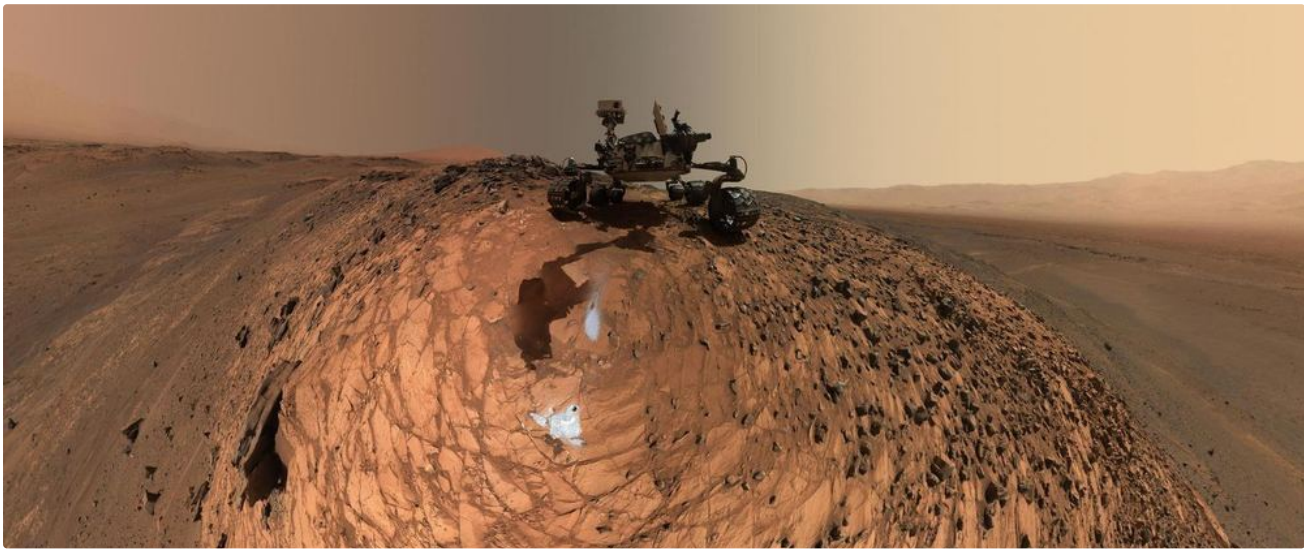


Image: NASA/JPL (NASA/JPL-Caltech/MSSS)

A lander sees one spot; a rover can follow the clues. Since Lunokhod 1 rolled onto the Moon in 1970, rovers have crossed tens of kilometres of the Moon and Mars, climbed mountains, drilled rocks, cached samples for return to Earth and found the strongest evidence yet that Mars was once habitable. In 2021 a 1.8 kg helicopter made the first powered flight on another planet, and a nuclear-powered rotorcraft is being built to fly through the dense air of Titan. This chapter explains how rovers move over soft ground, stay warm and powered, navigate by themselves, and carry laboratories across other worlds; it reviews every rover mission to date and those planned for the next decade.

IN THIS CHAPTER

28.1 Sixty Years of Rovers
28.2 Engineering a Rover
28.3 What Rovers Have Found
28.4 Aerial Explorers
28.5 Rovers at the Lunar Poles
28.6 The Next Generation

A **rover** is a mobile laboratory. Mobility lets one mission visit many sites, follow geological layers like the pages of a book, and reach targets first spotted from orbit. The price is complexity: a rover must cross unknown terrain, mostly on its own, with its power, temperature and communications all depending on where it stops. This chapter describes robotic rovers on the Moon and Mars, the vehicles that carried Apollo astronauts, and the newest class of explorer — aircraft that fly in other worlds' atmospheres.

28.1 Sixty Years of Rovers

28.1.1 The First Rovers: Lunokhod and the Apollo Lunar Roving Vehicle

The first rover on another world was the Soviet **Lunokhod 1**, landed by Luna 17 on 17 November 1970. The 756 kg, eight-wheeled vehicle was driven from Earth by a five-person crew who watched slow-scan television pictures arriving with a delay of several seconds; a polonium-210 heater kept it alive through the lunar nights. It operated for 11 lunar days and drove 10.5 km. **Lunokhod 2** (1973) drove 39 km in four months — the distance record for 41 years. Their landing sites have been photographed by NASA's Lunar Reconnaissance Orbiter (Figure 28.1).

FIGURE 28.1 The Lunokhod 2 rover (arrow), as seen by NASA's Lunar Reconnaissance Orbiter, at the end of its tracks in the crater Le Monnier, where it stopped in 1973. Image: NASA/JPL (NASA/GSFC/Arizona State University).

The last three Apollo missions (1971–72) carried the **Lunar Roving Vehicle (LRV)**, a 210 kg, battery-powered, four-wheel-drive car that folded into the side of the Lunar Module. With wire-mesh wheels and a top speed of about 13 km/h, the LRVs carried astronauts and their samples a combined 90 km and let them explore up to 7.6 km from the lander (Figure 28.2).



FIGURE 28.2 Apollo 15's Jim Irwin with the first Lunar Roving Vehicle at Hadley–Apennine in July 1971. Mount Hadley rises in the background. Image: NASA/JSC.

28.1.2 Mars Rovers

NASA's rovers have grown with each generation (Figure 28.3):

- **Sojourner** (1997), a 10.6 kg, solar-powered microrover carried by Mars Pathfinder, explored the rocks around the lander for 83 sols (Martian days) and travelled about 100 m.
- **Spirit and Opportunity** (2004), 185 kg each, were designed for 90 sols and 600 m of driving. Opportunity found layered rocks and spherules of hematite ("blueberries") formed in acidic groundwater. It operated for 14 years until a planet-wide dust storm in 2018 cut its power. Its 45.16 km is still the record for distance driven on another world. Spirit found almost pure silica deposited by ancient hot springs before it became stuck in soft soil in 2009.
- **Curiosity** (2012), 899 kg and powered by a radioisotope generator, landed in Gale crater and showed that it once held a long-lived lake whose water could have supported microbes. It has found organic molecules preserved in mudstones and seasonal changes in atmospheric methane, and has climbed more than 1.3 km up the layered Mount Sharp, driving more than 37 km by 2026.
- **Perseverance** (2021), 1 025 kg, explores Jezero crater, which contained a lake fed by a river delta 3.5 billion years ago. It is the first rover to **collect and seal samples for return to Earth**. By early 2026 it had filled 33 of its 43 sample tubes and left ten of them in a depot on the crater floor in 2022–23. It had also climbed out over the crater rim and driven more than 42 km. Its MOXIE experiment made 122 g of oxygen from the Martian carbon dioxide atmosphere.



FIGURE 28.3 Three generations of NASA Mars rovers at JPL: a flight spare of Sojourner (front), a Mars Exploration Rover test vehicle (left) and a Curiosity test rover (right). Image: NASA/JPL (NASA/JPL-Caltech).

China's **Zhurong** rover (240 kg) was landed by Tianwen-1 in May 2021 in Utopia Planitia. It drove 1.9 km and found evidence of water-altered minerals and buried layers with its ground-penetrating radar. In May 2022 it entered hibernation under a coating of dust and did not wake.

28.1.3 Rovers on the Moon Since 2013

China's **Yutu** (2013) drove 114 m before its mobility failed. **Yutu-2**, landed by Chang'e 4 on the far side in January 2019, became the longest-lived lunar rover. Its ground-penetrating radar mapped layers of debris tens of metres deep. It was still active, though nearly immobile, in 2025 after driving about 1.6 km. India's 26 kg **Pragyan**, carried by Chandrayaan-3 (2023), confirmed sulphur in the south polar regolith with its laser-induced breakdown spectrometer before going to sleep at the end of the lunar day. Japan's SLIM carried two tiny rovers in 2024: LEV-1, a hopping rover, and LEV-2 (SORA-Q), a 250 g sphere that unfolded into a two-wheeled rover and photographed SLIM lying on its nose. The UAE's first rover, **Rashid**, was lost when the Hakuto-R lander crashed in 2023.

TABLE 28.1 Planetary rovers to 2026.

Rover	Operator	Body	Years	Mass	Power	Distance
Lunokhod 1, 2	USSR	Moon	1970–71, 1973	756 kg, 836 kg	Solar; Po-210 heater	10.5 km, 39 km
Lunar Roving Vehicle (×3)	NASA	Moon	1971–72	210 kg	Batteries	90 km in total
Sojourner	NASA	Mars	1997	10.6 kg	Solar	≈ 0.1 km
Spirit, Opportunity	NASA	Mars	2004–10, 2004–18	185 kg	Solar	7.7 km, 45.2 km
Curiosity	NASA	Mars	2012–	899 kg	MMRTG	> 37 km
Yutu, Yutu-2	CNSA	Moon	2013–16, 2019–	140 kg	Solar, radioisotope heaters	0.1 km, 1.6 km

Rover	Operator	Body	Years	Mass	Power	Distance
Perseverance	NASA	Mars	2021–	1 025 kg	MMRTG	> 42 km
Zhurong	CNSA	Mars	2021–22	240 kg	Solar	1.9 km
Pragyan	ISRO	Moon	2023	26 kg	Solar	≈ 0.1 km

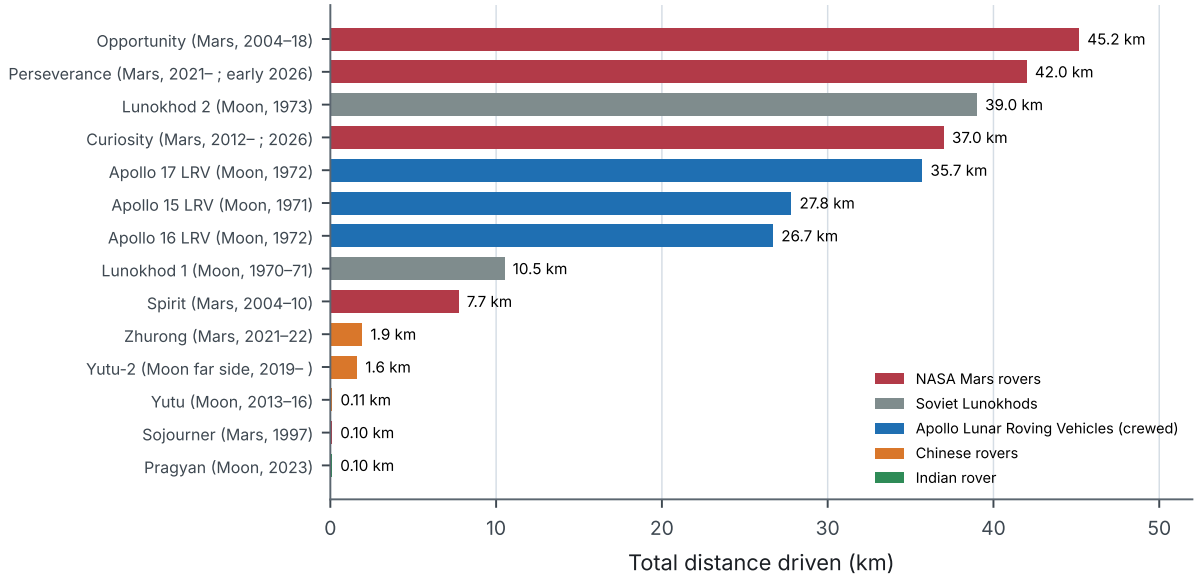


FIGURE 28.4 The distances driven by rovers on the Moon and Mars.

28.2 Engineering a Rover

28.2.1 Mobility

All NASA Mars rovers since Sojourner have used the **rocker-bogie** suspension (Figure 28.5). Each side has a large “rocker” link carrying the rear wheel (or front, depending on direction) and a smaller “bogie” carrying the other two wheels. The two rockers are connected through a differential, so the body pitches only half as much as either side. The system has no springs: all six wheels stay in contact with the ground, sharing the load and providing traction, and the rover can climb obstacles almost as high as the wheel diameter. Each wheel has its own drive motor, and the four corner wheels can be steered so the rover can turn in place.

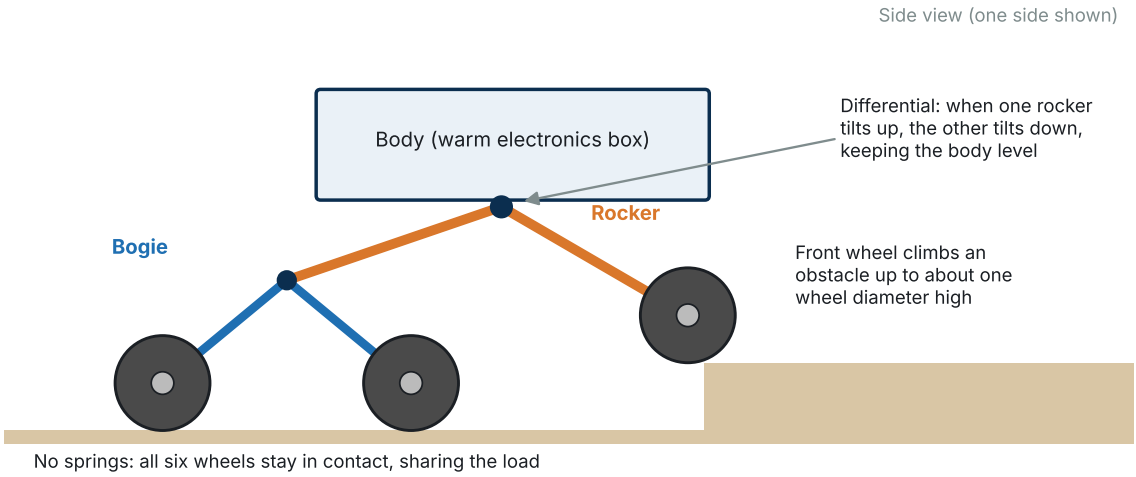
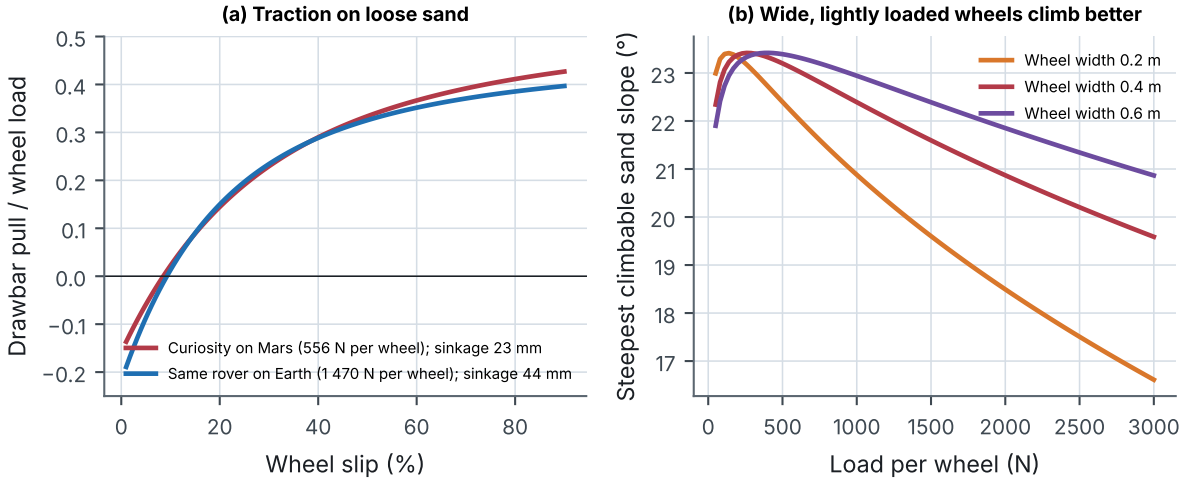


FIGURE 28.5 The rocker-bogie suspension used on NASA's Mars rovers.

The greatest mobility hazard is soft ground. A wheel sinks until the soil beneath it can support its load. It must then bulldoze and compact soil in front of it — the **compaction resistance** — while its grousers (tread cleats) shear the soil to generate thrust. The science of this, **terramechanics**, developed by M. G. Bekker and J. Y. Wong for military vehicles, predicts the net force available to pull the rover forward, the **drawbar pull**, as a function of wheel slip (Figure 28.6). On loose sand, even a well-designed rover can climb only about 20–25°; Spirit and Opportunity both became trapped in sand, Opportunity escaping after five weeks of careful driving in 2005, Spirit never.



Bekker–Wong terramechanics for a rigid wheel of 0.26 m radius on cohesionless sand (n = 1, kc = 1.4 kN/m², kφ = 820 kN/m³, c = 0.17 kPa, φ = 35°). Representative values.

FIGURE 28.6 Wheel–soil interaction. (a) Drawbar pull rises with slip; the same rover on Earth sinks deeper, which is why engineers test Mars rovers with lightened copies. (b) Wider wheels and lighter loads let a rover climb steeper sandy slopes.

EXAMPLE: TESTING A MARS ROVER ON EARTH

Curiosity has a mass of 899 kg. On Mars each of its six wheels carries 899 × 3.71 / 6 = 556 N; on Earth, 899 × 9.81 / 6 = 1 470 N.

1. With the soil model of Figure 28.6, the wheels sink about 23 mm on Mars but 44 mm on Earth, and the compaction resistance rises from 16 % to 22 % of the wheel load.
2. A rover tested at full mass on Earth would therefore perform worse than on Mars — misleading the engineers.
3. JPL instead tests mobility with a lightened rover, nicknamed “Scarecrow”, that has the same wheels and suspension. It weighs on Earth what Curiosity weighs on Mars (about 340 kg mass), with no computer or cameras on board.

Wheels must also survive rocks. Curiosity’s 0.5 m aluminium wheels, with skins only 0.75 mm thick, were punctured and torn by sharp, wind-carved rocks from 2013 (Figure 28.7). The team responded by planning routes over softer ground and, in 2017, uploading a **traction-control** algorithm that adjusts each wheel’s speed to reduce the forces on it. Perseverance’s wheels were made thicker and narrower, with more grousers.



FIGURE 28.7 Damage to one of Curiosity’s aluminium wheels after about 4.8 km of driving, in December 2013. Image: NASA/JPL (NASA/JPL-Caltech/MSSS).

28.2.2 Power and Temperature

A rover must keep its electronics and batteries above about −40 °C through nights that, on Mars, reach −90 °C. Solar-powered rovers house their electronics in an insulated **warm electronics box** heated by electric heaters and radioisotope heater units (RHUs); Spirit and Opportunity carried eight RHUs each. Dust gradually covers solar arrays, reducing power by a fraction of a per cent per sol. Opportunity’s arrays were repeatedly cleaned by gusts of wind and dust devils, which is why it lasted 14 years. Zhurong’s power, by contrast, was not restored. Curiosity and Perseverance use **multi-mission radioisotope thermoelectric generators** (MMRTGs). These provide about 110 W of electricity at launch — around 2.5 kWh per sol, day and night. Their waste heat warms the rover through a pumped fluid loop. Batteries supply the peaks when the rover drives, drills or runs its laboratories.

28.2.3 Communications and Operations

Rovers can talk directly to Earth through high-gain X-band antennas, but most of their data — hundreds of megabits per sol — is relayed by orbiters passing overhead (Chapter 25). Lunar far-side rovers depend on relay satellites such as Queqiao. Because of the light-time (4–24 minutes each way at Mars) and the limited contact windows, rovers cannot be joysticked. Instead, the operations team plans each sol's activities — drives, images, arm placements, drilling — and uplinks them as a command sequence in the Martian morning. The rover executes them and sends back the results by the afternoon relay pass. The team then plans the next sol. A rover team works on "Mars time" during the first months of a mission, with its working day shifting 40 minutes later each Earth day.

28.2.4 Autonomous Navigation

To drive further than the operators can see, rovers navigate themselves (Figure 28.8). Stereo cameras build a three-dimensional map of the terrain ahead, and software assesses each patch for slope, rocks and steps relative to the rover's dimensions. The software chooses the path that makes most progress towards the goal set by the team. Meanwhile, **visual odometry** — tracking features between successive images — measures how far the rover has really moved, correcting for wheel slip. Opportunity's patterned tracks record this process (Figure 28.9). Spirit and Opportunity had to stop to think after each short step. Perseverance's faster computer and dedicated vision processor let it plan "while driving". It can cover more than 100 m per hour across favourable terrain — several times faster than Curiosity.

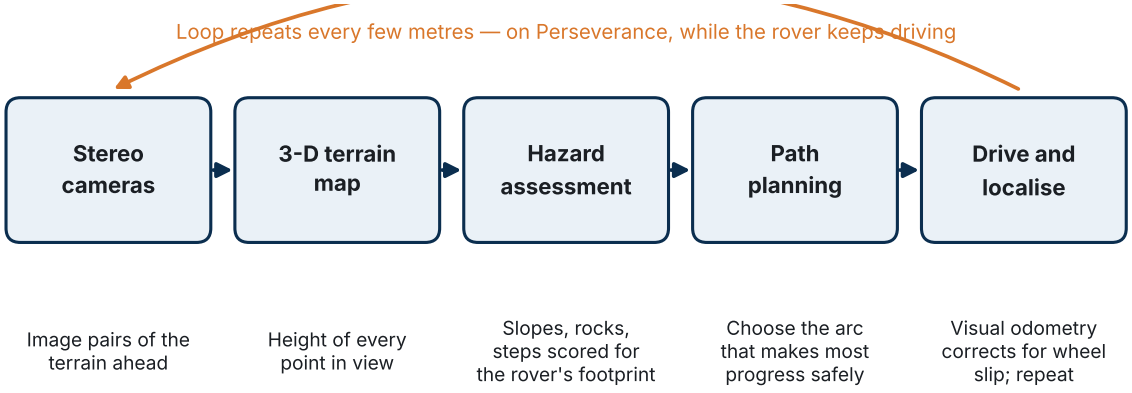


FIGURE 28.8 The autonomous navigation loop of a Mars rover.

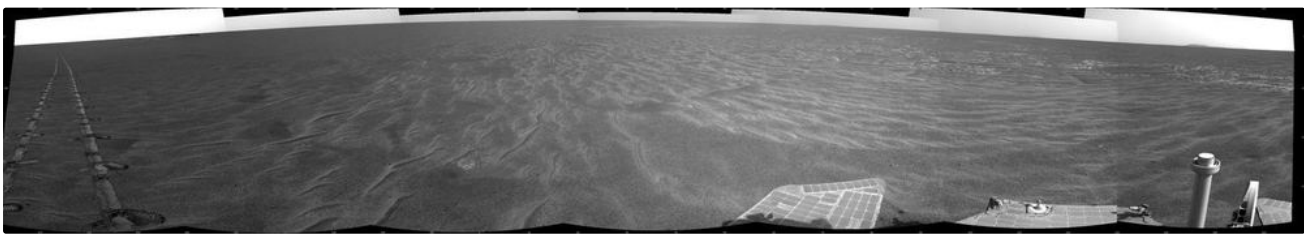


FIGURE 28.9 Opportunity's tracks on Meridiani Planum in 2011. The pattern of short arcs records the rover stopping to check for hazards autonomously. Image: NASA/JPL (NASA/JPL-Caltech).

28.2.5 A Laboratory on Wheels

Rover science payloads combine remote-sensing instruments on the mast, contact instruments on a robotic arm, and laboratories inside the body (Table 28.2).

TABLE 28.2 Science instruments of Curiosity and Perseverance.

Location	Instrument type	Curiosity	Perseverance
Mast	Colour stereo cameras with zoom	Mastcam	Mastcam-Z
Mast	Laser-induced breakdown spectroscopy (zaps rocks from up to 7 m)	ChemCam	SuperCam (also Raman and infrared spectroscopy, microphone)
Arm	Hand lens imager	MAHLI	WATSON
Arm	X-ray spectrometer for chemistry	APXS	PIXL (maps elements at 0.1 mm scale)
Arm	Ultraviolet Raman and fluorescence for organics	—	SHERLOC
Arm	Rock drill	Powder drill	Coring drill producing intact cores
Body	Mineralogy by X-ray diffraction	CheMin	—
Body	Organic and isotope analysis by gas chromatography–mass spectrometry	SAM	—
Body	Sample sealing and caching	—	43 tubes
Body or deck	Weather, radiation, ground-penetrating radar, oxygen production	REMS, RAD, DAN	MEDA, RIMFAX, MOXIE

28.3 What Rovers Have Found

The Mars rovers have turned the question “was Mars ever habitable?” into a resounding “yes”. Opportunity found that Meridiani Planum was once soaked in acidic, salty groundwater, and Spirit found the deposits of hot springs. Curiosity found that Gale crater held fresh, neutral water for long periods, with all the chemical ingredients needed by microbes. Perseverance has studied the ancient delta and the volcanic rocks of Jezero’s floor. In July 2024, it drilled a rock nicknamed **Cheyava Falls** with millimetre-sized “leopard spots” — pale patches with dark rims rich in iron phosphate and iron sulphide minerals (Figure 28.10). On Earth, such reactions are often driven by microbes. In September 2025 the team reported in *Nature* that the sample, named Sapphire Canyon, contains a **potential biosignature** — a feature that might have a biological origin but could also have formed without life. Only analysis in laboratories on Earth can settle the question.

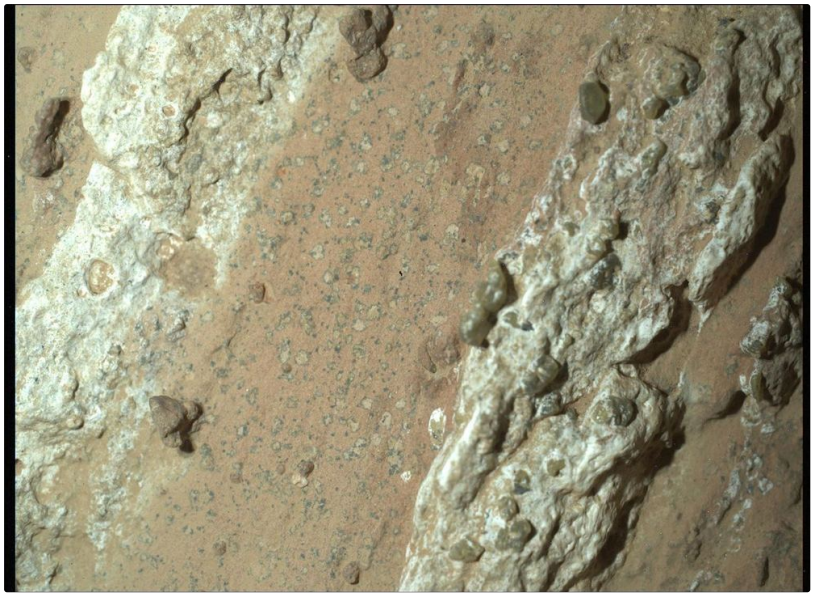


FIGURE 28.10 The “leopard spots” in the Cheyava Falls rock, imaged by Perseverance’s WATSON camera in July 2024. The sample drilled here may preserve a potential biosignature. Image: NASA/JPL (NASA/JPL-Caltech/MSSS).

That question is now tied to the fate of **Mars Sample Return**. The NASA–ESA plan to collect Perseverance’s tubes with a lander, a small rocket and an Earth-return orbiter grew to an estimated cost of about US$11 billion with return no earlier than 2040. In 2025–26 the US Congress provided no money for the existing programme, leaving the samples on Mars with no retrieval plan. Meanwhile, China’s **Tianwen-3** prepares to launch in 2028 and return samples from a different site by 2031 (Chapter 29).

28.4 Aerial Explorers

28.4.1 Ingenuity: First Flight on Another World

On 19 April 2021, NASA’s **Ingenuity** helicopter, carried to Mars beneath Perseverance, rose 3 m above Jezero crater, hovered and landed in a flight lasting 39 seconds. This was the first powered, controlled flight of an aircraft on another planet (Figure 28.11). Flying on Mars is hard: the atmosphere has 1.4 % of the density of Earth’s at sea level. Ingenuity had a mass of 1.8 kg and two counter-rotating rotors 1.2 m across. They spun at about 2 500 rpm — five times faster than a terrestrial helicopter’s rotor. This speed was limited by the need to keep the blade tips below the speed of sound, which is lower on Mars than on Earth (Figure 28.12).

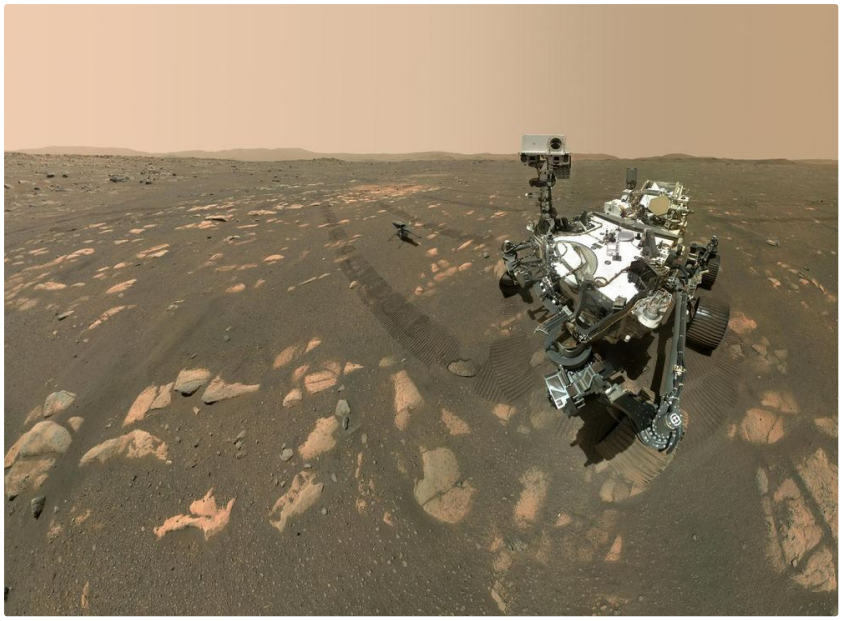
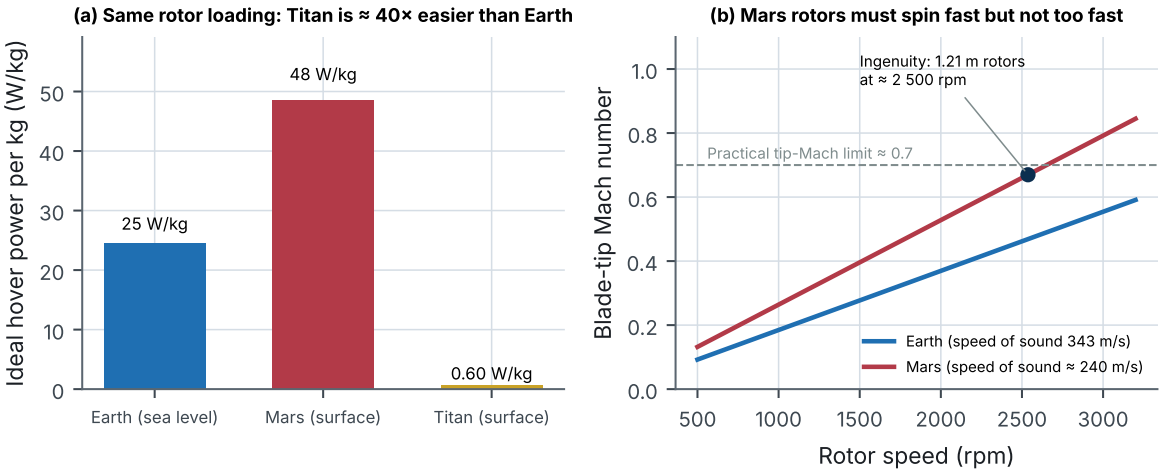


FIGURE 28.11 Perseverance's selfie with Ingenuity in April 2021, shortly before the helicopter's first flight. Image: NASA/JPL (NASA/JPL-Caltech/MSSS).



Momentum theory: ideal hover power per unit mass = g^1.5 √(disk loading / 2ρ), here for Ingenuity's disk loading (1.8 kg on a 1.21 m disk). Real power is roughly twice the ideal.

FIGURE 28.12 The physics of flying on other worlds. (a) Hover power per kilogram for the same rotor loading. (b) Rotor tip Mach number: Mars's lower speed of sound limits rotor speed.

Planned as a five-flight technology demonstration, Ingenuity flew 72 times over nearly three years, covering about 17 km in a total of 129 minutes. It reached altitudes of 24 m and speeds of 10 m/s, and scouted routes for Perseverance. It navigated by tracking features on the ground with a downward-looking camera. On its 72nd flight, on 18 January 2024, over featureless sand ripples, the navigation system lost track of the ground and the helicopter landed hard, breaking its rotor blades (Figure 28.13). It remains on Mars as a weather station.

FIGURE 28.13 Ingenuity's own navigation camera reveals the shadow of a rotor blade with its tip missing after its final flight in January 2024. Image: NASA/JPL (NASA/JPL-Caltech).

EXAMPLE: HOVERING ON MARS

Ingenuity weighs $T = 1.8 \times 3.71$ = 6.7 N on Mars. Its rotor disk area is $A = \pi \times 0.605^2$ = 1.15 m², and the air density at Jezero is about 0.017 kg/m³.

1. Momentum theory gives the ideal hover power $P = T^{3/2}/\sqrt{2\rho A}$ = 6.7^1.5 / √(2 × 0.017 × 1.15) = 17.3 / 0.198 ≈ **88 W**.
2. The same helicopter on Earth would weigh 17.7 N and need only $17.7^{1.5}/\sqrt{2 \times 1.225 \times 1.15}$ = 74.5 / 1.68 ≈ 44 W: half the power, because the dense air makes the rotor far more effective.
3. Real rotors have a figure of merit (ideal/actual power) of about 0.5–0.6, and the coaxial rotors interfere with each other. Ingenuity drew about 350 W in flight from six lithium-ion cells, so each flight was limited to about 2–3 minutes.

28.4.2 Balloons

The only balloons flown on another planet were the two **Vega** balloons of 1985, released into the atmosphere of Venus at about 54 km altitude. They drifted there for two days, tracked by a worldwide network of radio telescopes that measured the winds. Venus's cloud layer, where temperature and pressure are similar to Earth's surface, is attractive for long-lived balloons; NASA and others have tested variable-altitude "aerobot" prototypes that could rise and sink through the clouds.

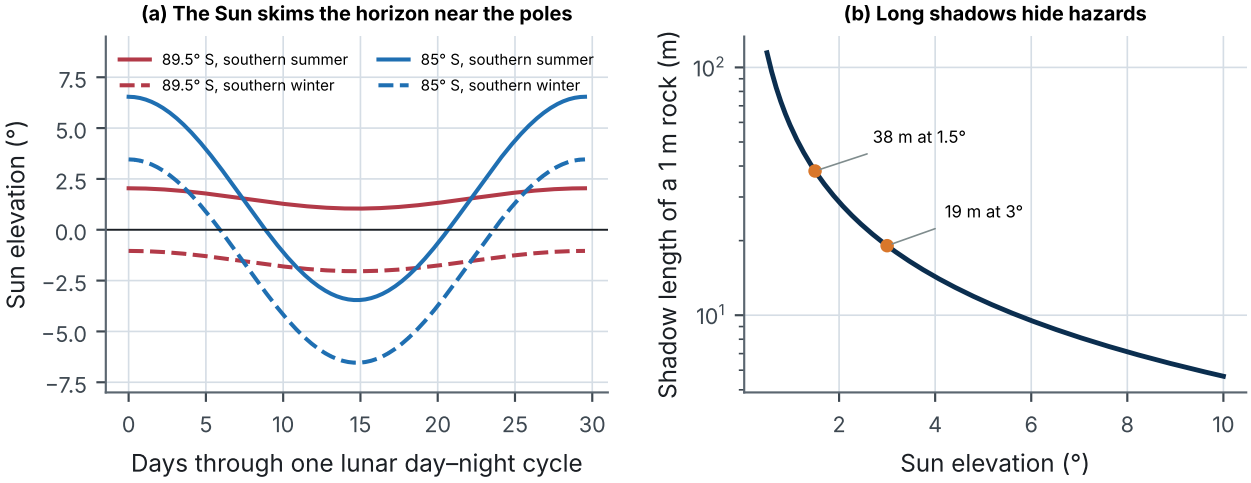
28.4.3 Dragonfly: Flying on Titan

Titan is the easiest place in the Solar System to fly. Its atmosphere is four times denser than Earth's and its gravity one-seventh, so hovering takes about 40 times less power per kilogram than on Earth (Figure 28.12). NASA's **Dragonfly**, a car-sized rotorcraft lander with eight rotors and an MMRTG, is scheduled for launch on a Falcon Heavy in July 2028 and arrival in 2034. It will land first in the Shangri-La dune fields. It will then fly from site to site — eventually more than 175 km in total — towards the 80 km Selk impact crater. There, liquid water and organic molecules may once have mixed. Its mass spectrometer and drill will analyse the surface for prebiotic chemistry. Dragonfly will recharge its batteries from the MMRTG between flights, making flights of several kilometres at a time, roughly once per Titan day (16 Earth days).

28.5 Rovers at the Lunar Poles

The next generation of lunar rovers is heading for the south pole, where water ice is thought to be trapped in **permanently shadowed regions**. These are crater floors that have not seen sunlight for billions of years and stay below −200 °C. The Moon's spin axis is tilted only 1.54° to the plane of its orbit around the Sun. As a result, the Sun near the poles never rises more than a few degrees above the horizon. It circles the sky once a month, casting shadows tens of times longer than the objects that cast them (Figure 28.14). This creates unique engineering problems:

- **Power and thermal.** Solar arrays must be vertical, and they must turn to follow the Sun around the horizon. A rover that drives into a shadow loses its power and starts to freeze within hours; NASA's VIPER was planned to travel between "safe havens" of sunlight, entering shadowed craters for short excursions only.
- **Navigation.** Low sun angles hide rocks and holes in long shadows and dazzle cameras pointed towards the Sun. Polar rovers carry their own headlights, and stereo vision and lidar must work in near-total darkness.
- **Communications.** Earth, too, hangs low on the horizon and is often hidden by terrain, so rovers need relay satellites or carefully planned routes to stay in contact.



The Moon's spin axis is tilted only 1.54° to the ecliptic, so near the poles the Sun circles the horizon and crater floors can stay dark for billions of years.

FIGURE 28.14 Lighting at the lunar south pole. (a) Sun elevation over one lunar day–night cycle at two latitudes. (b) Shadow length of a 1 m boulder at low Sun angles.

EXAMPLE: HOW LONG ARE THE SHADOWS AT THE POLE?

At 89.5° S during southern summer, the Sun stays about 2° above a flat horizon.

1. A boulder of height h casts a shadow of length $h / \tan e$, where e is the Sun's elevation: for $e = 2°$, a 1 m boulder casts a shadow 29 m long.
2. A crater only 10 m deep, with walls rising 5° from its floor, is entirely in shadow when the Sun is lower than 5° — which, at this latitude, is always.
3. On polar summits a few kilometres high, the Sun may be visible for 80–90 % of the year. This is why these "peaks of near-eternal light" are favoured sites for future lunar bases and their solar power stations.

28.6 The Next Generation

TABLE 28.3 Rovers and aerial explorers in development in 2026.

Mission	Body	Planned	Features
VIPER (NASA)	Moon, south pole	Late 2027, on Blue Origin Blue Moon MK1	Cancelled in 2024 and revived in 2025; prospects for water ice with a 1 m drill and a neutron spectrometer; can raise and lower each wheel to free itself
CADRE (NASA)	Moon	2027, on IM-3	Three suitcase-sized rovers that explore cooperatively as a team
Chang'e 7 (CNSA)	Moon, south pole	2027	Rover and a hopping probe to search shadowed craters for ice
Lunar Terrain Vehicle (NASA)	Moon	From about 2028	Unpressurised rover for Artemis crews, also driven remotely between missions; competitors Astrolab, Lunar Outpost and Intuitive Machines
Lunar Cruiser (JAXA/Toyota)	Moon	Early 2030s	Pressurised rover in which two astronauts can live and drive for weeks
IDEFIX (CNES/DLR) on MMX (JAXA)	Phobos	Launch Oct 2026; landing 2029	25 kg rover, first on a Martian moon
Rosalind Franklin (ESA)	Mars, Oxia Planum	Launch 2028; landing 2030	Drill to 2 m depth to reach organics shielded from radiation; NASA provides radioisotope heaters and landing engines
Dragonfly (NASA)	Titan	Launch 2028; arrival 2034	Rotorcraft lander



FIGURE 28.15 A mobility test model of NASA's VIPER lunar rover on a tilting bed of simulated regolith at NASA Glenn Research Center. Image: NASA/GRC (NASA/GRC/Bridget Caswell).

AI ON ROVERS

Rovers are among the most autonomous robots ever built. Besides autonomous navigation, Curiosity and Perseverance use **AEGIS**, software that selects rocks in navigation-camera images and aims the laser spectrometer at them without waiting for the ground. Perseverance's PIXL uses **adaptive sampling**, deciding on board where to dwell longer on interesting mineral grains. An onboard scheduler rearranges activities when tasks run faster or slower than planned. On the ground, machine-learning classifiers of terrain type (sand, bedrock, sharp rocks) trained on orbital and rover images help plan routes that spare the wheels. Future rovers will combine these into "science autonomy" — recognising what is scientifically interesting and acting on it (Chapters 48 and 51).

KEY TAKEAWAYS

- Rovers have explored the Moon since 1970 and Mars since 1997; Opportunity holds the distance record (45.2 km), with Perseverance (> 42 km) and Curiosity (> 37 km) still driving in 2026.
- Rocker-bogie suspensions, terramechanics, wheel design and traction control determine where rovers can go; soft sand and sharp rocks are the main hazards.
- Solar rovers are limited by dust and night-time cold; MMRTG-powered rovers have steady power day and night.
- Light-time forces sol-by-sol planning and growing autonomy: hazard-avoiding navigation, visual odometry, autonomous targeting.
- Rovers established that Mars was once habitable; Perseverance has cached samples, one with a potential biosignature, whose return to Earth is uncertain.
- Ingenuity proved that aircraft can fly on Mars; Dragonfly will fly on Titan, where flight is easiest of all.

REVIEW QUESTIONS

1. Compare the Lunokhod rovers, the Apollo Lunar Roving Vehicle and the Mars rovers in purpose, mobility and power.
2. Explain how a rocker-bogie suspension works and why it suits rough terrain.
3. What are the advantages and disadvantages of solar power and MMRTG power for Mars rovers?
4. Why must rover operations be planned sol by sol? How have autonomous navigation and visual odometry increased driving distances?
5. What evidence have rovers found that Mars was once habitable?
6. For an ideal rotor, hover power scales as $(mg)^{3/2}/\sqrt{(\rho A)}$. Compare the hover power of the same vehicle on Titan ($\rho = 5.4$ kg/m^3, $g = 1.35$ m/s^2) with that on Earth ($\rho = 1.225$ kg/m^3, $g = 9.81$ m/s^2).
7. Why was Ingenuity's first flight a major engineering achievement given Mars's thin atmosphere?
8. What special challenges do rovers face at the lunar poles?

29 CHAPTER Sample-Return Missions



Image: NASA (NASA/Keegan Barber)

The most powerful instruments for studying another world are not on spacecraft but in laboratories on Earth. Sample-return missions bring pieces of the Moon, asteroids, a comet and even the solar wind home, where they can be dated, dissected atom by atom and preserved for scientists not yet born. This chapter explains how samples are collected — by drill, scoop, gas jet, bullet or aerogel — how they are launched off other worlds and carried back through Earth's atmosphere at up to 13 km/s, and how they are kept pristine. It reviews every sample-return mission from Apollo to OSIRIS-REx and Chang'e 6, what their samples have revealed, and the missions under way in 2026, including China's Tianwen-2 at the asteroid Kamoʻoalewa and the uncertain future of Mars Sample Return.

IN THIS CHAPTER

29.1 Sample-Return Missions So Far
29.2 Engineering a Sample Return
29.3 Mars Sample Return and the Next Decade

Flight instruments must be small, light, low-powered and able to survive launch. Laboratories on Earth have none of these limits. They can measure the age of a rock to within a few million years from the decay of radioactive isotopes, and image individual atoms with transmission electron microscopes. They can also analyse organic molecules at concentrations of parts per billion, and probe grains with synchrotron X-rays. Above all, samples can be **kept**: techniques that do not yet exist can be applied decades later. In 2022, NASA opened a tube of lunar soil that Apollo 17 astronauts had sealed in a vacuum on the Moon in 1972, so that it could be studied with instruments invented since.

TABLE 29.1 What laboratories on Earth can do that flight instruments cannot.

Capability	Spacecraft instrument	Laboratory on Earth

Capability	Spacecraft instrument	Laboratory on Earth
Radiometric age	Rough estimates only (one experimental dating on Curiosity)	Precise ages from many isotope systems (U–Pb, Rb–Sr, Ar–Ar, Sm–Nd)
Spatial resolution	≈ 0.1 mm (PIXL on Perseverance)	Below 1 nm (electron microscopes, atom probes)
Organic analysis	A few target molecules	Thousands of compounds; chirality; isotopes of individual molecules
Number of techniques	A few, fixed at launch	Hundreds, improving every year
Re-analysis	Impossible once the mission ends	Samples curated for future generations

29.1 Sample-Return Missions So Far

Nine missions or programmes have returned extraterrestrial material to Earth (Figure 29.1). Their returns range from 382 kg of lunar rock to less than a milligram of comet dust and a fraction of a milligram of solar-wind atoms.

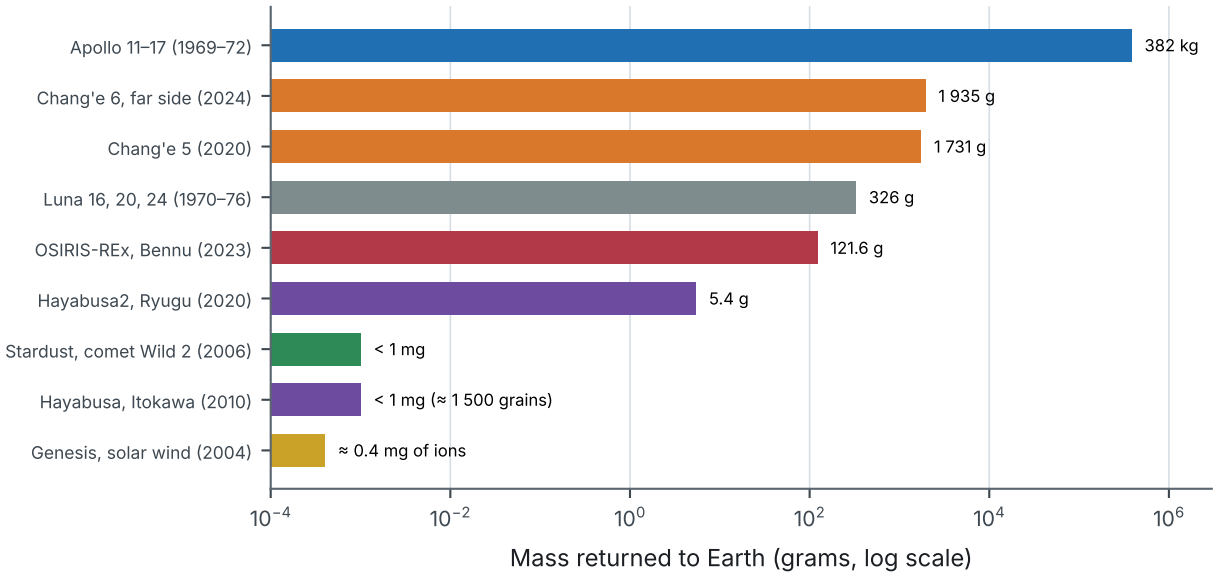


FIGURE 29.1 Mass of extraterrestrial material returned by each mission (log scale).

29.1.1 The Moon

The six **Apollo** landings (1969–72) brought back 382 kg of rocks, soil and cores — 2 196 samples in all — collected by astronauts trained in field geology (Figure 29.2). Three robotic Soviet landers, **Luna 16, 20 and 24** (1970–76), returned a total of 326 g, including a 2 m drill core. Their analyses showed that the Moon formed hot, with a global magma ocean whose floating crystals created the bright highland crust. They also supported the theory that it formed from debris thrown out when a Mars-sized body collided with the young Earth. Lunar samples also calibrated the **crater-counting chronology** used to estimate the ages of surfaces throughout the Solar System.



FIGURE 29.2 Apollo 17 geologist-astronaut Harrison Schmitt beside a large split boulder at Taurus–Littrow in December 1972. Image: NASA/JSC.

After 44 years, China resumed lunar sample return. **Chang'e 5** (November–December 2020) landed in Oceanus Procellarum and drilled and scooped 1 731 g of soil. It launched the soil into lunar orbit in an ascent stage that docked automatically with the orbiter — the first robotic rendezvous in lunar orbit. The sample was then returned to Inner Mongolia using a "skip" re-entry that bounced off the upper atmosphere to reduce heating. Its basalts are only about 2.0 billion years old, showing that the Moon was volcanically active far later than Apollo samples had suggested. **Chang'e 6** (May–June 2024) returned 1 935 g from the Apollo crater inside the South Pole–Aitken basin, the first samples from the Moon's far side, relaying its signals through the Queqiao-2 satellite. They include basalts about 2.8 billion years old and fragments from volcanism 4.2 billion years ago, and suggest that the far-side mantle was drier than the near side's.

29.1.2 A Comet and the Sun

NASA's **Stardust** flew 236 km from the nucleus of comet Wild 2 in January 2004 at 6.1 km/s. It captured thousands of dust particles in **aerogel** — a silica foam less than 1 % as dense as glass. The aerogel slowed them over a few millimetres without vaporising them (Figure 29.4). Its capsule re-entered Earth's atmosphere in January 2006 at 12.9 km/s, the fastest of any human-made object, protected by the first PICA (phenolic-impregnated carbon ablator) heat shield. The comet dust contained the amino acid glycine and, surprisingly, high-temperature minerals that could only have formed close to the young Sun. This was proof that material was mixed across the whole protoplanetary disk before the comets formed.

FIGURE 29.3 The Stardust collector: blocks of aerogel in an aluminium grid, like an ice-cube tray, before launch. Image: NASA/JPL (NASA/JPL).

FIGURE 29.4 Carrot-shaped tracks of comet particles that struck the aerogel at 6.1 km/s; the particles themselves lie at the ends of the tracks. Image: NASA/JPL (NASA/JPL).

NASA's **Genesis** spent 27 months at the Sun–Earth L1 point (2001–04) exposing wafers of ultra-pure silicon, sapphire, gold and diamond to the solar wind, collecting about 0.4 mg of solar atoms. On its return in September 2004 its parachute failed to open — an acceleration sensor had been installed upside down — and the capsule hit the Utah desert at more than 300 km/h. Many collectors survived, and their analysis showed that the Sun contains a higher proportion of the oxygen isotope ^{16}O than Earth, Moon and Mars, a clue to how the planets formed.

29.1.3 Asteroids

Japan's **Hayabusa** survived failed reaction wheels, a fuel leak, seven weeks without contact and failing ion engines to return about 1 500 tiny grains of the asteroid Itokawa in June 2010. They proved that the most common type of asteroid (S-type) is the source of the most common meteorites (ordinary chondrites), linking telescope spectra with laboratory samples.

Hayabusa2 reached the 900 m carbon-rich asteroid **Ryugu** in June 2018. It released two hopping rovers and the MASCOT lander, and touched down to collect surface grains in February 2019. It then fired a 2 kg copper projectile into the asteroid at 2 km/s to blast an artificial crater. In July 2019 it touched down beside the crater to collect material excavated from below the surface. Its capsule landed at Woomera, Australia, in December 2020 with 5.4 g of Ryugu. The samples are among the most primitive materials ever studied. They are chemically similar to the rare CI

chondrite meteorites and to the composition of the Sun, but altered by liquid water. They contain more than 20 amino acids and the RNA nucleobase uracil. The spacecraft continues: it flew past the asteroid Torifune (2001 CC21) in July 2026 and will rendezvous with the small, fast-spinning asteroid 1998 KY26 in 2031.

NASA's **OSIRIS-REx** reached the 490 m asteroid **Bennu** in December 2018 and found a surface far rougher than expected, covered in boulders (Figure 29.5), and occasionally ejecting small particles into space. On 20 October 2020 it descended to the site called Nightingale and made a **touch-and-go (TAG)** contact lasting about six seconds. Its sampler head released a burst of nitrogen gas that blew loose material into a collection chamber. The head sank almost half a metre into the surface, which turned out to behave like a pit of plastic balls. Had the spacecraft not fired its thrusters to back away, it would have sunk further into the asteroid. So much material was collected that rocks wedged the sampler's flap open and particles leaked out, and the team stowed the sample early.

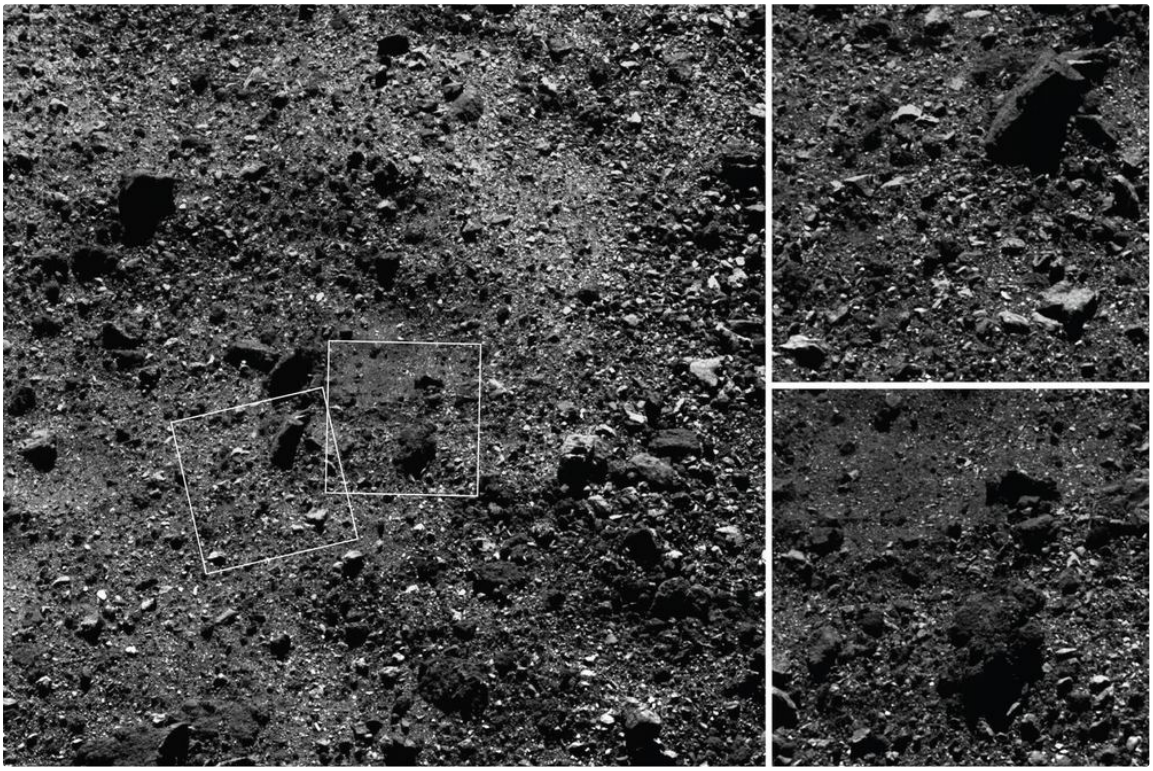


FIGURE 29.5 Bennu's northern hemisphere seen by OSIRIS-REx in 2019. The asteroid's surface is covered in boulders, making a safe sampling site hard to find. Image: NASA/GSFC.

The capsule landed in Utah on 24 September 2023 (Figure 29.6) with 121.6 g — twice the mission requirement. Two of the 35 fasteners on the sampler head resisted removal, and the curation team had to design new tools before they could open it fully in January 2024 (Figure 29.7). In January 2025 scientists reported that the Bennu samples contain 14 of the 20 amino acids used by life on Earth and all five nucleobases of DNA and RNA. The samples also contain abundant ammonia and a complete sequence of eleven salt minerals that crystallised as salty water evaporated on Bennu's parent body. The spacecraft, renamed **OSIRIS-APEX**, will reach the asteroid Apophis just after its close pass by Earth in April 2029.



FIGURE 29.6 The OSIRIS-REx sample return capsule shortly after landing in the Utah desert on 24 September 2023. Image: NASA (NASA/Keegan Barber).

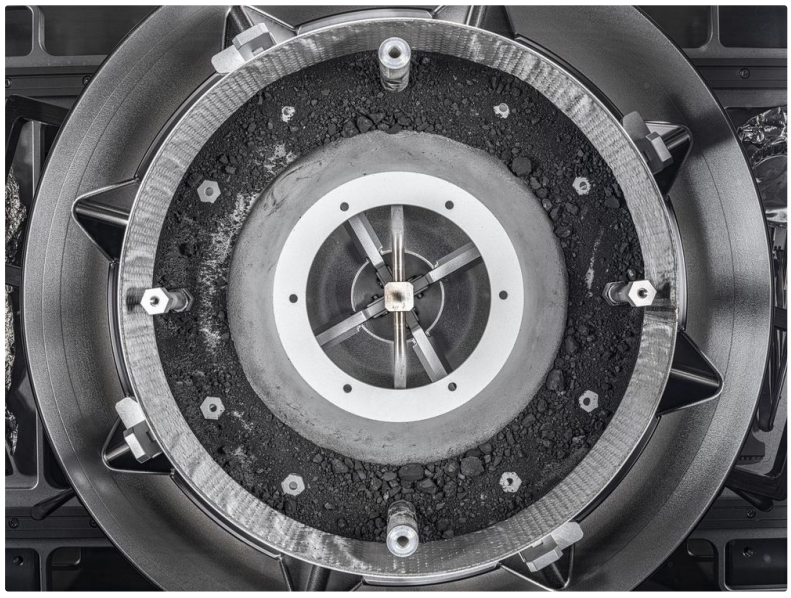


FIGURE 29.7 The OSIRIS-REx sampler head (TAGSAM) opened in the curation laboratory at NASA's Johnson Space Center in January 2024, with dark Bennu material visible around its rim. Image: NASA/JSC (Erika Blumenfeld & Joseph Aebers).

TABLE 29.2 Sample-return missions completed by 2026.

Mission	Agency	Target	Launch	Return	Sample	Collection method
Apollo 11–17	NASA	Moon	1969–72	1969–72	382 kg	Astronauts
Luna 16, 20, 24	USSR	Moon	1970–76	1970–76	326 g	Drill
Stardust	NASA	Comet Wild 2	1999	2006	< 1 mg	Aerogel during flyby
Genesis	NASA	Solar wind	2001	2004	≈ 0.4 mg	Collector wafers
Hayabusa	JAXA	Asteroid Itokawa	2003	2010	≈ 1 500 grains	Touchdown (projectile failed to fire)

Mission	Agency	Target	Launch	Return	Sample	Collection method
Hayabusa2	JAXA	Asteroid Ryugu	2014	2020	5.4 g	Touchdown with projectile, two sites
OSIRIS-REx	NASA	Asteroid Bennu	2016	2023	121.6 g	Touch-and-go with nitrogen gas
Chang'e 5	CNSA	Moon	2020	2020	1 731 g	Drill and scoop
Chang'e 6	CNSA	Moon, far side	2024	2024	1 935 g	Drill and scoop

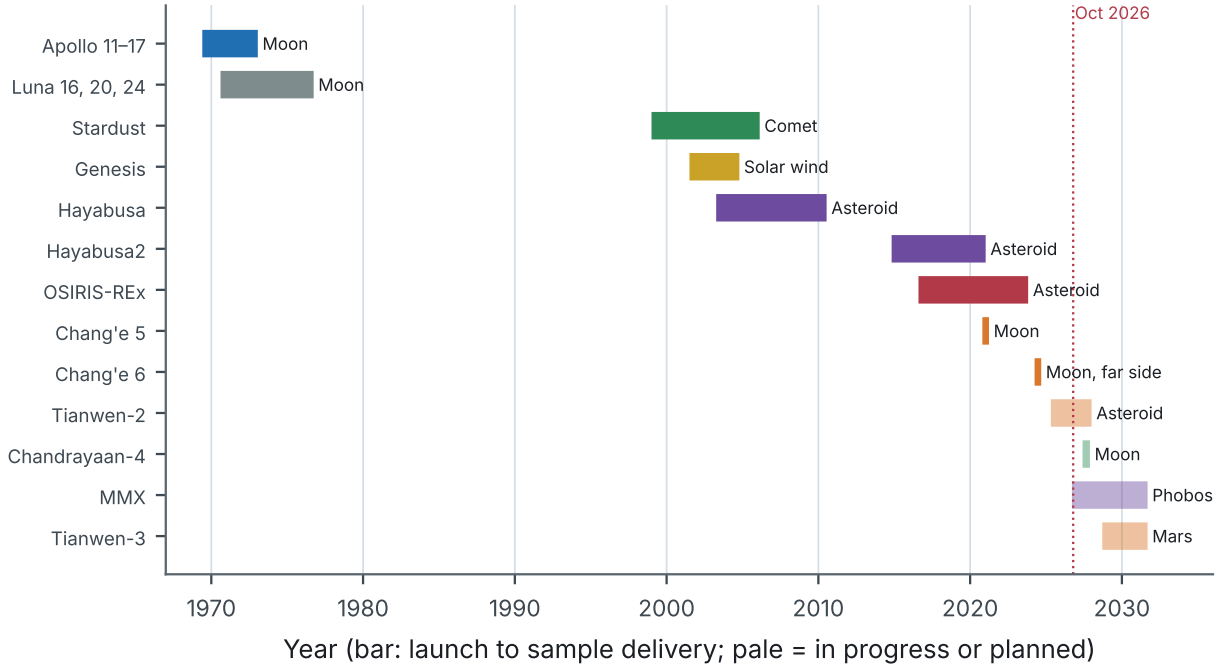


FIGURE 29.8 Sample-return missions from launch to delivery, including those under way or planned.

29.2 Engineering a Sample Return

A sample-return mission must do everything a lander does — and then reverse it: collect material, leave the surface, travel home and survive a high-speed entry into Earth's atmosphere, all without contaminating the sample.

29.2.1 Collecting

The method depends on the target (Figure 29.9). On the Moon, drills and scoops on robotic arms collect regolith and cores. On small asteroids with almost no gravity, a spacecraft cannot stand on the surface and push, so **touch-and-go** techniques are used. Hayabusa's sampler horn touches the surface for about a second while a 5 g tantalum bullet fired at 300 m/s throws fragments up into a catcher. OSIRIS-REx's TAGSAM used nitrogen gas to stir the regolith into its collector. Stardust captured dust during a fast flyby in aerogel, and Genesis simply exposed collectors to the solar wind.

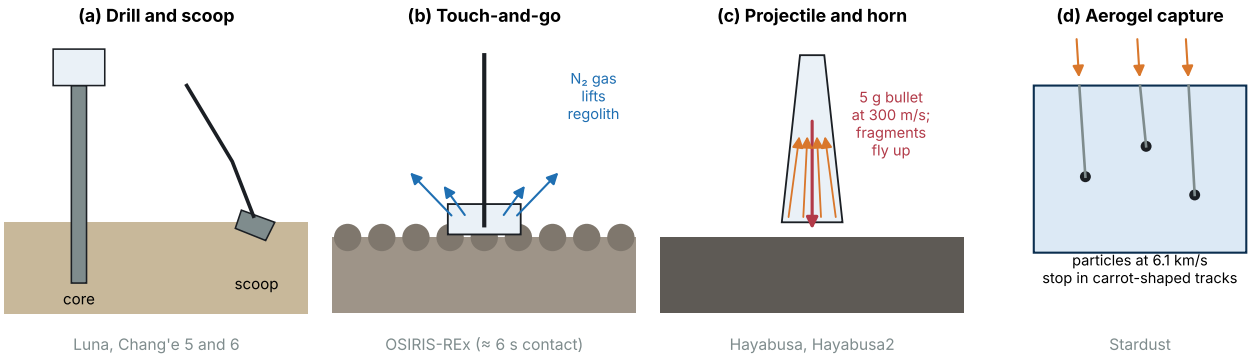
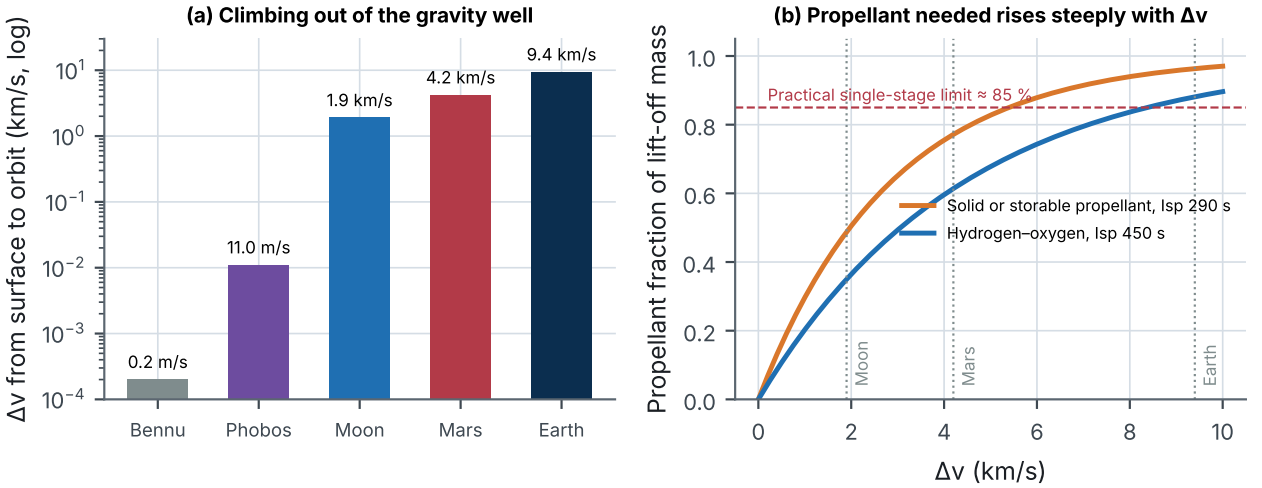


FIGURE 29.9 Sample-collection techniques.

Missions must also confirm what they have collected. OSIRIS-REx planned to measure its sample by spinning the spacecraft and measuring the change in its moment of inertia. When it found it had collected too much, it instead photographed the sampler head and stowed it quickly to minimise losses.

29.2.2 Leaving the Surface

Lifting a sample back into orbit requires a rocket — a small one on the Moon and a substantial one on Mars. On a small asteroid almost nothing is needed, as a gentle push exceeds the escape speed (Figure 29.10). Chang'e 5 and 6 used ascent stages that docked with the orbiter in lunar orbit and transferred the sample container, the same technique Apollo used with crews. For Mars, a **Mars Ascent Vehicle** must reach about 4.2 km/s, including losses, while surviving months on the surface at Martian temperatures. NASA's design was a two-stage solid rocket of about 400 kg lifting a 16 kg sample container — the first launch from another planet.



Δv includes typical gravity and drag losses. Phobos and Bennu: escape speeds; a sample can be lifted off with a gentle push.

FIGURE 29.10 Leaving the surface. (a) Ascent Δv to orbit, from a gentle push on an asteroid to 4.2 km/s on Mars. (b) Propellant fraction needed for two classes of propellant.

EXAMPLE: SIZING A MARS ASCENT VEHICLE

The ascent vehicle must deliver Δv = 4.2 km/s using solid motors with I_{sp} = 290 s (v_e = 2 844 m/s). Assume each stage's structure, nozzle and casing weigh 12 % of the stage's loaded mass (structural ratio ε = 0.12).

1. **Single stage.** Mass ratio $R = e^{4\,200/2\,844}$ = 4.38, so 77 % of the lift-off mass is propellant. The payload fraction is $(1/R - \varepsilon)/(1 - \varepsilon)$ = (0.228 − 0.12)/0.88 = **12.3 %**.
2. **Two equal stages,** each providing 2.1 km/s: R = 2.09 per stage; payload fraction per stage (0.478 − 0.12)/0.88 = 0.407; overall 0.407^2 = **16.5 %**.
3. Staging increases the payload by about a third and leaves margin for the real penalties. These are guidance, thermal protection, the launch structure and the need to fire after months in the cold. In NASA's design they reduced the payload fraction to about 4 %.
4. For comparison, a lunar ascent needs only 1.9 km/s: with storable propellants ($I_{sp} \approx$ 315 s), propellant is 46 % of the ascent stage's mass.

29.2.3 Coming Home

From the Moon, the return trip takes a few days; from asteroids and Mars, a year or more. Shortly before arrival the spacecraft releases a small **sample-return capsule** on a trajectory that intercepts Earth directly, and then diverts itself to fly past. The capsule enters at 11–13 km/s, much faster than a spacecraft returning from Earth orbit at 7.8 km/s. Because the heating rate rises roughly as the cube of the speed, a capsule entering at 12.4 km/s (OSIRIS-REx) experiences about four times the peak heat flux of one returning from low Earth orbit. Capsules use ablative heat shields of PICA or carbon phenolic and deploy parachutes at a few kilometres' altitude. They land in remote, closely controlled areas — the US Air Force's Utah Test and Training Range, the Woomera Range in South Australia, or Inner Mongolia in China. There, recovery teams wait with helicopters and clean containers.

29.2.4 Keeping Samples Pristine

A sample is only as valuable as its purity. Contamination by terrestrial organics, water or even fingerprints can ruin the search for prebiotic chemistry. Spacecraft that touch samples are built from a short list of approved, well-characterised materials, and **witness plates** fly alongside the samples to record any contamination. Capsules are purged with dry nitrogen immediately after landing, and the samples are opened and processed in glove boxes filled with pure nitrogen. This is done at dedicated **curation facilities**, such as NASA's at the Johnson Space Center (Figure 29.11) and JAXA's at Sagamihara. Fractions are shared among agencies: NASA received 10 % of the Ryugu sample, and JAXA and the Canadian Space Agency received portions of Bennu.



FIGURE 29.11 Scientists in clean-room suits examining asteroid Ryugu particles at NASA's Johnson Space Center, part of the sample exchanged with JAXA. Image: NASA/JSC (ROBERT MARKOWITZ NASA-JSC).

29.2.5 Planetary Protection for Returned Samples

International planetary protection policy places samples from the Moon, comets and most asteroids in the **unrestricted Earth return** category: they are not thought to pose any biological risk. Samples from Mars, Europa or Enceladus — places that might harbour life — fall into **restricted Earth return** (Category V). The mission must "break the chain of contact" with the target body, so that nothing that has touched Mars can reach Earth's environment outside a sealed container. The samples must be opened in a high-containment receiving facility comparable to a biosafety-level-4 laboratory until they are shown to be safe. The capsule designed for Mars Sample Return was to land without a parachute. A parachute that might fail was considered a greater risk than a deliberate impact onto a crushable, energy-absorbing structure designed to keep the sample container intact (Figure 29.12).



FIGURE 29.12 Artist's concept of the Earth Entry System designed for Mars Sample Return approaching Earth. Image: NASA/JPL (NASA/ESA/JPL-Caltech/GSFC).

29.3 Mars Sample Return and the Next Decade

Mars Sample Return (MSR) was planned as a NASA–ESA campaign to bring Perseverance's sealed tubes to Earth (Figure 29.13). A Sample Retrieval Lander would receive the tubes and launch them into Mars orbit on the Mars Ascent Vehicle; ESA's Earth Return Orbiter would capture the container and carry it home. As the design matured, the estimated cost rose to about US$ 8–11 billion and the return date slipped to the late 2030s or 2040. NASA studied cheaper alternatives in 2024–25, but the US budget for fiscal year 2026 provided no money for the existing programme. In October 2026 the samples, including one with a potential biosignature (Chapter 28), remain on Mars without a funded plan to retrieve them.

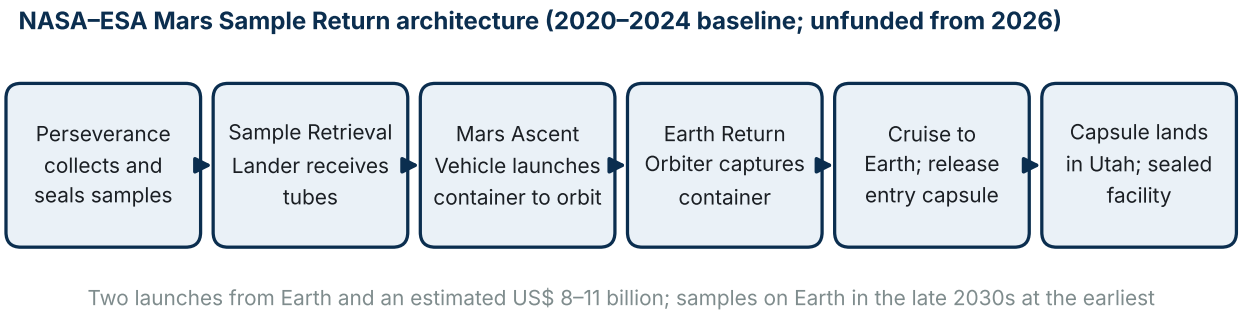


FIGURE 29.13 The NASA–ESA Mars Sample Return architecture and China's simpler Tianwen-3 approach.

China plans to reach the goal first. **Tianwen-3**, to be launched on two Long March 5 rockets in 2028, will land, drill and scoop at its own site, launch the samples to a waiting orbiter and return them to Earth in 2031. Other sample-return missions under way or planned include:

- **Tianwen-2** (CNSA): launched in May 2025, it arrived at the small near-Earth asteroid **Kamoʻoalewa**, a quasi-satellite of Earth that spins once every 28 minutes and may be a fragment of the Moon, in mid-2026. It is mapping the asteroid before sampling, and is due to return its capsule to Earth in late 2027 before continuing to the main-belt comet 311P.
- **MMX** (JAXA): scheduled for launch in October 2026 on an H3 rocket, it will orbit Mars in 2027. It will land on Phobos with the French–German IDEFIX rover in 2029 and collect at least 10 g of Phobos regolith with a coring sampler and pneumatic sampler. It will return the samples in 2031.
- **Chandrayaan-4** (ISRO): approved in 2024, it will be launched in two parts on two LVM-3 rockets. Its five modules will dock in Earth and lunar orbit to return about 3 kg from near the south pole, around 2027–28.

- **Artemis**: astronauts landing near the lunar south pole from the late 2020s will collect samples from the oldest and most volatile-rich regions of the Moon (Chapters 31 and 33).

AI IN SAMPLE-RETURN MISSIONS

Touch-and-go sampling depends on autonomous navigation: OSIRIS-REx used **Natural Feature Tracking**, comparing images taken during its descent with an onboard catalogue of Bennu's boulders. This let it update its trajectory and abort automatically if it was heading for a hazard. Hayabusa2 navigated using target markers it dropped on the surface. On Earth, machine learning segments X-ray computed-tomography scans of thousands of particles to classify minerals and find rare grains without destroying them. It also helps curators track every fragment as samples are subdivided among hundreds of laboratories (Chapter 51).

KEY TAKEAWAYS

- Laboratories on Earth can date, image and analyse samples far beyond the capability of flight instruments, and curated samples serve future generations.
- Nine missions or programmes have returned extraterrestrial material: Apollo, Luna, Stardust, Genesis, Hayabusa, Hayabusa2, OSIRIS-REx, Chang'e 5 and Chang'e 6.
- Collection techniques are matched to gravity: drills and scoops on the Moon, touch-and-go with gas or projectiles on asteroids, aerogel during a comet flyby.
- Leaving the surface costs 1.9 km/s from the Moon and 4.2 km/s from Mars; return capsules enter Earth's atmosphere at 11–13 km/s and need ablative heat shields.
- Samples revealed the Moon's violent origin and late volcanism, the mixing of the early Solar System, and abundant prebiotic molecules on carbon-rich asteroids.
- Mars samples require "restricted Earth return". NASA's Mars Sample Return lost its funding in 2026, while China's Tianwen-3 aims to return Martian samples in 2031; Tianwen-2, MMX and Chandrayaan-4 are under way or in preparation.

REVIEW QUESTIONS

1. Give three measurements that a laboratory on Earth can make on a returned sample but a flight instrument cannot, and explain why curation for future generations matters.
2. Why did Stardust use aerogel, and what did its comet particles reveal about mixing in the early Solar System?
3. Compare the sampling techniques of Hayabusa2 and OSIRIS-REx. Why are touch-and-go methods used on small asteroids?
4. Using the rocket equation, estimate the propellant fraction for a lunar ascent stage ($\Delta v = 1.9$ km/s, $I_{sp} = 315$ s). How does this compare with a Mars ascent?
5. A capsule returns from an asteroid at 12.4 km/s. By roughly what factor is its peak heating higher than that of a capsule returning from low Earth orbit at 7.8 km/s?
6. What is "restricted Earth return", and why did the Mars Sample Return capsule design omit a parachute?
7. What did the Chang'e 5 and Chang'e 6 samples reveal about the Moon's volcanic history?
8. List the organic molecules and minerals reported in the Bennu samples, and explain why they are significant for the origin of life.